

 Devin AI Export - PDF Version

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Source: Devin AI Platform

Overview

Relevant source files

- [README.md](<https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md>))
- [src/main/java/com/techx/tradex/notification/configurations/AppConf.java](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java>))
- [src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java>))
- [src/main/java/com/techx/tradex/notification/services/SocketClusterService.java](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java>))
- [src/main/resources/application.yaml](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml>))

Purpose and Scope

This notification service is a comprehensive multi-channel communication platform built with Spring Boot that processes and delivers notifications across email, SMS, push notifications, and real-time messaging channels. The service acts as a centralized dispatcher that receives notification requests via Kafka and HTTP endpoints, then routes them to appropriate delivery channels based on the specified method.

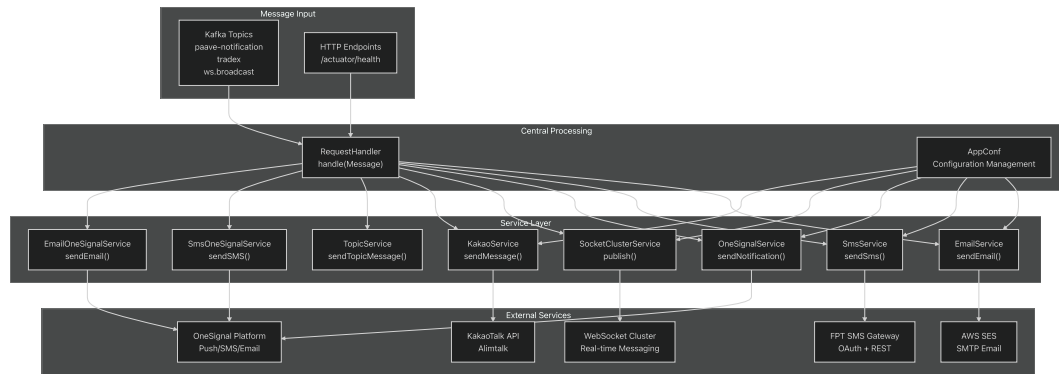
The service supports multiple external integrations including OneSignal for push notifications, KakaoTalk for Korean market messaging, FPT SMS for Vietnamese SMS delivery, and AWS SES for email. It includes a sophisticated template system with multi-language support (English, Vietnamese, Korean) and dynamic content generation using FreeMarker.

For detailed information about specific notification channels, see [Email Notification System](#), [SMS Notification System](#), and [Push Notification System](#). For template management details, see [Template System](#).

System Architecture Overview

The notification service follows a hub-and-spoke architecture with `RequestHandler` serving as the central message dispatcher. The system receives messages from Kafka topics and routes them to specialized service classes based on the notification method.

Core Architecture Components



Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L23-L139> [src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java23-139](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L23-L139) <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L23-L139> [\[](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L23-L139)

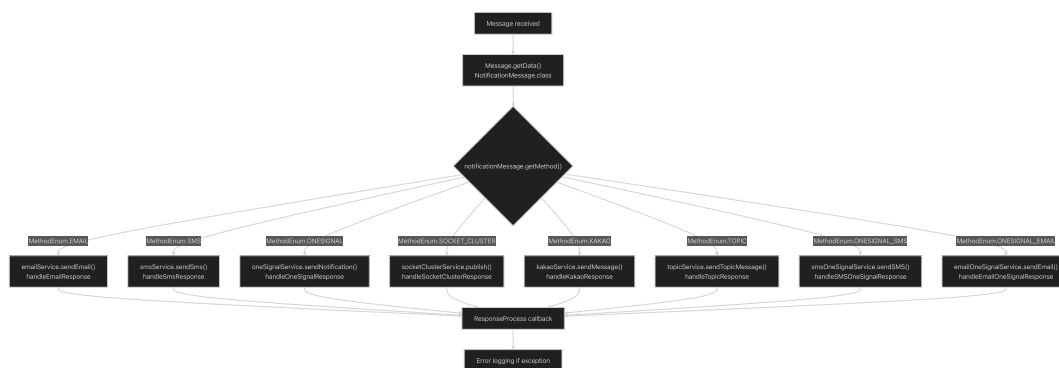
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[README.md24-36](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L24-L36) <https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L24-L36> <https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L24-L36> [\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L24-L36)

Message Processing Flow

The `RequestHandler` class extends `KafkaRequestHandlerAndSender` and implements a method-based routing system. Each incoming `NotificationMessage` contains a `MethodEnum` that determines which service will process the request.

Request Routing Logic



Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L108-L138> [src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java108-138](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L108-L138) <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L108-L138> [\[](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L108-L138)

[src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java36-75](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L36-L75) <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L36-L75> <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L36-L75> [\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L36-L75)

Configuration Management

The `AppConf` class provides centralized configuration for all notification channels and external service integrations. It uses Spring Boot's `@ConfigurationProperties` annotation to bind YAML configuration to strongly-typed Java objects.

Configuration Structure

Configuration Section	Purpose	Key Properties
<code>app.email</code>	SMTP email configuration	<code>endpoint</code> , <code>port</code> , <code>smtpUsername</code> , <code>sender</code>
<code>app.oneSignal</code>	OneSignal API configuration	<code>appId</code> , <code>apiKey</code>
<code>app.oneSignalMap</code>	Multi-tenant OneSignal configs	Keyed by domain (e.g., <code>nhsv</code>)
<code>app.smsServerMap</code>	SMS provider configurations	<code>url</code> , <code>type</code> , <code>user</code> , <code>senderName</code>
<code>app.kakao</code>	KakaoTalk integration	<code>url</code> , <code>companyCode</code> , <code>templateCodes</code>
<code>app.fptSms</code>	FPT SMS OAuth configuration	<code>authUrl</code> , <code>clientId</code> , <code>clientSecret</code>
<code>app.socketCluster</code>	WebSocket configuration	<code>hostname</code> , <code>port</code> , <code>usingKafka</code> , <code>kafkaTopic</code>
<code>app.templatesMap</code>	Template mappings by domain	Domain-specific template configurations

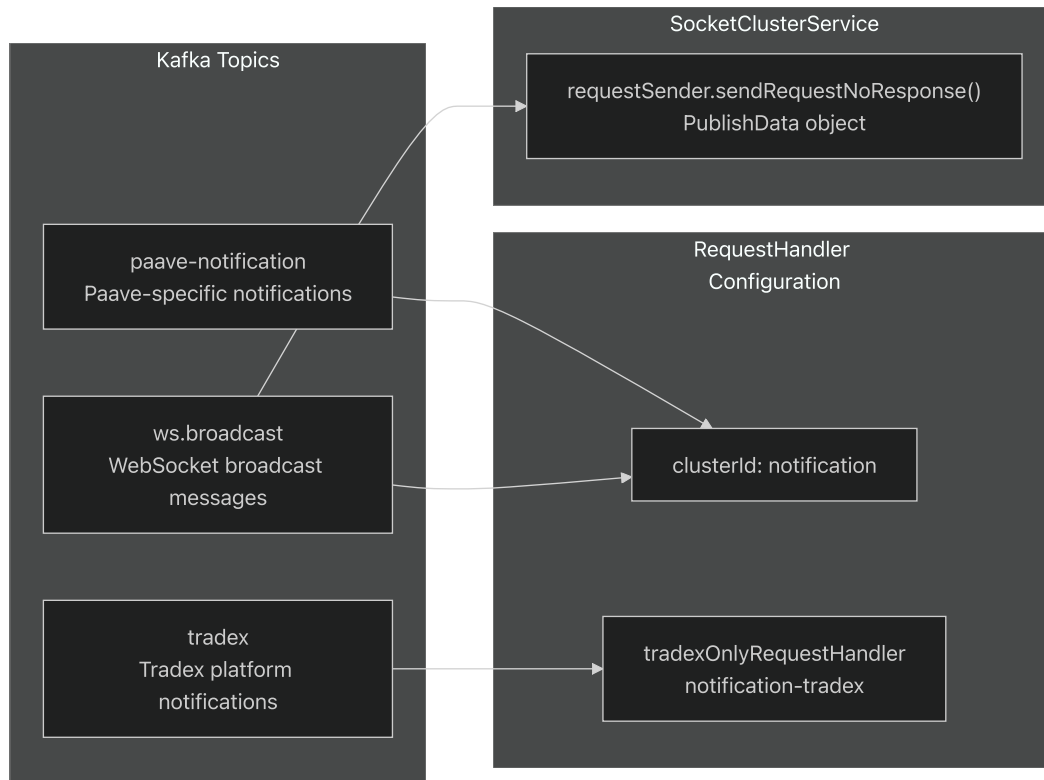
The configuration supports multi-tenant deployments where different domains can have separate configurations for the same services.

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L10-L138>
[src/main/java/com/techx/tradex/notification/configurations/AppConf.java10-138](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L10-L138) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L10-L138>)
[src/main/resources/application.yaml30-83](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L30-L83)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L30-L83>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L30-L83>)

Kafka Integration and Topics

The service integrates with Apache Kafka for reliable message processing and real-time communication. The `RequestHandler` subscribes to multiple Kafka topics and can also publish messages for WebSocket broadcasting.

Kafka Topic Configuration



The `RequestHandler` initializes with domain-specific topic subscriptions, adding the `tradex`-only handler when the domain equals "tradex". The `SocketClusterService` can publish to the `ws.broadcast` topic when `usingKafka` is enabled.

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L99-L105> [src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L99-L105](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L99-L105) [\[https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L99-L105\]](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L99-L105) [src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L110-L122](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L110-L122) [\[https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L110-L122\]](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L110-L122) [%5Bhttps://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L110-L122">%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L110-L122) [src/main/resources/application.yaml#L34-L39](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L34-L39) [\https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L34-L39 [\]\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L34-L39\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L34-L39)

Technology Stack

The notification service is built on the following technology foundation:

Component	Technology	Version	Purpose
Runtime	Java	8	Core platform
Framework	Spring Boot	2.0.0.RELEASE	Application framework
Build	Maven	3.6+	Dependency management
Database	MySQL	5.7+	Notification history storage
Cache	Redis	5.0+	Performance optimization
Message Queue	Apache Kafka	2.3.1	Reliable message processing
Template Engine	FreeMarker	2.3.26	Dynamic content generation
Web Services	Spring WS	2.x	SOAP service integration
Testing	JUnit	4 & 5	Unit and integration testing

The service requires external dependencies including `tradex-common-java` which provides shared models and Kafka utilities used throughout the notification system.

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L40-L48>)

[README.md#L40-L48](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L40-L48) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L40-L48>)

README.md#L134-L148 (<https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L134-L148>

<https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L134-L148>))

System Architecture

Relevant source files

- [[README.md](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md>))
- [[src/main/java/com/techx/tradex/notification/configurations/AppConf.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java>))
- [[src/main/java/com/techx/tradex/notification/constants/TopicMessageType.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/constants/TopicMessageType.java)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/constants/TopicMessageType.java> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/constants/TopicMessageType.java>))
- [[src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java>))

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src/main/java/com/techx/tradex/notification/dao/ZaloDao.java](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/ZaloDao.java>)

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src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java>)

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src/main/java/com/techx/tradex/notification/services/SocketClusterService.java](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java>)

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src/main/java/com/techx/tradex/notification/services/TopicService.java](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java>)

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src/main/resources/application.yaml](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml>)

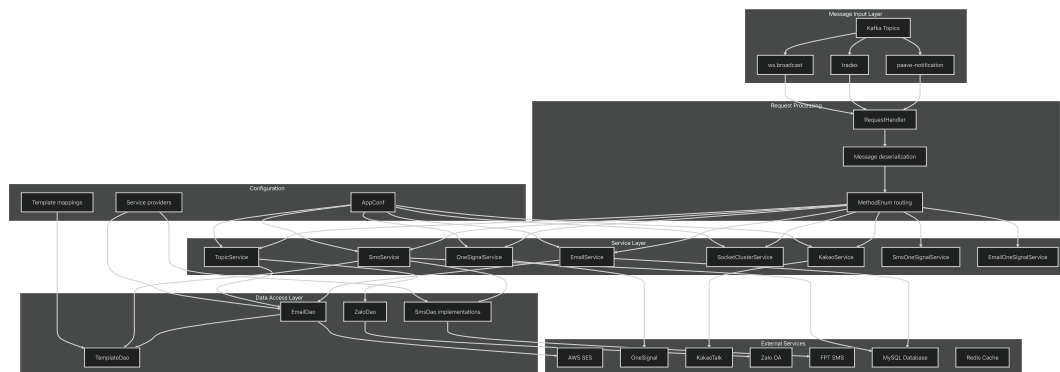
This document describes the high-level architecture of the notification service, focusing on message processing flows, channel routing, and external service integrations. It covers the core system components and their interactions but does not detail individual notification channels - for specific channel implementations, see [Email Notification System](#), [SMS Notification System](#), [Push Notification System](#), and [Messaging Platform Integrations](#). For template system details, see [Template System](#).

System Overview

The notification service is a Spring Boot microservice that processes messages from Kafka topics and routes them through multiple communication channels. The system supports email, SMS, push notifications, real-time messaging via WebSocket, and regional messaging platforms like KakaoTalk and Zalo.

Core Architecture

High-Level System Components



Sources: <https://github.com/nhsv->

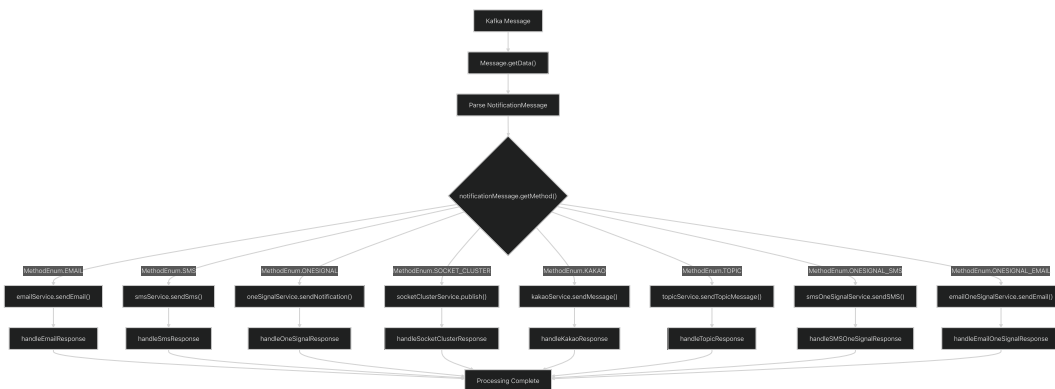
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<src/main/java/com/techx/tradex/notification/services/TopicService.java#L1-L162> <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L1-L162> [%5B">https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L1-L162](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L1-L162)

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Message Processing Flow



Sources: <https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L108-L139>

<src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L108-L139> <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/controllers/RequestHandler.java#L108-L139>

Configuration Architecture

The system uses a hierarchical configuration structure through `AppConf` that supports multiple providers and environments.

Configuration Structure

Configuration Section	Purpose	Key Components
<code>app.email</code>	Email service settings	<code>endpoint</code> , <code>port</code> , <code>sender</code> , <code>smtpUsername</code> , <code>smtpPassword</code>
<code>app.oneSignal</code>	OneSignal API configuration	<code>appId</code> , <code>apiKey</code>
<code>app.oneSignalMap</code>	Multi-tenant OneSignal configs	Brand-specific <code>appId</code> and <code>apiKey</code> mappings
<code>app.smsServerMap</code>	SMS provider configurations	Provider-specific <code>url</code> , <code>type</code> , <code>user</code> , <code>pass</code> settings
<code>app.kakao</code>	KakaoTalk API settings	<code>url</code> , <code>companyCode</code> , <code>templateCodes</code>
<code>app.zalo</code>	Zalo OA configuration	<code>accessToken</code> , <code>sendMessageUrl</code>
<code>app.socketCluster</code>	WebSocket configuration	<code>hostname</code> , <code>port</code> , <code>usingKafka</code> , <code>kafkaTopic</code>
<code>app.fptSms</code>	FPT SMS OAuth settings	<code>authUrl</code> , <code>sendUrl</code> , <code>clientId</code> , <code>clientSecret</code>

Sources: [_\(https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L10-L138)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L10-L138\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L10-L138)

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[src/main/resources/application.yaml30-83\]](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L30-L83) [_\(https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L30-L83)

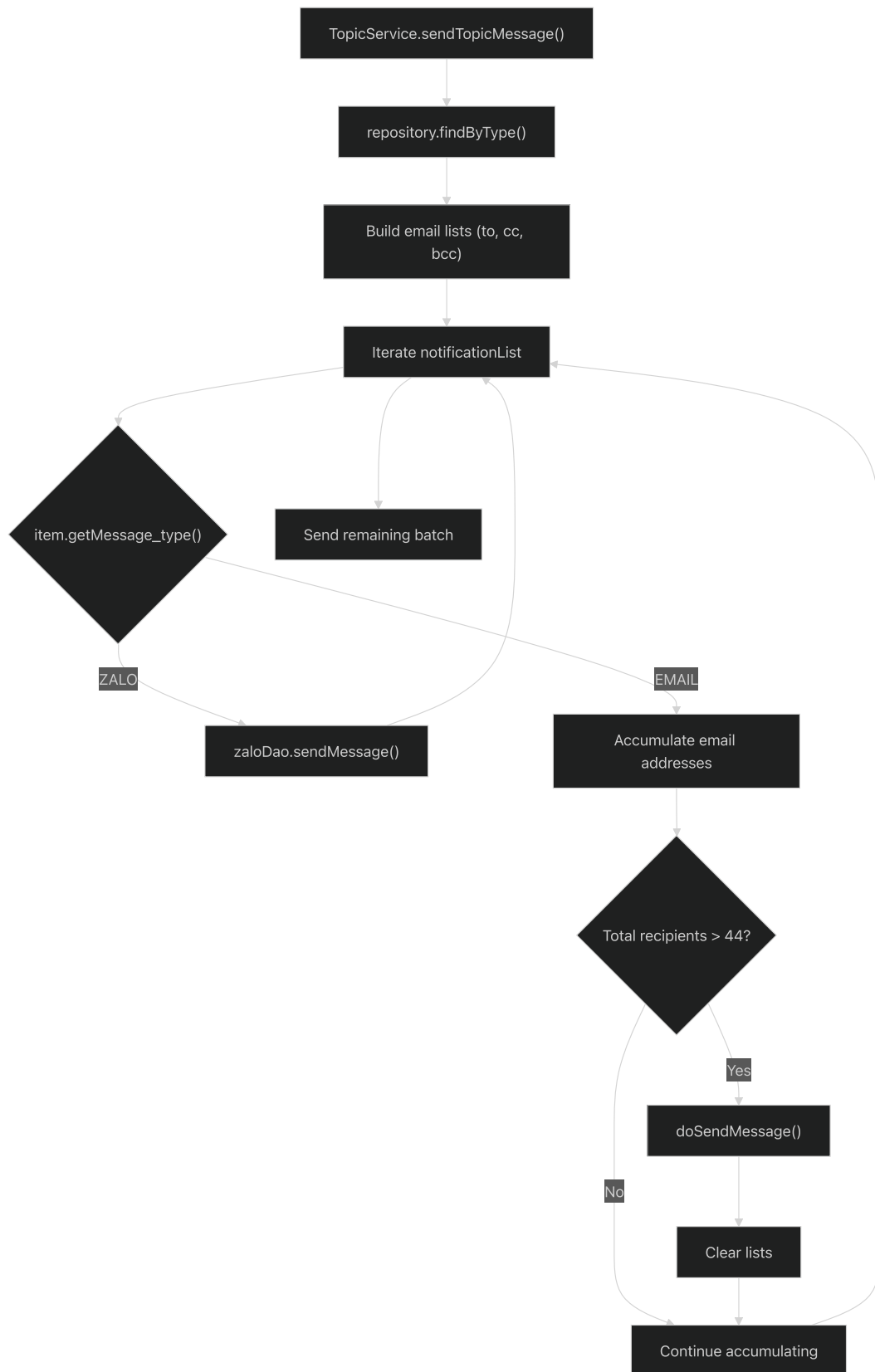
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[tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L30-L83\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L30-L83))

Topic-Based Bulk Messaging

The `TopicService` handles bulk message distribution by querying recipient lists from the database and routing messages through appropriate channels.

Topic Message Processing Flow



Sources: <https://github.com/nhsv->

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L52-L86](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L52-L86)

<https://github.com/nhsv->

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L52-L86](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L52-L86)

<https://github.com/nhsv->

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L95-118](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L95-118)

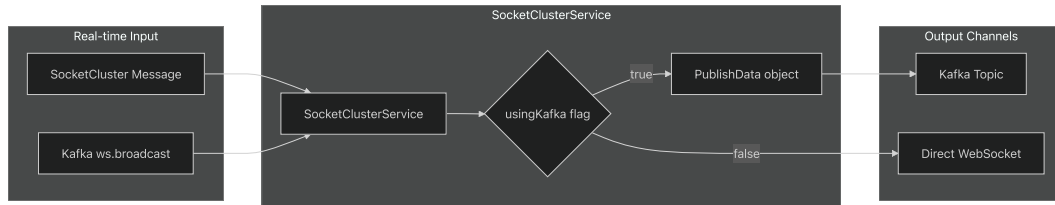
[L118](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L95-L118) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L95-L118)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L95-L118](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L95-L118)))

Real-Time Messaging Architecture

The `SocketClusterService` supports both direct WebSocket connections and Kafka-based message broadcasting for real-time notifications.

WebSocket Integration



Sources: ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L124-L156)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L124-L156](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L124-L156)).

[src/main/java/com/techx/tradex/notification/services/SocketClusterService.java124-156](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L124-L156) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L124-L156)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L124-L156](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L124-L156))[

[src/main/java/com/techx/tradex/notification/services/SocketClusterService.java55-104](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L55-104)]([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L55-104)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L55-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L55-104)

[L104](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L55-104) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L55-104)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L55-L104](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/SocketClusterService.java#L55-104)))

Database Integration

The system uses MySQL for persistence and Redis for caching, with JPA repositories for data access.

Data Access Pattern

Component	Repository/DAO	Purpose
<code>TopicService</code>	<code>NotificationListRepository</code>	Query bulk notification recipients
<code>EmailService</code>	<code>EmailDao</code>	Send email messages via SMTP
<code>SmsService</code>	Multiple SMS DAOs	Send SMS via different providers
<code>ZaloService</code>	<code>ZaloDao</code>	Send Zalo OA messages
Template rendering	<code>TemplateDao</code>	Load and process FreeMarker templates

Sources: ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L39-L41)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L39-L41](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L39-L41)).

[src/main/java/com/techx/tradex/notification/services/TopicService.java39-41](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L39-L41) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L39-L41)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L39-L41](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L39-L41))[

[src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java17-18](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L17-18)]([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L17-18)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L17-L18](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L17-18)

(<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L17-L18>))

External Service Integration Patterns

The system integrates with external services using different protocols and authentication methods:

Integration Protocols

Service	Protocol	Authentication	DAO Implementation
AWS SES	SMTP	Username/Password	EmailDao
OneSignal	REST API	API Key	OneSignalService
KakaoTalk	HTTP POST	Company Code	KakaoService
Zalo OA	REST API	Access Token	ZaloDao
FPT SMS	OAuth + REST	Client Credentials	SMS DAOs
SOAP SMS	SOAP/WSDL	Username/Password	SMS DAOs

Sources: ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L36-L61)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L36-L61](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L36-L61)).

[src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java36-61](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L36-L61) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L36-L61)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L36-L61](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L36-L61))[

[src/main/resources/application.yaml41-75](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L41-L75)]([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L41-L75)

[tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L41-L75](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L41-L75) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L41-L75)

[tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L41-L75](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L41-L75)))

Email Notification System

Overview

Relevant source files

- [[src/main/resources/templates/email_fss_warning_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html)
- [[src/main/resources/templates/email_fss_warning_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_vi.html)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_vi.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_vi.html)
- [[src/main/resources/templates/email_warning_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html)
- [[src/main/resources/templates/email_warning_ko.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_ko.html)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_ko.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_ko.html)

[tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_ko.html](#)))

- [

 src/main/resources/templates/email_warning_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_vi.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_vi.html))
- [

 src/main/resources/templates/welcome_to_paave_ekyc_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html))
- [

 src/main/resources/templates/welcome_to_paave_ekyc_ko.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_ko.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_ko.html))
- [

 src/main/resources/templates/welcome_to_paave_ekyc_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_vi.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_vi.html))
- [

 src/main/resources/templates/welcome_to_paave_kis_acc_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_en.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_en.html))
- [

 src/main/resources/templates/welcome_to_paave_kis_acc_ko.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_ko.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_ko.html))
- [

 src/main/resources/templates/welcome_to_paave_kis_acc_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_vi.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_vi.html))

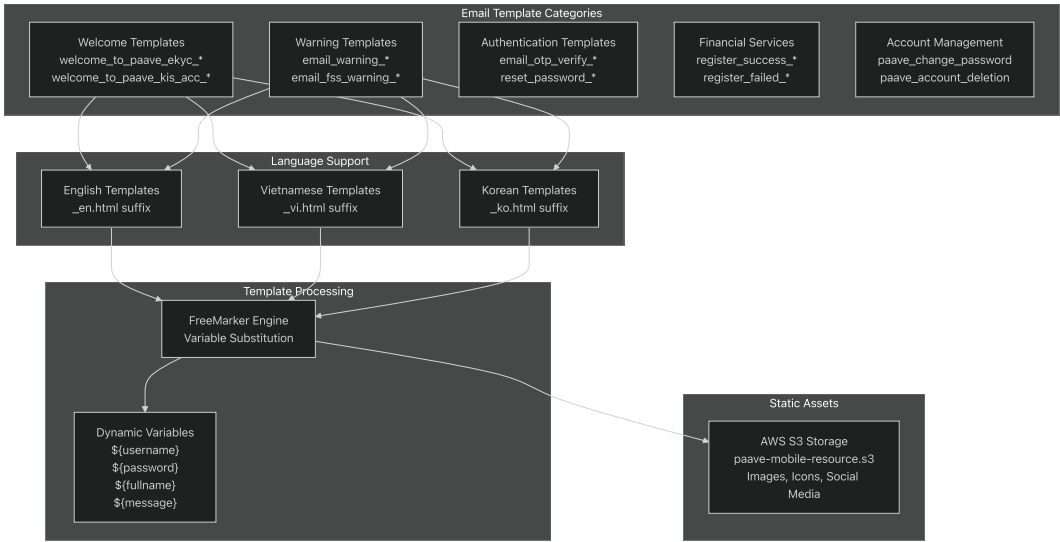
Purpose and Scope

The Email Notification System is a comprehensive template-based email delivery platform that supports multi-language communication across various business scenarios. It provides HTML email templates with dynamic content substitution using FreeMarker template engine, supports three languages (English, Vietnamese, Korean), and integrates with AWS S3 for static assets.

For information about SMS notifications, see [SMS Notification System](#). For push notifications, see [Push Notification System](#). For template system architecture details, see [Template System](#).

System Overview

The email notification system operates as a template-driven service that processes dynamic content through FreeMarker templating engine. It maintains separate HTML templates for different email types and languages, with consistent branding and styling across all communications.



Sources: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L1-L188).

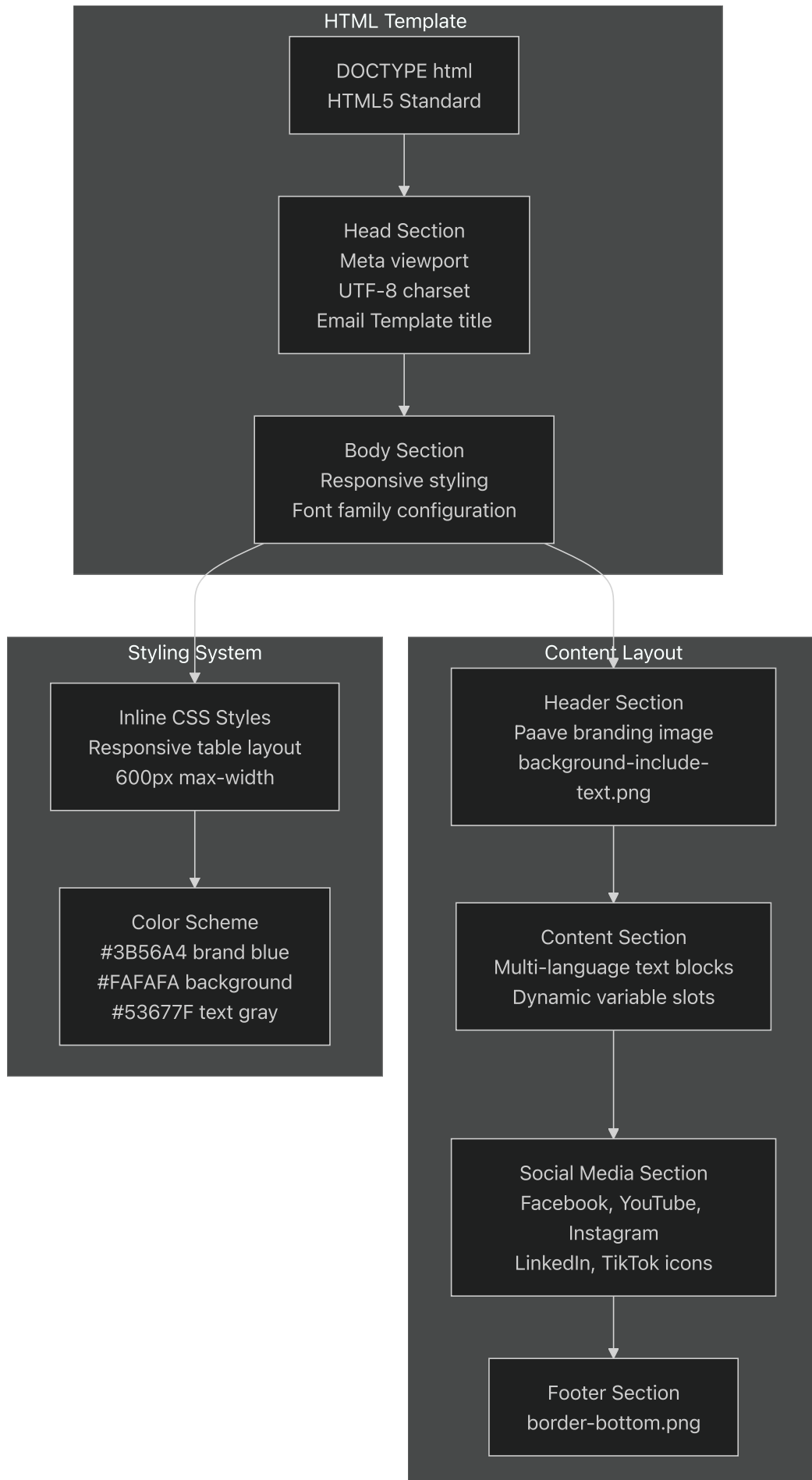
src/main/resources/templates/welcome_to_paave_ekyc_en.html1-188 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L1-L188) [https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L1-L188] ([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L1-L188)).

src/main/resources/templates/welcome_to_paave_ekyc_vi.html1-188] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_vi.html#L1-L188) [https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_vi.html#L1-L188] ([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_vi.html#L1-L188)).

Template Architecture

The email templates follow a consistent HTML structure with responsive design principles. Each template includes header branding, content sections, social media integration, and footer elements.

Template Structure Analysis



Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L1-L20)

src/main/resources/templates/welcome_to_paave_ekyc_en.html#L1-L20 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L1-L20)

src/main/resources/templates/welcome_to_paave_ekyc_en.html#L115-L173 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L115-L173) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L115-L173)

Multi-Language Support

The system provides comprehensive multi-language support through language-specific template files. Each email type maintains separate templates for English, Vietnamese, and Korean languages.

Language Template Organization

Email Type	English	Vietnamese	Korean
Welcome eKYC	<code>welcome_to_paave_ekyc_en.html</code>	<code>welcome_to_paave_ekyc_vi.html</code>	<code>welcome_to_paave_ekyc_ko.html</code>
Welcome KIS Account	<code>welcome_to_paave_kis_acc_en.html</code>	<code>welcome_to_paave_kis_acc_vi.html</code>	<code>welcome_to_paave_kis_acc_ko.html</code>
Warning Messages	<code>email_warning_en.html</code>	<code>email_warning_vi.html</code>	<code>email_warning_ko.html</code>
FSS Warnings	<code>email_fss_warning_en.html</code>	<code>email_fss_warning_vi.html</code>	-

The templates include bilingual content within single files, displaying both Vietnamese and English text sections for comprehensive user understanding.

Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L37-L72)

src/main/resources/templates/welcome_to_paave_ekyc_en.html#L37-L72 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L37-L72)

src/main/resources/templates/welcome_to_paave_ekyc_en.html#L77-L109 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L77-L109) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L77-L109)

Template Categories

Welcome and Onboarding Templates

Welcome templates support user onboarding for both eKYC and KIS account registration flows. These templates include account credentials and platform introduction content.

eKYC Welcome Templates:

- Display username and password credentials in highlighted sections
- Include platform mission statement and value proposition
- Provide support contact information and user manual links

KIS Account Welcome Templates:

- Personalized greeting using `${fullname}` variable
- Focus on investment platform benefits
- Exclude credential display (account-specific flow)

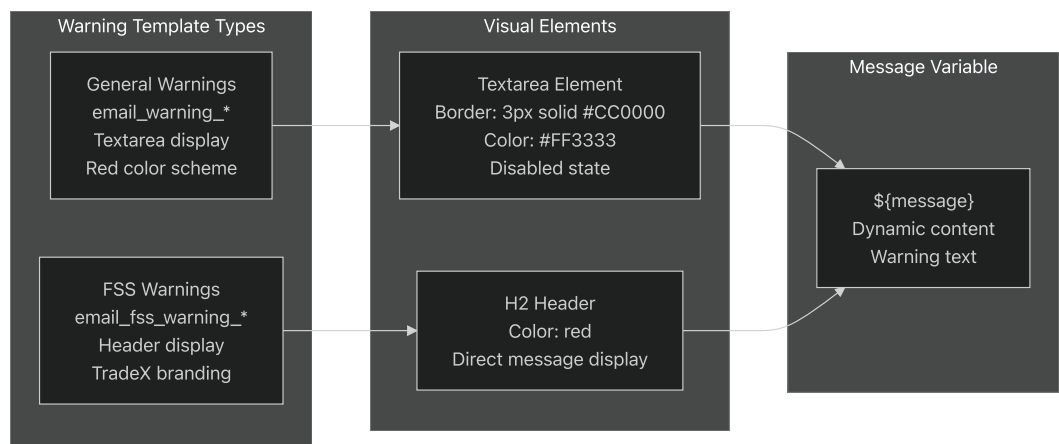
Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L50-L59

[src/main/resources/templates/welcome_to_paave_ekyc_en.html#L50-L59](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L50-L59) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L50-L59)

[src/main/resources/templates/welcome_to_paave_kis_acc_en.html#L37-L39](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_en.html#L37-L39) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_en.html#L37-L39)

Warning and Alert Templates

Warning templates handle system alerts and FSS (Financial Supervisory Service) specific notifications. These templates use distinct visual styling to emphasize urgency.



Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L39-L40

[src/main/resources/templates/email_warning_en.html#L39-L40](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L39-L40) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L39-L40)

[src/main/resources/templates/email_fss_warning_en.html#L29-L29](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html#L29-L29) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html#L29-L29)

Dynamic Content and Variables

The template system uses FreeMarker variables for dynamic content substitution. Common variables include user credentials, personalization data, and message content.

Variable Usage Patterns

Variable	Purpose	Templates	Example Usage
<code>\${username}</code>	User login credentials	Welcome eKYC	Account details section
<code>\${password}</code>	User login credentials	Welcome eKYC	Account details section
<code>\${fullname}</code>	User personalization	Welcome KIS	Greeting personalization
<code>\${message}</code>	Dynamic content	Warning templates	Alert message display

The variables are styled with consistent formatting:

- Blue color (#3B56A4) for credential highlights
- Bold font weight for emphasis
- Proper line height for readability

Sources: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L54-L57)

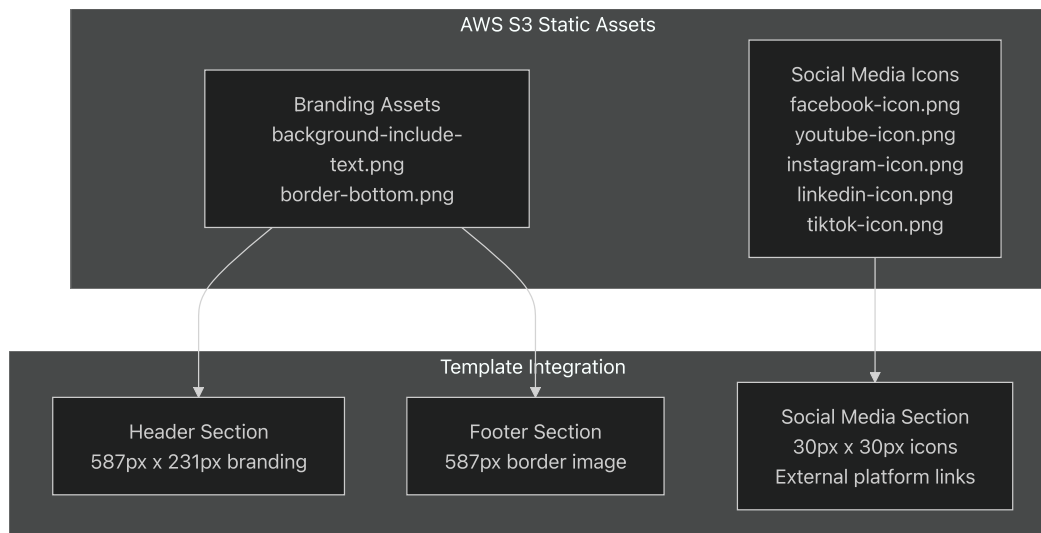
src/main/resources/templates/welcome_to_paave_ekyc_en.html#L54-L57 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L54-L57)

src/main/resources/templates/welcome_to_paave_kis_acc_en.html#L37-L39 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_en.html#L37-L39)

Static Asset Integration

The email templates integrate with AWS S3 for static asset delivery, including branding images and social media icons. All assets are hosted on the `paave-mobile-resource.s3.ap-southeast-1.amazonaws.com` domain.

Asset Categories



Social Media Integration

The templates include comprehensive social media integration with links to:

- Facebook: <https://www.facebook.com/paave.io>
- YouTube: <https://www.youtube.com/channel/UCIcUElAt7uhXVKkt4TJUtdA>
- Instagram: <https://www.instagram.com/paave.io/>

- LinkedIn: <https://www.linkedin.com/company/difisoft>
- TikTok: <https://www.tiktok.com/@paave.io>

Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L17-L18)

src/main/resources/templates/welcome_to_paave_ekyc_en.html#L17-L18 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L17-L18)

src/main/resources/templates/welcome_to_paave_ekyc_en.html#L132-L165 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L132-L165) ([%5B">https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L132-L165](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L132-L165))

src/main/resources/templates/welcome_to_paave_ekyc_en.html#L180-L181 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L180-L181) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L180-L181)

Welcome and Onboarding Emails

Relevant source files

- [src/main/resources/templates/welcome_to_paave_ekyc_en.html] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html)
- [src/main/resources/templates/welcome_to_paave_ekyc_ko.html] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_ko.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_ko.html)
- [src/main/resources/templates/welcome_to_paave_ekyc_vi.html] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_vi.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_vi.html)
- [src/main/resources/templates/welcome_to_paave_kis_acc_en.html] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_en.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_en.html)
- [src/main/resources/templates/welcome_to_paave_kis_acc_ko.html] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_ko.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_ko.html)
- [src/main/resources/templates/welcome_to_paave_kis_acc_vi.html] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_vi.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_vi.html)

This document covers the welcome and onboarding email templates used by the notification service for user registration and account setup workflows. These templates are part of the broader email notification system and handle the critical first impression emails sent to new users.

For authentication-related emails like OTP verification and password reset, see [Authentication Emails](#). For financial services registration confirmations, see [Financial Services Emails](#).

Purpose and Scope

Welcome and onboarding emails serve as the initial communication touchpoint for new users registering with Paave services. The system provides specialized templates for two distinct onboarding flows:

1. **Paave eKYC Registration** - Complete account setup emails including credentials
2. **KIS Account Welcome** - Welcome messages for KIS-integrated account registration

All templates support multi-language localization (English, Vietnamese, Korean) and include consistent branding, social media integration, and user guidance links.

Template Categories and Structure

Paave eKYC Welcome Templates

The eKYC welcome templates handle complete account provisioning workflows where users receive their initial login credentials.

Template File	Language	Variables Used
<code>welcome_to_paave_ekyc_en.html</code>	English/Vietnamese	<code>\${username}</code> , <code>\${password}</code>
<code>welcome_to_paave_ekyc_vi.html</code>	Vietnamese/English	<code>\${username}</code> , <code>\${password}</code>
<code>welcome_to_paave_ekyc_ko.html</code>	Korean/English	<code>\${username}</code> , <code>\${password}</code>

Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L1-L188

[src/main/resources/templates/welcome_to_paave_ekyc_en.html1-188](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L1-L188) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L1-L188)

[src/main/resources/templates/welcome_to_paave_ekyc_vi.html1-188](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_vi.html#L1-L188) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_vi.html#L1-L188)

[src/main/resources/templates/welcome_to_paave_ekyc_ko.html1-188](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_ko.html#L1-L188) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_ko.html#L1-L188)

KIS Account Welcome Templates

The KIS welcome templates provide personalized welcome messages for users completing KIS-integrated registration without exposing credentials.

Template File	Language	Variables Used
<code>welcome_to_paave_kis_acc_en.html</code>	English/Vietnamese	<code>\${fullname}</code>
<code>welcome_to_paave_kis_acc_vi.html</code>	Vietnamese/English	<code>\${fullname}</code>
<code>welcome_to_paave_kis_acc_ko.html</code>	Korean/English	<code>\${fullname}</code>

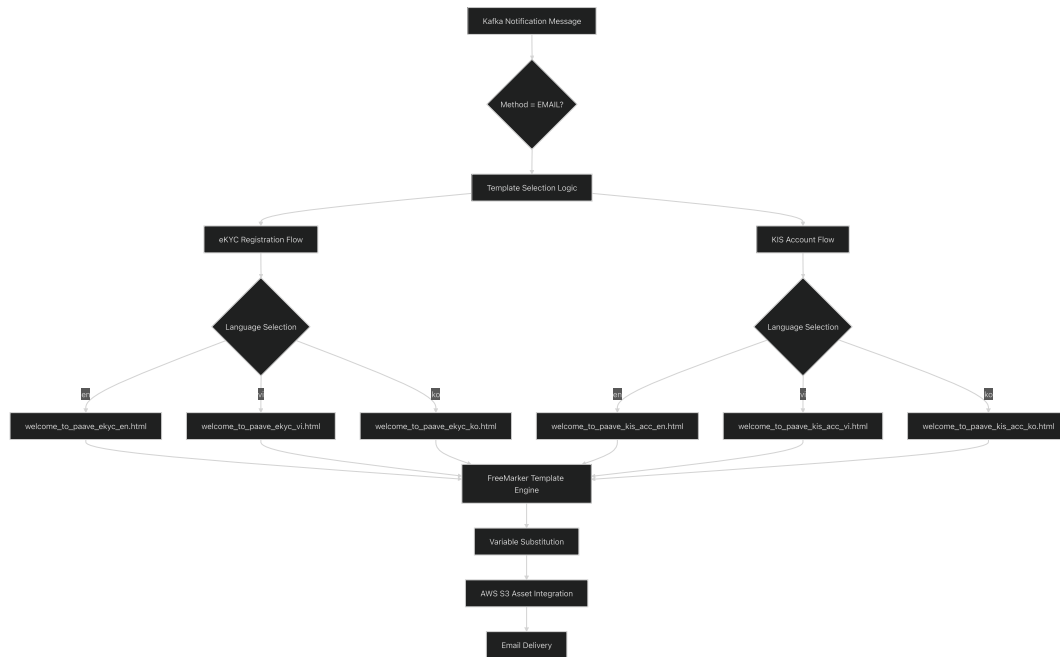
Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_en.html#L1-L174

[src/main/resources/templates/welcome_to_paave_kis_acc_en.html#L1-L174](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_en.html#L1-L174)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_en.html#L1-L174)]

[src/main/resources/templates/welcome_to_paave_kis_acc_vi.html#L1-L174](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_vi.html#L1-L174)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_vi.html#L1-L174)]

[src/main/resources/templates/welcome_to_paave_kis_acc_ko.html#L1-L174](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_ko.html#L1-L174)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_ko.html#L1-L174)]

Template Processing Flow



Sources: Template structure analysis from all welcome template files

FreeMarker Variable Usage

eKYC Template Variables

The eKYC templates utilize credentials-focused variables for complete account setup:

- `${username}` - User's login username[

[src/main/resources/templates/welcome_to_paave_ekyc_en.html#L54-L93](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L54-L93)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L54-L93)]

- `${password}` - Initial account password[

[src/main/resources/templates/welcome_to_paave_ekyc_en.html#L57-L96](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L57-L96)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L57-L96)]

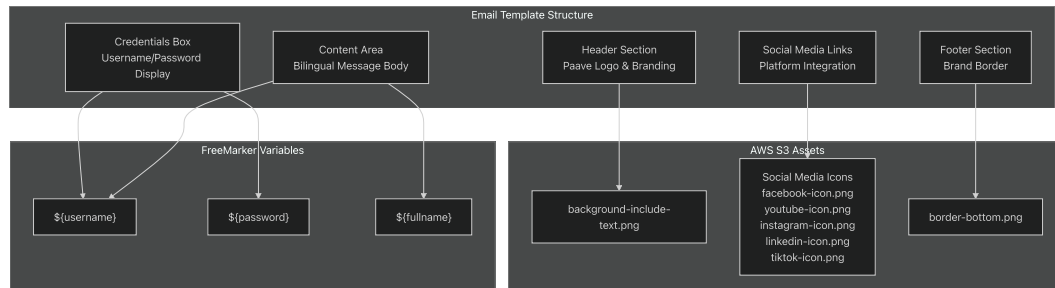
KIS Template Variables

The KIS templates use personalization variables for welcome messaging:

- `${fullname}` - User's complete name for personalization[

src/main/resources/templates/welcome_to_paave_kis_acc_en.html39-73](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_en.html#L39-L73)(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_en.html#L39-L73)

Email Template Architecture



Sources: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L17-L18)

src/main/resources/templates/welcome_to_paave_ekyc_en.html17-18 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L17-L18)

src/main/resources/templates/welcome_to_paave_ekyc_en.html17-18 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L17-L18)

src/main/resources/templates/welcome_to_paave_ekyc_en.html132-165](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L132-L165)

src/main/resources/templates/welcome_to_paave_ekyc_en.html132-165](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L132-L165)

src/main/resources/templates/welcome_to_paave_ekyc_en.html180-181](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L180-L181)

Multi-Language Implementation

All welcome templates implement a bilingual approach with both Vietnamese and English content sections within the same template file. This differs from single-language templates used elsewhere in the system.

Language Content Structure

Each template contains two main content blocks:

1. **Primary Language Section** - Usually Vietnamese content[

src/main/resources/templates/welcome_to_paave_ekyc_en.html34-73](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L34-L73)

2. **Secondary Language Section** - English translation[

src/main/resources/templates/welcome_to_paave_ekyc_en.html74-110](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L74-L110)

The Korean templates maintain the same dual-language structure with Korean as primary and English as secondary.

AWS S3 Asset Integration

All welcome templates integrate with AWS S3 for static asset delivery through the `paave-mobile-resource.s3.ap-southeast-1.amazonaws.com` bucket:

Asset Type	File Name	Usage
Header Logo	background-include-text.png	Main branding header
Social Icons	facebook-icon.png , youtube-icon.png , etc.	Social media links
Footer Border	border-bottom.png	Email footer styling

Sources: [_\(https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L18-L18)[tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L18-L18\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L18-L18)[src/main/resources/templates/welcome_to_paave_ekyc_en.html#L18-L18](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L18-L18) [_\(https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L18-L18)[tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L18-L18\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L18-L18) [\[](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L18-L18)[src/main/resources/templates/welcome_to_paave_ekyc_en.html#L134-L163\]](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L134-L163) [_\(https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L134-L163)
[tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L134-L163\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L134-L163) [\[](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L134-L163)
[_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L134-L163\)%5B\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L134-L163)[src/main/resources/templates/welcome_to_paave_ekyc_en.html#L181\]](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L181-L181) [_\(https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L181-L181)
[tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L181-L181](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L181-L181)
[_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L181-L181\)\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L181-L181)**User Experience Elements****Account Information Display**

eKYC templates feature a prominently styled credentials box for username and password display:

```

<div style="background-color: #EDF0F4; border-radius: 10px; padding: 15px; margin-bottom: 10px;">
  <p>Username: ${username}</p>
  <p>Password: ${password}</p>
</div>

```

Support and Documentation Links

All templates include consistent support resources:

- Email support: support@paave.io
- User manual links with language-specific URLs
- Social media platform links for community engagement

Sources: [_\(https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L62-L66)[tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L62-L66\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L62-L66)[src/main/resources/templates/welcome_to_paave_ekyc_en.html#L62-L66](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L62-L66) [_\(https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L62-L66)[tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L62-L66\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L62-L66) [\[](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L62-L66)[src/main/resources/templates/welcome_to_paave_ekyc_en.html#L100-L105\]](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L100-L105) [_\(https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L100-L105)
[tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L100-L105](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L100-L105)
[_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L100-L105\)\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L100-L105)**Integration with Notification System**

These welcome templates integrate with the broader notification service through the template selection and rendering pipeline. The system uses template naming conventions to route appropriate templates based on:

- Registration type (eKYC vs KIS)

- User language preference
- Brand context (Paave vs partner integrations)

For details on template selection logic and the FreeMarker integration, see [Template System](#).

Authentication Emails

Relevant source files

- [

 src/main/resources/templates/email_otp_verify_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html)
 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html))
- [

 src/main/resources/templates/email_otp_verify_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html)
 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html))
- [

 src/main/resources/templates/email_reset_password_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_en.html)
 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_en.html))
- [

 src/main/resources/templates/email_reset_password_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_vi.html)
 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_vi.html))
- [

 src/main/resources/templates/paave_email_otp_update_profile_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_update_profile_en.html)
 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_update_profile_en.html))
- [

 src/main/resources/templates/paave_email_otp_verify_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html)
 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html))
- [

 src/main/resources/templates/paave_email_otp_verify_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_vi.html)
 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_vi.html))
- [

 src/main/resources/templates/paave_reset_password_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_en.html)
 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_en.html))
- [

 src/main/resources/templates/reset_password_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_vi.html)
 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_vi.html))

Purpose and Scope

This document covers the authentication email templates and functionality within the notification service, specifically focusing on OTP verification and password reset email flows. Authentication emails are critical for user security, account verification, and password recovery across multiple brands and languages.

For general email template structure and multi-language support, see [Template System](#). For account lifecycle emails like welcome messages, see [Welcome and Onboarding Emails](#).

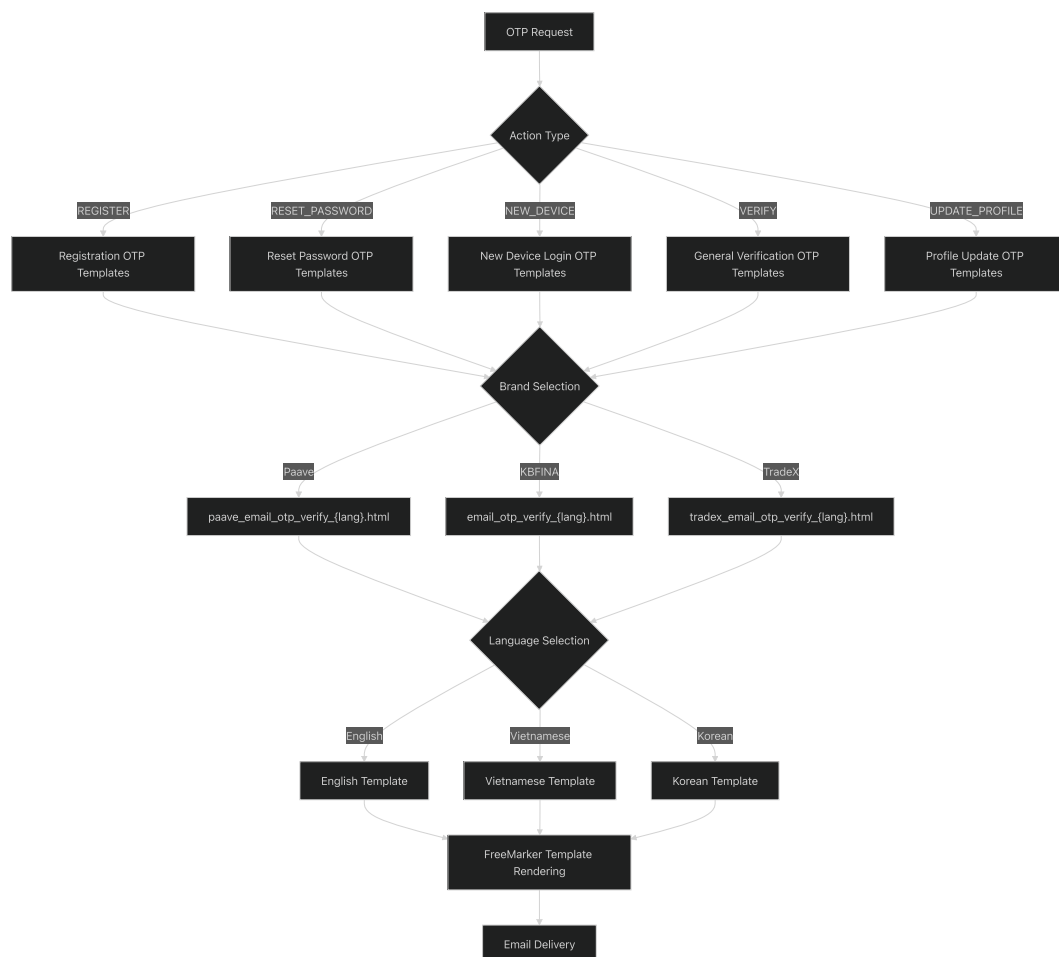
Authentication Email Categories

The notification service supports two primary categories of authentication emails, each with distinct template patterns and variable requirements:

Category	Purpose	Template Pattern	Key Variables
OTP Verification	Account registration, profile updates, login verification	<code>{brand}_email_otp_{action}_{lang}.html</code>	<code>\${otp}</code> , <code>\${action}</code> , <code>\${expiredInMinute}</code>
Password Reset	Account password recovery	<code>{brand}_reset_password_{lang}.html</code>	<code>\${resetCode}</code> , <code>\${resetPasswordUrl}</code> , <code>\${username}</code>

OTP Verification Email Templates

Template Structure and Flow



Paave OTP Templates

Paave OTP templates feature dual-language content within single templates and rich branding with social media integration:

Registration OTP Template: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L37-L53

[src/main/resources/templates/paave_email_otp_verify_en.html#L37-L53](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L37-L53) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L37-L53

- Includes both Vietnamese and English content sections
- Uses branded header image from AWS S3: `paave-mobile-resource.s3.ap-southeast-1.amazonaws.com`
- Features social media footer with platform links

Profile Update OTP Template: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_update_profile_en.html#L38-L48

[src/main/resources/templates/paave_email_otp_update_profile_en.html#L38-L48](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_update_profile_en.html#L38-L48) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_update_profile_en.html#L38-L48

- Specialized messaging for profile updates
- Same dual-language structure as registration template
- 5-minute expiration notice

KBFINA OTP Templates

KBFINA templates use conditional logic for action-specific messaging:

- 5-minute OTP expiration
- Branded social media footer

TradeX URL-Based Reset

TradeX implements clickable URL-based password reset:

Simple URL Reset: (<https://github.com/nhsv->

tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_vi.html#L54-L64)

[src/main/resources/templates/reset_password_v1.html54-64](#) (<https://github.com/nhsv>

tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_vi.html#L54-L64)

- Single reset button with `${resetPasswordUrl}` variable
- Fallback URL display for accessibility

Code + URL Hybrid: (<https://github.com/nhsv->

tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_en.html#L50-L62)

src/main/resources/templates/email_reset_password_en.html50-62 (<https://github.com/nhsv->

tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_en.html#L50-L62)

- Displays both reset code and clickable URL
- Expiration timestamp: `${expirationTime}`
- Constructed URL: `${baseUrl}/resetPassword/${username}/${resetCode}`

Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_en.html#L1-L155

[src/main/resources/templates/paave_reset_password_en.html1-155](#) (<https://github.com/nhsv->

tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_en.html#L1-L155]

src/main/resources/templates/reset_password_vi.html1-117](https://github.com/nhsv-

[tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_vi.html#L1-L117](#)]

(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_vi.html#L1-L117)%5B)

src/main/resources/templates/email_reset_password_en.html1-110](https://github.com/nhsv-

| [tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_en.html#L1-L110](#) | |

(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_en.html#L1-L110)%5B).

src/main/resources/templates/email reset password vi.html1-110](<https://github.com/nhsv->

[tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_vj.html#L1-L110](#)

(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_vi.html#L1-L110))

Template Variables and Localization

Core Authentication Variables



Multi-Language Support

The authentication email system supports three primary languages with different implementation approaches:

Language	Template Suffix	Implementation	Example
English	<code>_en.html</code>	Single or dual-language	<code>paave_email_otp_verify_en.html</code>
Vietnamese	<code>_vi.html</code>	Single or dual-language	<code>email_otp_verify_vi.html</code>
Korean	<code>_ko.html</code>	Single language	<code>paave_reset_password_ko.html</code>

Dual-Language Templates: Some Paave templates include both Vietnamese and English content within the same file, providing fallback language support.

Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_en.html#L37-L64)

[src/main/resources/templates/paave_reset_password_en.html#L37-L64](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_en.html#L37-L64) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_en.html#L37-L64) [https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_en.html#L37-L64]
[src/main/resources/templates/email_otp_verify_en.html#L96-L97](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html#L96-L97)] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html#L96-L97) [https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html#L96-L97]([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html#L96-L97))

[src/main/resources/templates/email_reset_password_en.html#L50-L77](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_en.html#L50-L77)] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_en.html#L50-L77) [https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_en.html#L50-L77]([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_en.html#L50-L77))

Brand-Specific Template Features

Paave Brand Features

AWS S3 Asset Integration: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L17-L18)

[src/main/resources/templates/paave_email_otp_verify_en.html#L17-L18](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L17-L18) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L17-L18)

- Header banner: `paave-mobile-resource.s3.ap-southeast-1.amazonaws.com/static/background-include-text.png`
- Social media icons: Facebook, YouTube, Instagram, LinkedIn, TikTok

Social Media Footer: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L88-L131)

[src/main/resources/templates/paave_email_otp_verify_en.html#L88-L131](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L88-L131) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L88-L131)

- Responsive social media link grid
- Platform-specific branded icons from S3

Tradex Brand Features

Minimalist Design: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_vi.html#L22-L24)

[src/main/resources/templates/reset_password_vi.html#L22-L24](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_vi.html#L22-L24) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_vi.html#L22-L24)

- Simple text-based logo
- Clean layout with single action button
- Subdued color scheme (#1ca70b green branding)

KBFINA Brand Features

Conditional Action Logic: [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L96-L96\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L96-L96)
[src/main/resources/templates/email_otp_verify_vi.html#L96-L96](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L96-L96) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L96-L96\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L96-L96)

- Dynamic subject line based on action type
- Vietnamese action translation mapping
- Consistent red header accent (#da4145)

Sources: [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L17-L131\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L17-L131)

[src/main/resources/templates/paave_email_otp_verify_en.html#L17-L131](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L17-L131) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L17-L131\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L17-L131)

[src/main/resources/templates/reset_password_vi.html#L22-L24](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_vi.html#L22-L24) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_vi.html#L22-L24\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_vi.html#L22-L24) [%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_vi.html#L22-L24) [\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_vi.html#L22-L24)

[src/main/resources/templates/email_otp_verify_vi.html#L96-L96](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L96-L96) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L96-L96\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L96-L96)

Integration with Notification Service

Authentication emails integrate with the broader notification service through the FreeMarker template engine and multi-channel routing system. Templates are loaded from the `src/main/resources/templates/` directory and processed with dynamic variable substitution based on the authentication context and user preferences.

The template selection logic considers brand affiliation, language preference, and authentication action type to deliver appropriately localized and branded email content for secure user authentication flows.

Warning and Alert Emails

Relevant source files

- [src/main/resources/templates/email_fss_warning_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html)
- [src/main/resources/templates/email_fss_warning_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_vi.html) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_vi.html\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_vi.html)
- [src/main/resources/templates/email_warning_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html)
- [src/main/resources/templates/email_warning_ko.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_ko.html) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_ko.html\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_ko.html)

- [

src/main/resources/templates/email_warning_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_vi.html (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_vi.html))

Purpose and Scope

This document covers the warning and alert email functionality within the notification system, which provides critical system notifications and compliance alerts to users. The system supports two distinct types of warning emails: general system warnings and Financial Supervisory Service (FSS) specific warnings, each with multi-language template support.

For information about other email types such as authentication emails, see [Authentication Emails](#). For account management notifications, see [Account Management Emails](#).

Warning Email Types

The notification system implements two categories of warning emails, each designed for specific use cases and regulatory requirements:

Warning Type	Template Prefix	Languages	Purpose
General Warning	email_warning_	English, Vietnamese, Korean	System alerts, security notifications, general user warnings
FSS Warning	email_fss_warning_	English, Vietnamese	Financial regulatory compliance alerts for Korean market

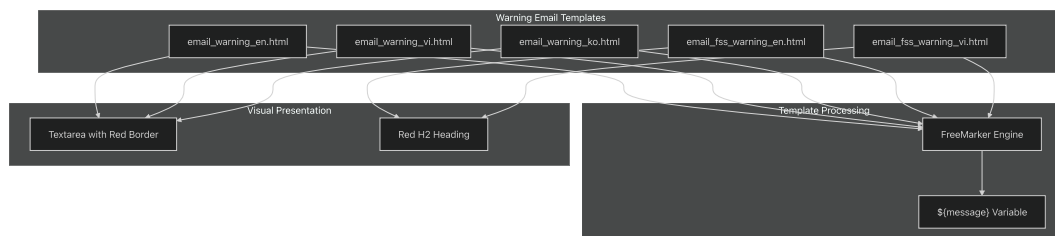
General Warning Emails

General warning emails use a distinctive visual design with prominent red styling to ensure critical messages are immediately noticeable. These templates utilize a textarea element with red borders and text to display warning content.

FSS Warning Emails

FSS (Financial Supervisory Service) warning emails are specifically designed for financial regulatory compliance in the Korean market. These use a simpler presentation format with red heading text, suitable for formal regulatory communications.

Template Architecture



Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L1-L66.

src/main/resources/templates/email_warning_en.html#L1-L66 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L1-L66)

src/main/resources/templates/email_warning_vi.html#L1-L66 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_vi.html#L1-L66)

([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_vi.html#L1-L66))

src/main/resources/templates/email_warning_ko.html1-66](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_ko.html#L1-L66)[

([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_ko.html#L1-L66))

src/main/resources/templates/email_fss_warning_en.html1-54](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html#L1-L54)[

([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html#L1-L54))

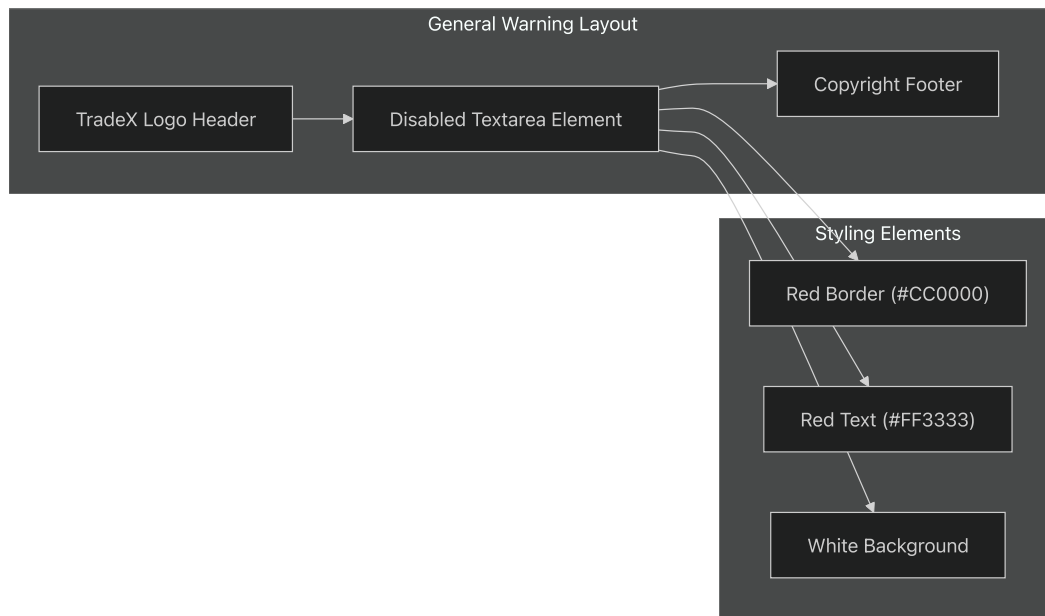
src/main/resources/templates/email_fss_warning_vi.html1-54](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_vi.html#L1-L54

(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_vi.html#L1-L54))

Template Structure and Styling

General Warning Template Design

General warning templates employ a visually striking design to ensure critical messages receive immediate attention:



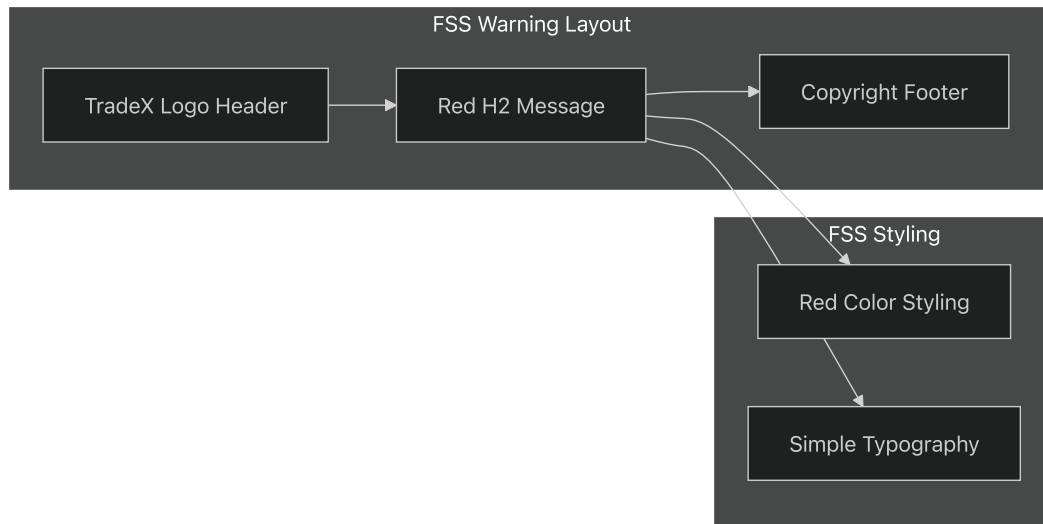
The general warning templates use specific CSS styling defined in (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L8-L16).

[src/main/resources/templates/email_warning_en.html8-16](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L8-L16) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L8-L16):

- Textarea border: `3px solid #CC0000` (red)
- Text color: `#FF3333` (red)
- Background: `#ffffff` (white)
- Disabled state to prevent user interaction

FSS Warning Template Design

FSS warning templates use a simpler, more formal presentation appropriate for regulatory communications:



The FSS templates display the warning message as a red H2 heading element as shown in https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html#L29-L29.

[src/main/resources/templates/email_fss_warning_en.html#L29-L29](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html#L29-L29) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html#L29-L29.

Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L8-L16.

[src/main/resources/templates/email_warning_en.html#L8-L16](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L8-L16) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L8-L16]

[src/main/resources/templates/email_warning_en.html#L39-L41](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L39-L41) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L39-L41]([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L39-L41)).

[src/main/resources/templates/email_fss_warning_en.html#L29-L29](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html#L29-L29) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html#L29-L29 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html#L29-L29).

Multi-Language Support

The warning email system provides comprehensive language support across different markets:

Language	General Warning Template	FSS Warning Template	Market Coverage
English	<code>email_warning_en.html</code>	<code>email_fss_warning_en.html</code>	International
Vietnamese	<code>email_warning_vi.html</code>	<code>email_fss_warning_vi.html</code>	Vietnamese market
Korean	<code>email_warning_ko.html</code>	Not available	Korean market

Language-Specific Template Features

All templates maintain consistent structure while supporting UTF-8 character encoding for proper display of non-Latin characters. The charset is explicitly set in the HTML meta tag: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L6-L6.

[src/main/resources/templates/email_warning_en.html#L6-L6](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L6-L6) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L6-L6.

Dynamic Content Variables

Both warning email types utilize the FreeMarker template engine with a single dynamic variable for message content:

Variable	Usage	Description
<code>\${message}</code>	Content placeholder	Dynamic warning message inserted at runtime

Variable Implementation

- **General warnings:** Variable rendered within textarea element[

src/main/resources/templates/email_warning_en.html40](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L40-L40)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L40-L40)

- **FSS warnings:** Variable rendered within H2 heading element[

src/main/resources/templates/email_fss_warning_en.html29](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html#L29-L29)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html#L29-L29)

The `${message}` variable allows for runtime injection of specific warning content while maintaining consistent visual presentation across all warning types.

Common Template Elements

All warning email templates share consistent branding and layout elements:

Header Section

- TradeX logo link pointing to `http://tradex.vn`
- Green brand color: `#1ca70b`
- Consistent typography using Arial font family

Footer Section

- Copyright notice: "© TradeX"
- Centered alignment with consistent styling
- Gray text color: `#839197`

Layout Structure

- XHTML 1.0 Transitional DOCTYPE
- Table-based layout for email client compatibility
- Responsive meta viewport tag
- 570px content width for optimal display

Sources: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L22-L24)

src/main/resources/templates/email_warning_en.html22-24 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L22-L24)

src/main/resources/templates/email_warning_en.html39-53](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L39-L53)
([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_warning_en.html#L39-L53))

src/main/resources/templates/email_fss_warning_en.html22-24](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html#L22-L24)
([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html#L22-L24))

src/main/resources/templates/email_fss_warning_en.html39-41](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html#L39-L41)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_fss_warning_en.html#L39-L41))

Financial Services Emails

Relevant source files

- [

 src/main/resources/templates/email_kis_ekyc_register_failed_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_en.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_en.html))
- [

 src/main/resources/templates/email_kis_ekyc_register_failed_ko.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_ko.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_ko.html))
- [

 src/main/resources/templates/email_kis_ekyc_register_failed_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html))
- [

 src/main/resources/templates/email_kis_ekyc_register_success_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_en.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_en.html))
- [

 src/main/resources/templates/email_kis_ekyc_register_success_ko.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_ko.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_ko.html))
- [

 src/main/resources/templates/email_kis_ekyc_register_success_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_vi.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_vi.html))
- [

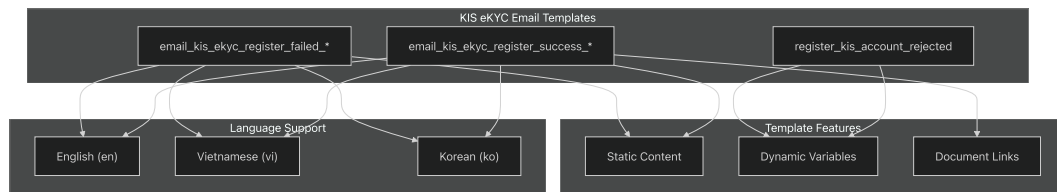
 src/main/resources/templates/register_kis_account_rejected.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/register_kis_account_rejected.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/register_kis_account_rejected.html))

This document covers the KIS eKYC (Korea Investment & Securities electronic Know Your Customer) email templates used for financial services compliance notifications. These templates handle the complete lifecycle of KIS account registration through the Paave platform, including success confirmations, failure notifications, and rejection messages with specific reasons.

For general authentication emails like OTP verification and password reset, see [Authentication Emails](#). For other account lifecycle emails, see [Account Management Emails](#).

Template Categories

The financial services email system consists of three main categories of KIS eKYC-related templates:



Sources: [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_en.html)

[tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_en.html)

[src/main/resources/templates/email_kis_ekyc_register_success_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_en.html) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_en.html

[src/main/resources/templates/email_kis_ekyc_register_failed_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html

[src/main/resources/templates/register_kis_account_rejected.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/register_kis_account_rejected.html) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/register_kis_account_rejected.html https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/register_kis_account_rejected.html

Registration Success Templates

The success templates notify users when their KIS account registration request has been successfully submitted. These templates are simple, informational emails that provide next steps and contact information.

Template File	Language	Purpose
<code>email_kis_ekyc_register_success_en.html</code>	English	Success notification for English users
<code>email_kis_ekyc_register_success_vi.html</code>	Vietnamese	Success notification for Vietnamese users
<code>email_kis_ekyc_register_success_ko.html</code>	Korean	Success notification for Korean users

Template Content Structure

All success templates contain:

- Confirmation message that the account opening request was submitted successfully
- Timeline expectation (account details within 1 day)
- Required document completion instructions
- KIS contact information for support
- Links to account opening contracts

The templates include both Vietnamese and English contract links:

- Vietnamese: [Hợp đồng mở tài khoản](https://drive.google.com/file/d/19K5Kd1r2x9UA7IDBe3la4CKoGQgbkRsF/view?usp=sharing) <https://drive.google.com/file/d/19K5Kd1r2x9UA7IDBe3la4CKoGQgbkRsF/view?usp=sharing>
- English: [Account Opening Contract](https://drive.google.com/file/d/14dY9zcBYRp8odEaJMNHVZVse2IM4V8Tf/view?usp=sharing) <https://drive.google.com/file/d/14dY9zcBYRp8odEaJMNHVZVse2IM4V8Tf/view?usp=sharing>

Sources: [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_en.html#L12-L18)

[tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_en.html#L12-L18](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_en.html#L12-L18)

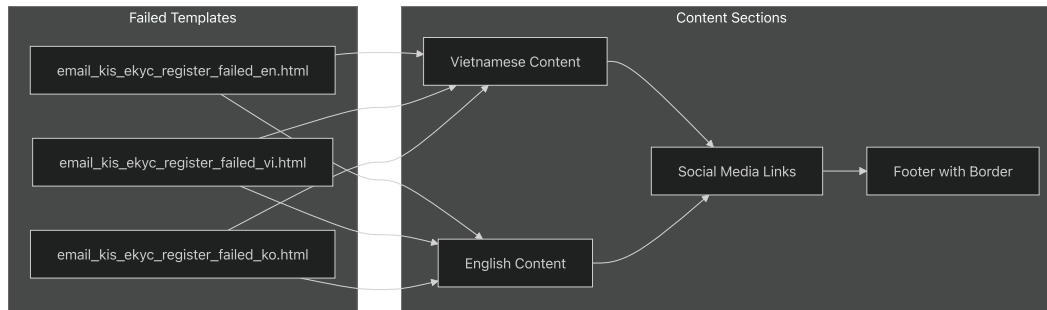
[src/main/resources/templates/email_kis_ekyc_register_success_en.html12-18](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_en.html#L12-L18) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_en.html#L12-L18

[src/main/resources/templates/email_kis_ekyc_register_success_vi.html21-23](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_vi.html#L21-L23) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_vi.html#L21-L23

[L23](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_vi.html#L21-L23) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_vi.html#L21-L23))

Registration Failed Templates

The failed templates notify users when their KIS account registration has been rejected without specific reason details. These templates provide general guidance for next steps.



Dual Language Support

Each failed template contains both Vietnamese and English content sections within the same email:

Vietnamese Section. (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L34-L41)

src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L34-L41 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L34-L41):

- "Bạn thân mến" greeting
- Rejection notification message
- Alternative options (retry or use virtual trading)
- Support contact information

English Section. (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L46-L52)

src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L46-L52 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L46-L52):

- "Dear Customer" greeting
- Parallel content in English
- Same support and alternative options

Sources: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_en.html#L31-L54)

src/main/resources/templates/email_kis_ekyc_register_failed_en.html#L31-L54 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_en.html#L31-L54)

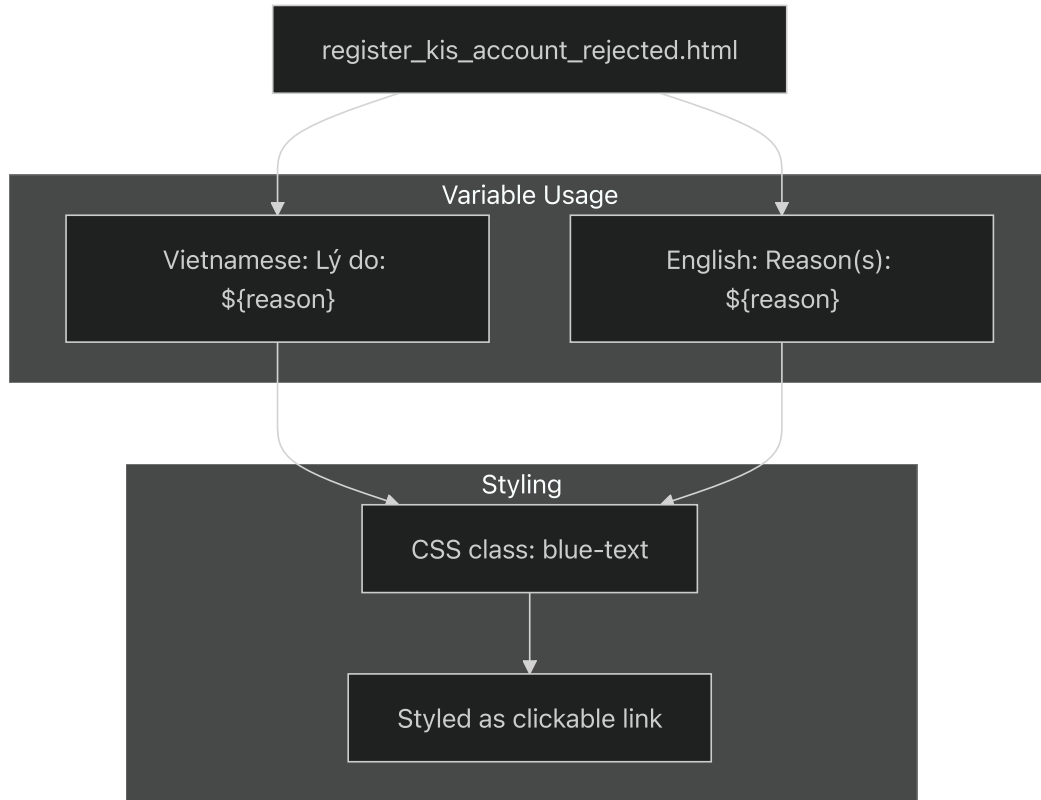
src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L31-L54 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L31-L54)

Registration Rejected Template

The `register_kis_account_rejected.html` template provides detailed rejection notifications with specific reasons using FreeMarker variables.

Dynamic Variable Support

The template uses the `${reason}` variable to display specific rejection reasons:



The reason variable is displayed with special styling using the `blue-text` CSS class:

- Vietnamese: Lý do: `<a class="blue-text">${reason}</a>` [
 [src/main/resources/templates/register_kis_account_rejected.html312](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/register_kis_account_rejected.html#L312-L312)
https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/register_kis_account_rejected.html#L312-L312
- English: Reason(s): `<a class="blue-text">${reason}</a>` [
 [src/main/resources/templates/register_kis_account_rejected.html324](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/register_kis_account_rejected.html#L324-L324)
https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/register_kis_account_rejected.html#L324-L324

Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/register_kis_account_rejected.html#L312-L312

[src/main/resources/templates/register_kis_account_rejected.html312](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/register_kis_account_rejected.html#L312-L312) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/register_kis_account_rejected.html#L312-L312

[src/main/resources/templates/register_kis_account_rejected.html324](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/register_kis_account_rejected.html#L324-L324) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/register_kis_account_rejected.html#L324-L324

Visual Design Structure

All KIS eKYC templates follow the standard Paave email design pattern:

Header Section

- Paave banner header image from AWS S3

- Consistent branding across all templates

Content Areas

- Background color: #FAFAFA [

src/main/resources/templates/email_kis_ekyc_register_failed_vi.html31](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L31-L31)(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L31-L31))

- Rounded corners: border-radius: 10px
- Padding: 15px for content spacing

Social Media Integration

All templates include social media links with platform-specific icons:

- Facebook: https://www.facebook.com/paave.io
- YouTube: https://www.youtube.com/channel/UCIcUElAt7uhXVKkt4TJUtdA
- Instagram: https://www.instagram.com/paave.io/
- LinkedIn: https://www.linkedin.com/company/difisoft
- TikTok: https://www.tiktok.com/@paave.io

Footer

- Border bottom image from AWS S3
- Consistent visual closure

Sources: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L14-L15).

src/main/resources/templates/email_kis_ekyc_register_failed_vi.html14-15(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L14-L15)[

src/main/resources/templates/email_kis_ekyc_register_failed_vi.html75-108](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L75-L108)[(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L75-L108)%5B).

src/main/resources/templates/email_kis_ekyc_register_failed_vi.html134-136](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L134-L136)(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L134-L136)).

Contact Information

All templates provide consistent support contact information:

- Primary support email: support@paave.io
- KIS customer support phone: 0283914 85 85 - Nhấn phím số 2 (Press 2)

The templates emphasize contacting support for login issues or questions about the registration process.

Sources: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_en.html#L17-L18).

src/main/resources/templates/email_kis_ekyc_register_success_en.html17-18(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_success_en.html#L17-L18)[

src/main/resources/templates/email_kis_ekyc_register_failed_vi.html37-39](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L37-L39)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_kis_ekyc_register_failed_vi.html#L37-L39))

Account Management Emails

Relevant source files

- [

src/main/resources/templates/email_invite_paave_admin_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_invite_paave_admin_en.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_invite_paave_admin_en.html))
- [

src/main/resources/templates/email_invite_paave_admin_ko.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_invite_paave_admin_ko.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_invite_paave_admin_ko.html))
- [

src/main/resources/templates/email_invite_paave_admin_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_invite_paave_admin_vi.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_invite_paave_admin_vi.html))
- [

src/main/resources/templates/email_reset_pass_paave_admin_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_pass_paave_admin_en.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_pass_paave_admin_en.html))
- [

src/main/resources/templates/email_reset_pass_paave_admin_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_pass_paave_admin_vi.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_pass_paave_admin_vi.html))
- [

src/main/resources/templates/paave_account_deleted_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deleted_en.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deleted_en.html))
- [

src/main/resources/templates/paave_account_deletion_init_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deletion_init_en.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deletion_init_en.html))
- [

src/main/resources/templates/paave_account_deletion_init_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deletion_init_vi.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deletion_init_vi.html))
- [

src/main/resources/templates/paave_change_password.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_change_password.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_change_password.html))
- [

src/main/resources/templates/paave_change_password_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_change_password_en.html)(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_change_password_en.html)

- [

src/main/resources/templates/paave_deleting_account.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_deleting_account.html)(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_deleting_account.html)

- [

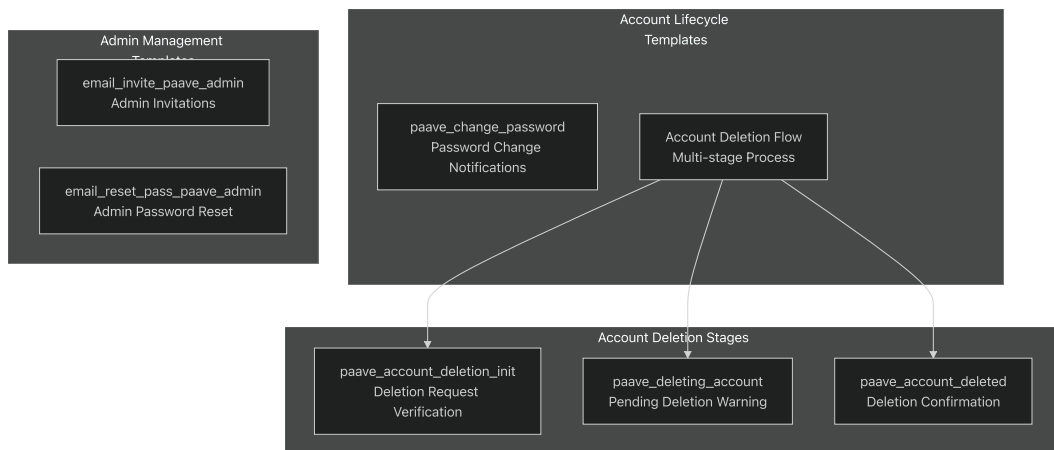
src/main/resources/templates/paave_deleting_account_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_deleting_account_en.html)(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_deleting_account_en.html)

This document covers the email templates and functionality for managing user account lifecycle events within the Paave notification service. These templates handle password changes, account deletion processes, and administrative account management.

For user onboarding and welcome emails, see [Welcome and Onboarding Emails](#). For authentication-related emails like OTP verification, see [Authentication Emails](#).

Template Categories and Organization

The account management email system is organized into three primary categories based on user journey and security requirements:



Sources: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_change_password.html#L1-L165).

src/main/resources/templates/paave_change_password.html#L1-L165 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_change_password.html#L1-L165)

src/main/resources/templates/paave_account_deletion_init_en.html#L1-L160](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deletion_init_en.html#L1-L160)([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deletion_init_en.html#L1-L160))

src/main/resources/templates/email_invite_paave_admin_en.html#L1-L50](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_invite_paave_admin_en.html#L1-L50)(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_invite_paave_admin_en.html#L1-L50)

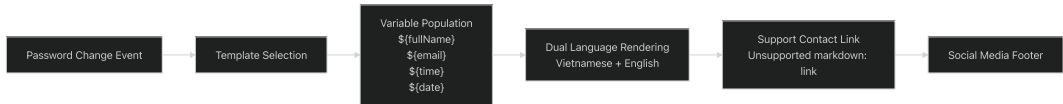
Password Change Notifications

The password change notification system provides immediate security alerts when user passwords are modified.

Template Structure

Template File	Language	Variables Used
paave_change_password.html	Bilingual (Vietnamese/English)	<code>\${fullName}</code> , <code>\${email}</code> , <code>\${time}</code> , <code>\${date}</code>
paave_change_password_en.html	English-focused	<code>\${fullName}</code> , <code>\${email}</code> , <code>\${time}</code> , <code>\${date}</code>

The templates include both Vietnamese and English content sections within the same file, ensuring users receive notifications in both languages for better comprehension of security events.

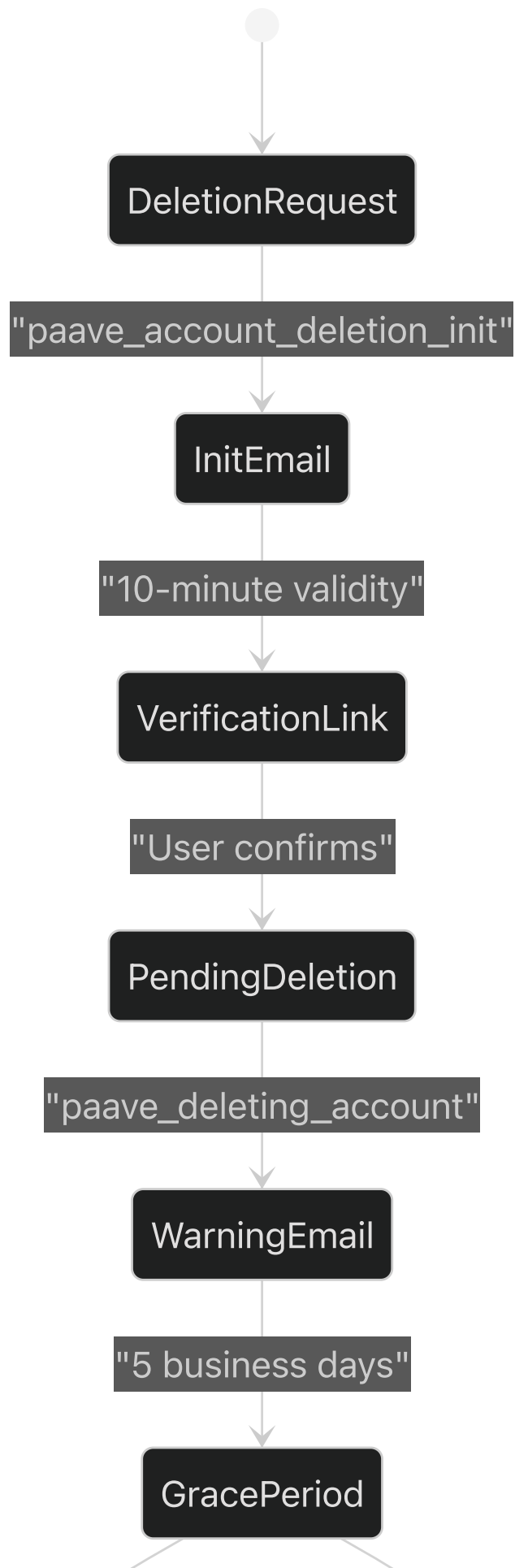


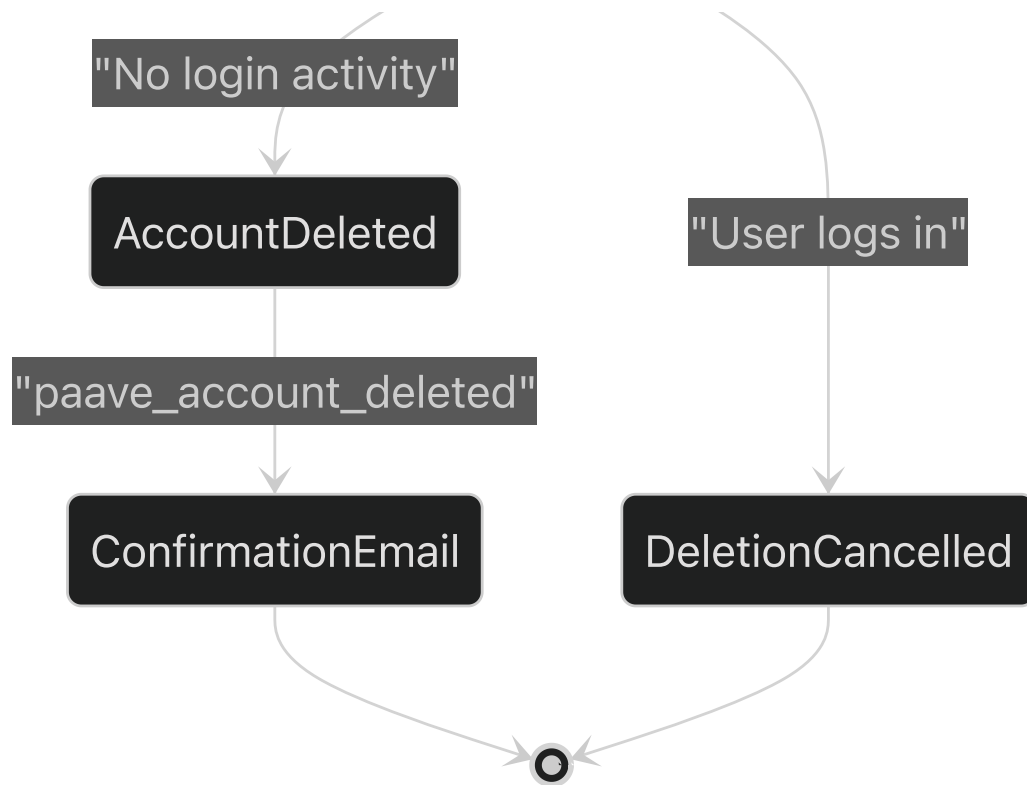
Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_change_password.html#L36-L78)
[src/main/resources/templates/paave_change_password.html36-78](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_change_password.html#L36-L78) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_change_password.html#L36-L78) [
[src/main/resources/templates/paave_change_password_en.html36-78](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_change_password_en.html#L36-L78)] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_change_password_en.html#L36-L78)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_change_password_en.html#L36-L78))

Account Deletion Workflow

The account deletion process implements a multi-stage security workflow with verification steps and grace periods.

Deletion Process Stages





Template Variables and Security Features

Stage	Template	Key Variables	Security Features
Initiation	paave_account_deletion_init_*	<code>\${fullName}</code> , <code>\${urlDeleteVi}</code> , <code>\${urlDeleteEn}</code>	10-minute link expiry
Warning	paave_deleting_account*	<code>\${fullName}</code>	5-day grace period
Confirmation	paave_account_deleted*	<code>\${fullName}</code> , <code>\${date}</code>	Final notification

Sources: [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deletion_init_en.html#L44-L64)

[tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deletion_init_en.html#L44-L64](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deletion_init_en.html#L44-L64)

[src/main/resources/templates/paave_account_deletion_init_en.html44-64](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deletion_init_en.html#L44-L64) [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deletion_init_en.html#L44-L64)

[tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deletion_init_en.html#L44-L64](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deletion_init_en.html#L44-L64) [

[src/main/resources/templates/paave_deleting_account_en.html56-67](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_deleting_account_en.html#L56-L67)] [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_deleting_account_en.html#L56-L67)
[tradex/notification/blob/c316bffb/src/main/resources/templates/paave_deleting_account_en.html#L56-L67](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_deleting_account_en.html#L56-L67)] [
[%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_deleting_account_en.html#L56-L67)]

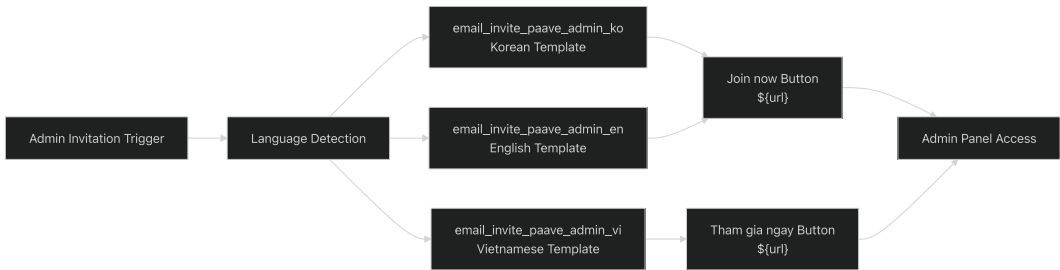
[src/main/resources/templates/paave_account_deleted_en.html55-62](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deleted_en.html#L55-L62)] [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deleted_en.html#L55-L62)
[tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deleted_en.html#L55-L62](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deleted_en.html#L55-L62)
https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deleted_en.html#L55-L62)

Administrative Account Management

Administrative account templates handle privileged user lifecycle events with enhanced security considerations.

Admin Invitation System

The admin invitation templates support multi-language deployment across Vietnamese, English, and Korean markets:



Admin Password Reset

Admin password reset templates provide secure credential recovery for administrative accounts:

Template	Language	Key Features
email_reset_pass_paave_admin_en.html	English	10-minute URL validity
email_reset_pass_paave_admin_vi.html	Vietnamese	Minimal styling for security

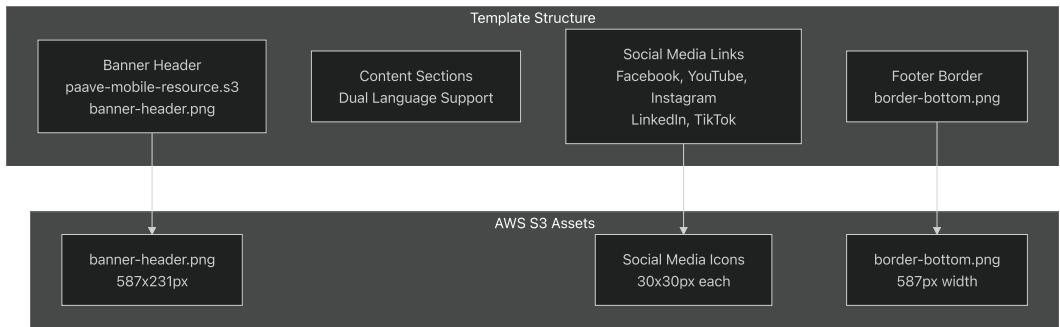
Both templates use the `${username}` and `${url}` variables with a standardized 10-minute expiration window for security compliance.

Sources: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_invite_paave_admin_en.html#L34-L44)
[src/main/resources/templates/email_invite_paave_admin_en.html34-44](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_invite_paave_admin_en.html#L34-L44) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_invite_paave_admin_en.html#L34-L44) [https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_invite_paave_admin_en.html#L34-L44]
src/main/resources/templates/email_reset_pass_paave_admin_en.html9-23](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_pass_paave_admin_en.html#L9-L23) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_pass_paave_admin_en.html#L9-L23)

Template Infrastructure and Styling

Common Design Elements

All account management templates share consistent infrastructure components:



Variable Substitution Patterns

The templates utilize FreeMarker variable substitution with consistent naming conventions:

Variable Pattern	Usage	Security Consideration
<code>\${fullName}</code>	User personalization	Sanitized display name
<code>\${email}</code>	Account identification	Verified email address
<code>\${time} / \${date}</code>	Temporal context	Server-side timestamp
<code>\${url*}</code>	Action links	Time-limited tokens

Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_change_password.html#L17-L18
[src/main/resources/templates/paave_change_password.html#L17-L18](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_change_password.html#L17-L18) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_change_password.html#L17-L18) [
[src/main/resources/templates/paave_account_deletion_init_en.html#L17-L18](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deletion_init_en.html#L17-L18)] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deletion_init_en.html#L17-L18) [
([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_account_deletion_init_en.html#L17-L18)),
[src/main/resources/templates/paave_deleting_account_en.html#L17-L18](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_deleting_account_en.html#L17-L18)] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_deleting_account_en.html#L17-L18)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_deleting_account_en.html#L17-L18)

Multi-Language Implementation Strategy

The account management templates implement language support through multiple approaches:

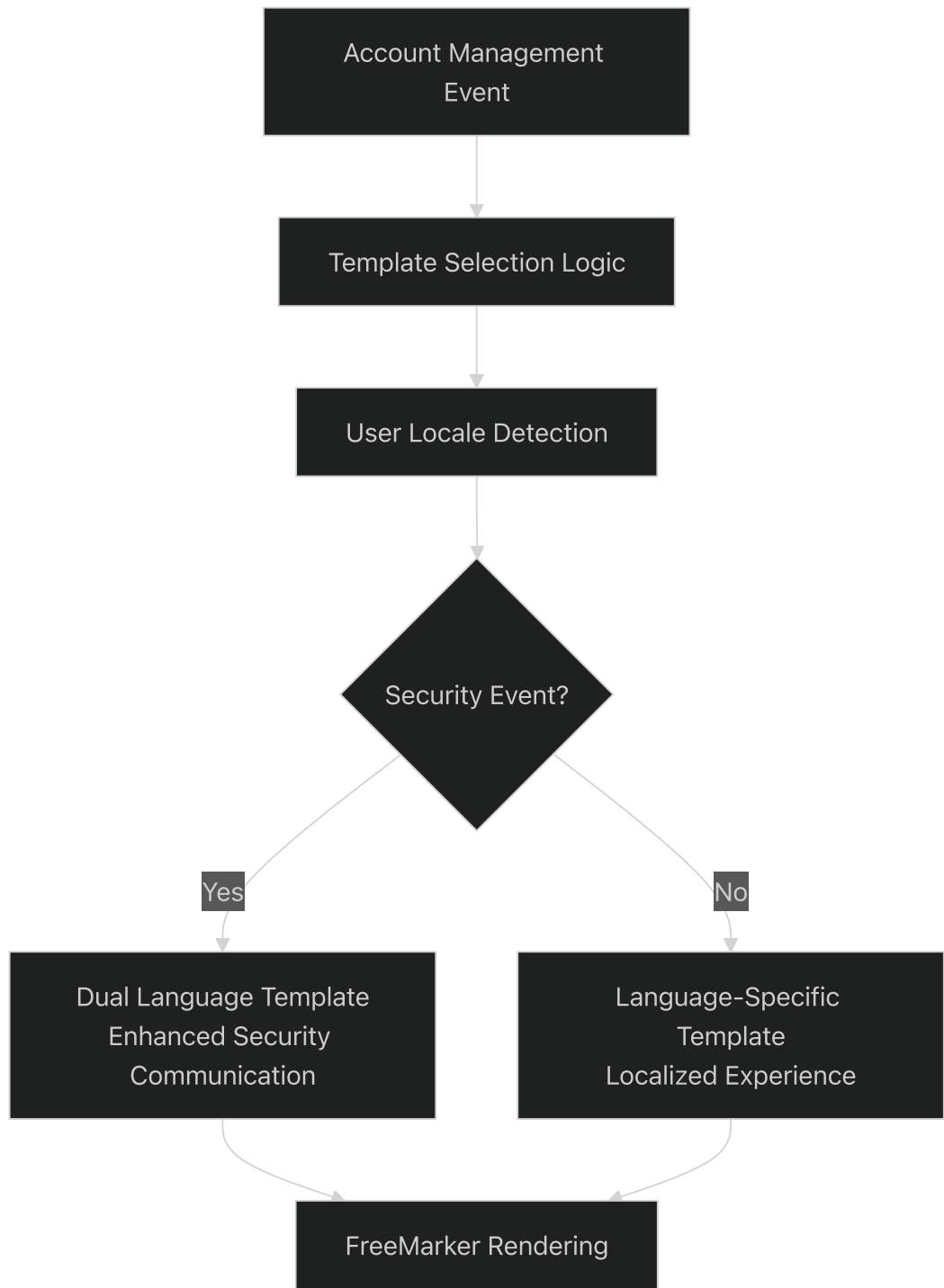
Dual-Language Templates

Templates like `paave_change_password.html` include both Vietnamese and English content within a single file, ensuring users receive security notifications in multiple languages for critical account events.

Language-Specific Templates

Admin templates are separated by language suffix (`_en`, `_vi`, `_ko`) to support targeted deployment in different geographic markets.

Template Selection Logic



Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_change_password.html#L36-L78)

src/main/resources/templates/paave_change_password.html#L36-L78 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_change_password.html#L36-L78)

src/main/resources/templates/email_invite_paave_admin_ko.html#L1-L50 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_invite_paave_admin_ko.html#L1-L50)

src/main/resources/templates/email_invite_paave_admin_vi.html#L1-L47 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_invite_paave_admin_vi.html#L1-L47)

[src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java1-75](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L1-L75) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L1-L75>)

[src/main/java/com/techx/tradex/notification/dao/EmailOneSignalDao.java1-9](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/EmailOneSignalDao.java#L1-L9)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/EmailOneSignalDao.java#L1-L9>) ([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/EmailOneSignalDao.java#L1-L9)).

[src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java1-43](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java#L1-L43)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java#L1-L43>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java#L1-L43>).

Core Components

EmailOneSignalService

The `EmailOneSignalService` class serves as the primary orchestrator for OneSignal email operations. It processes incoming `NotificationMessage` objects and coordinates with the template system and OneSignal API.

Component	Responsibility	Key Dependencies
<code>EmailOneSignalService</code>	Message processing and coordination	<code>AppConf</code> , <code>EmailOneSignalDao</code> , <code>TemplateDao</code>
<code>EmailOneSignalDao</code>	OneSignal API abstraction	<code>OneSignalDao</code> , <code>ObjectMapper</code>
<code>SmsOneSignalRequest</code>	Request model (shared with SMS)	None
<code>EmailSendFilter</code>	OneSignal-specific email format	None

Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L25-L39>)

[src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java25-39](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L25-L39) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L25-L39>)

Data Access Layer

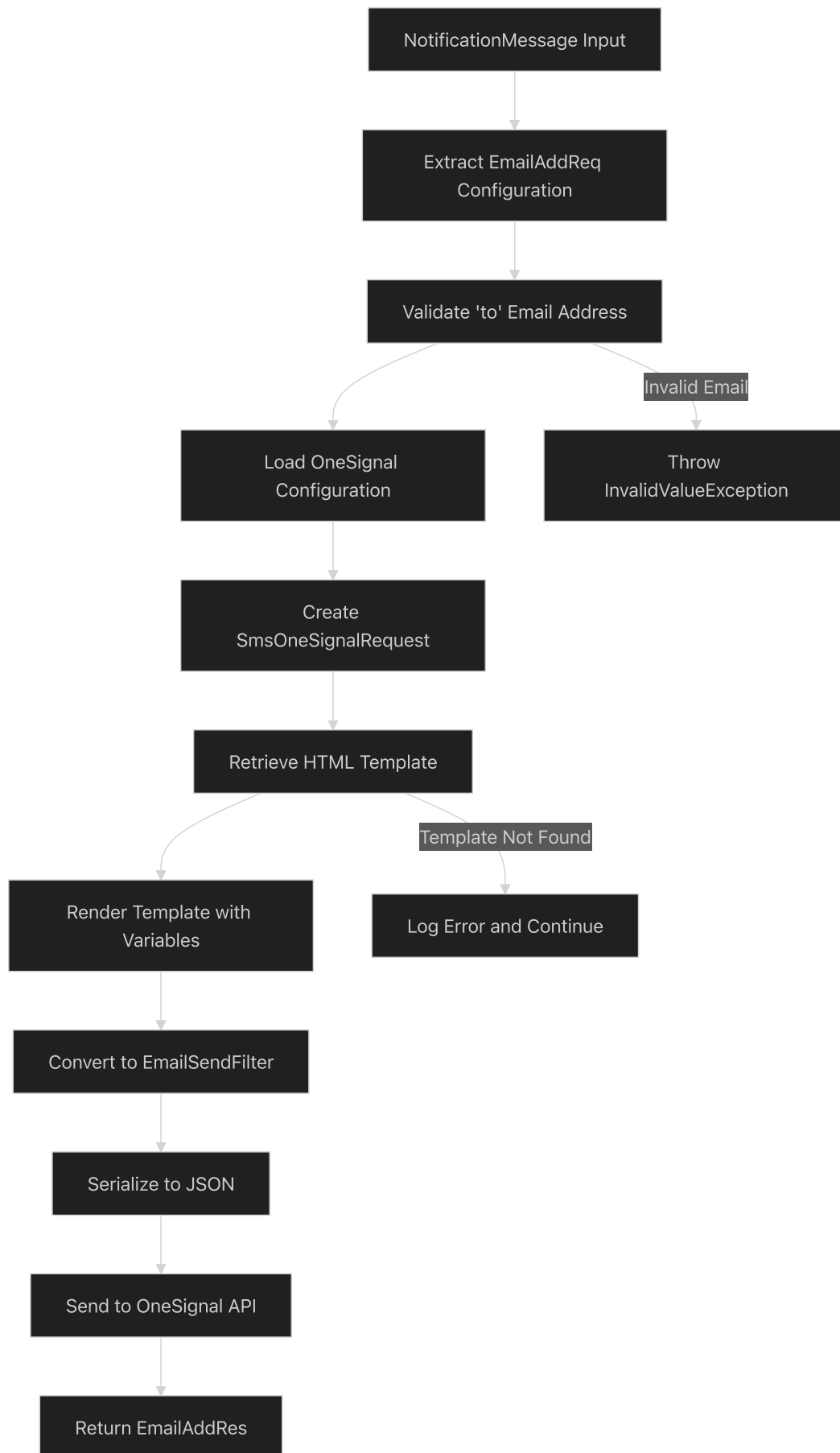
The data access layer implements the DAO pattern to abstract OneSignal API interactions. The `EmailOneSignalDao` interface defines the contract, while the implementation handles the conversion to OneSignal's expected format.

Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/EmailOneSignalDao.java#L5-L8>)

[src/main/java/com/techx/tradex/notification/dao/EmailOneSignalDao.java5-8](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/EmailOneSignalDao.java#L5-L8) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/EmailOneSignalDao.java#L5-L8>)

[src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java14-24](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java#L14-L24)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java#L14-L24>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java#L14-L24>).

Message Processing Flow



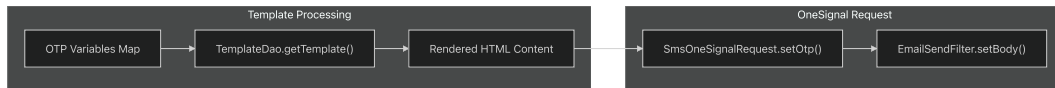
Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L41-L74>

[src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java41-74](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L41-L74) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L41-L74>)

[src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java26-41](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java#L26-L41)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java#L26-L41>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java#L26-L41>)

Template Integration

The OneSignal Email Integration specifically uses the `email_otp_verify.html` template for OTP verification emails. The template system integration follows this pattern:



The template processing uses a `LinkedHashMap` to provide the `otp` variable to the FreeMarker template engine:

Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L53-L67>)

[src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java53-67](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L53-L67) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L53-L67>)

Configuration Requirements

The OneSignal Email Integration requires several configuration parameters from `AppConf` :

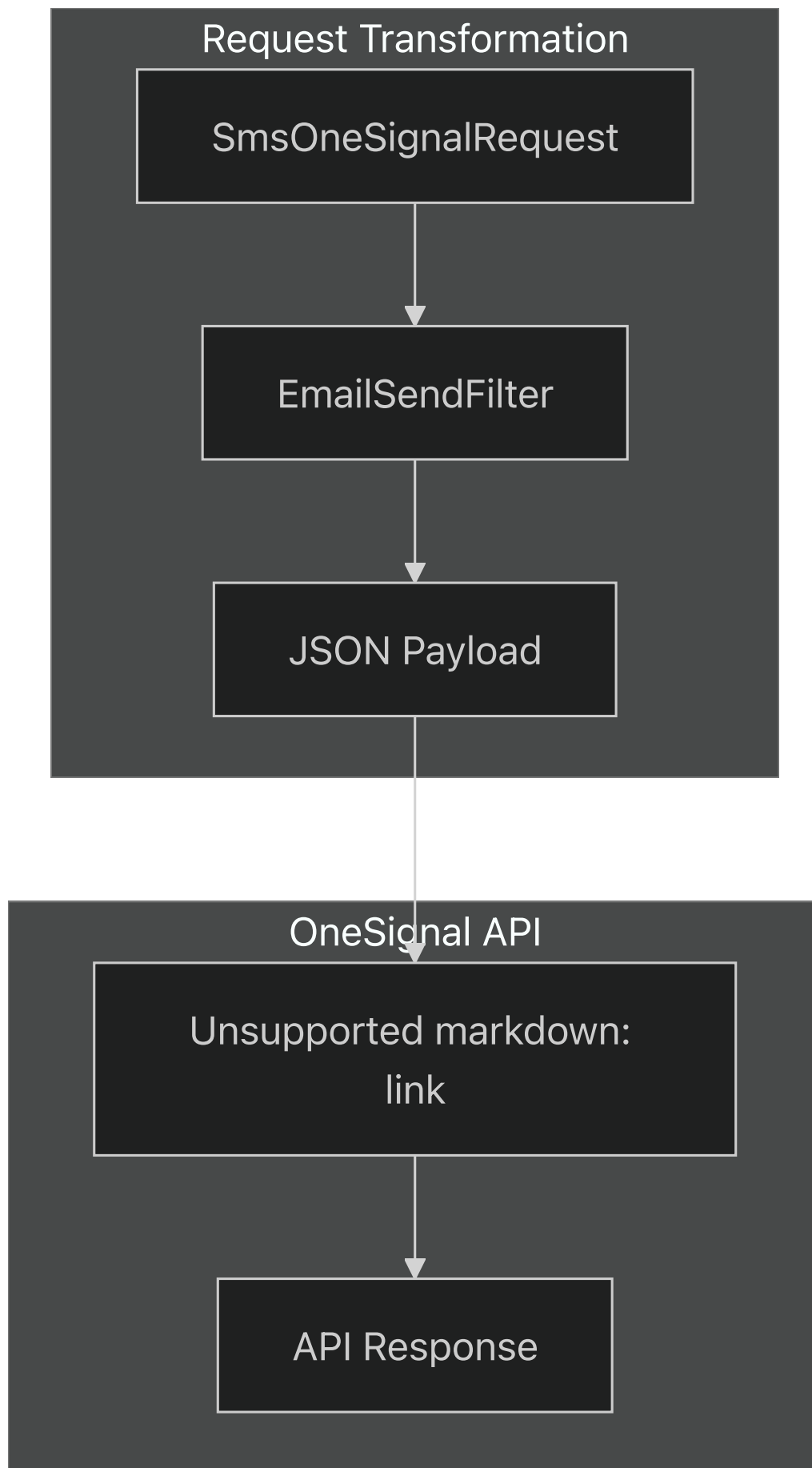
Configuration Path	Purpose	Usage
<code>appConf.getOneSignal().getAppId()</code>	OneSignal Application ID	API authentication
<code>appConf.getOneSignal().getApiKey()</code>	OneSignal API Key	API authentication
<code>appConf.getEmail().getSubject()</code>	Email subject line	Default email subject
<code>appConf.getEmail().getSender()</code>	From name display	Email sender identification

Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L48-L52>)

[src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java48-52](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L48-L52) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L48-L52>)

OneSignal API Integration

The integration communicates with OneSignal's notifications endpoint at `https://onesignal.com/api/v1/notifications`. The `EmailSendFilter` model structures the request according to OneSignal's email notification format:



Sources: [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java#L16-L40)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java#L16-L40](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java#L16-L40)

[src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java16-40](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java#L16-L40) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java#L16-L40)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java#L16-L40](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/EmailOneSignalDao.java#L16-L40))

Error Handling

The service implements comprehensive error handling for common failure scenarios:

- **Invalid Email Address:** Throws `InvalidValueException` when the `to` field is null or empty
- **Template Not Found:** Logs error but continues processing (graceful degradation)
- **General Exceptions:** Wrapped in `RuntimeException` with full stack trace logging

Sources: [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L45-L47)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L45-L47](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L45-L47)

[src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java45-47](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L45-L47) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L45-L47)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L45-L47](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L45-L47))

[src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java62-73](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L62-L73)

([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L62-L73)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L62-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L62-L73)

[L73](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L62-L73) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L62-L73)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L62-L73](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/EmailOneSignalService.java#L62-L73))

SMS Notification System

Overview

Relevant source files

- [[src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java)](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java>)
- [[src/main/java/com/techx/tradex/notification/dao/impl/KakaoDao.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/KakaoDao.java)](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/KakaoDao.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/KakaoDao.java>)
- [[src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java)](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java>)
- [[src/main/java/com/techx/tradex/notification/model/db/FptSmsHistory.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/db/FptSmsHistory.java)](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/db/FptSmsHistory.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/db/FptSmsHistory.java>)
- [[src/main/java/com/techx/tradex/notification/model/wSDL/SendMT.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/wSDL/SendMT.java)](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/wSDL/SendMT.java>)

(<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/wSDL/SendMT.java>)

Purpose and Scope

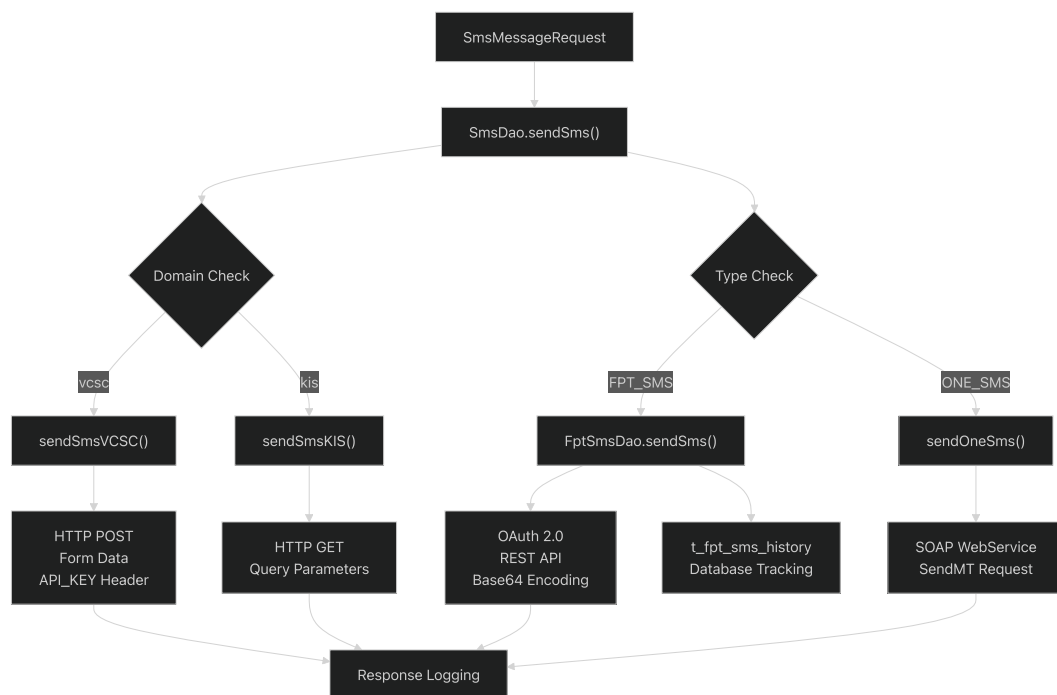
The SMS Notification System provides multi-provider SMS delivery capabilities through a unified interface. This system supports four distinct SMS providers (VCSC, KIS, FPT SMS, and OneSMS) with different communication protocols and authentication mechanisms. The system handles provider selection, message routing, authentication token management, and delivery tracking.

For push notifications and messaging platforms like KakaoTalk, see [Push Notification System](#) and [Messaging Platform Integrations](#). For OneSignal SMS integration, see [OneSignal SMS Integration](#).

SMS Provider Architecture

The SMS system uses a strategy pattern implemented in `SmsDao` to route messages to different providers based on configuration. Each provider implements a specific protocol and authentication method.

Provider Routing Architecture



Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L40-L51>)
[src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L40-L51](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L40-L51) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L40-L51>) [
[src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L53-L101](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L53-L101)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L53-L101>)
[src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L53-L101](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L53-L101)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L53-L101>)

Provider Implementation Details

VCSC SMS Provider

The VCSC provider uses HTTP POST with form-encoded data and a hardcoded API key for authentication.

Property	Value
Protocol	HTTP POST
Content Type	application/x-www-form-urlencoded
Authentication	API Key Header (<code>HU3uzCaqR0f0zJZ</code>)
Parameters	Mobile , Message

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L53-L71> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L53-L71>)

KIS SMS Provider

The KIS provider uses HTTP GET with query parameters, requiring no authentication headers.

Property	Value
Protocol	HTTP GET
Parameters	to (phone number), message (content)
Authentication	None

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L73-L87> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L73-L87>)

OneSMS Provider

The OneSMS provider uses SOAP WebService with WSDL-generated classes for structured messaging.

Property	Value
Protocol	SOAP WebService
Model Class	<code>SendMT</code>
Response Class	<code>SendMTResponse</code>
Authentication	Username/Password

The `SendMT` class contains the following fields:

- `user` - Authentication username
- `pass` - Authentication password
- `sms` - Message content
- `senderName` - Sender identifier
- `phone` - Recipient phone number

- `isFlash` - Flash SMS flag
- `isUnicode` - Unicode support flag

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L89-L101>

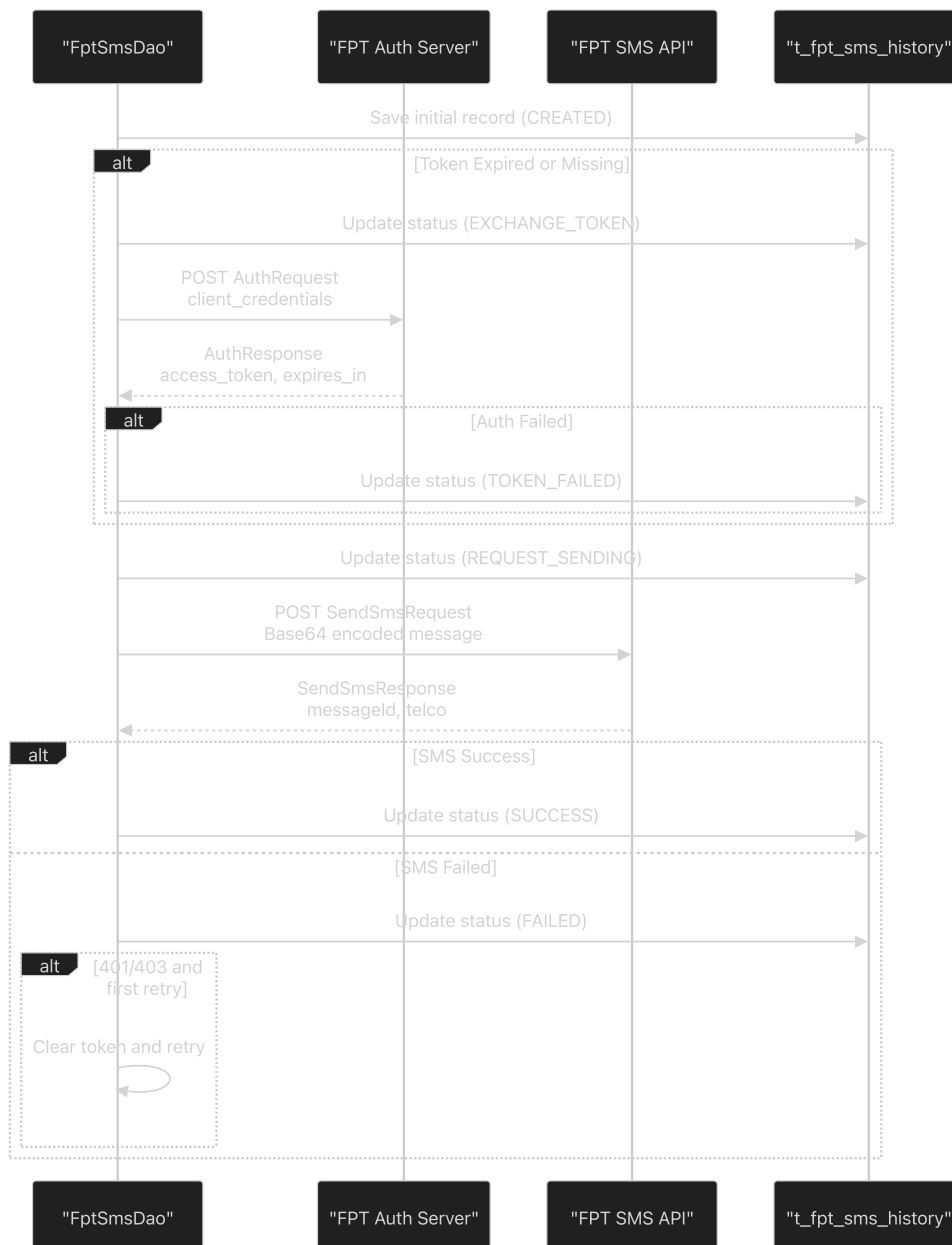
[src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java89-101](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L89-L101) <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L89-L101>

[src/main/java/com/techx/tradex/notification/model/wSDL/SendMT.java53-214](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/wSDL/SendMT.java#L53-L214) <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/wSDL/SendMT.java#L53-L214> <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/wSDL/SendMT.java#L53-L214>

FPT SMS Integration

The FPT SMS provider implements a sophisticated OAuth 2.0 authentication flow with automatic token refresh and comprehensive error handling.

FPT SMS OAuth Flow



Sources: <https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L108-L197>

<src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L108-L197> (<https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L108-L197>)

FPT SMS Data Models

The FPT SMS integration uses several data transfer objects for API communication:

Class	Purpose	Key Fields
AuthRequest	OAuth token request	clientId, clientSecret, grantType, scope
AuthResponse	OAuth token response	accessToken, expiresIn, tokenType
SendSmsRequest	SMS send request	accessToken, brandName, phone, message (Base64)
SendSmsResponse	SMS send response	messageId, telco, isSent

Sources: <https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L34-L74>

<src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L34-L74> (<https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L34-L74>)

FPT SMS History Tracking

The system maintains comprehensive tracking of FPT SMS operations in the `t_fpt_sms_history` table:

Field	Type	Purpose
brand_name	String	SMS brand identifier
phone_number	String	Recipient phone number
content	String	Message content
message_type	String	Message classification (CSKH)
status	String	Processing status
qty	Integer	Message segments (calculated by length)
telco	String	Recipient's telecom provider
fail_reason	String	Error details if failed

The status progression follows: `CREATED` → `EXCHANGE_TOKEN` → `REQUEST_SENDING` → `SUCCESS / FAILED`

Sources: <https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/db/FptSmsHistory.java#L9-L27>

<src/main/java/com/techx/tradex/notification/model/db/FptSmsHistory.java#L9-L27> (<https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/db/FptSmsHistory.java#L9-L27>)

<src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L111-121>] (<https://github.com/nhsv->

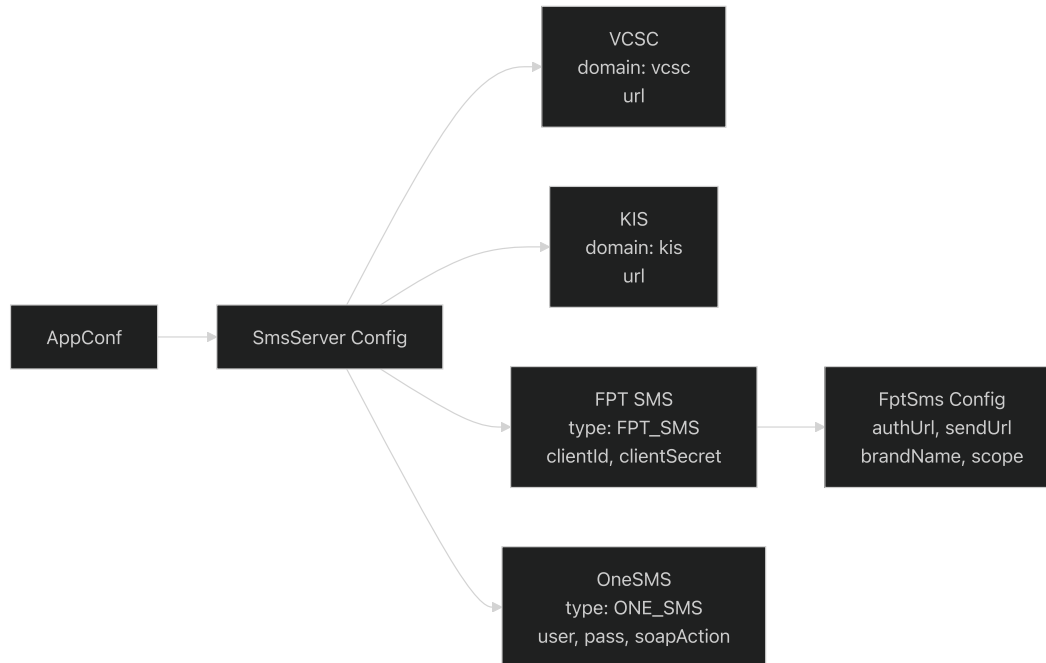
[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L111-](tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L111-121)

[L121](https://github.com/nhsv-) (<https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L111-L121>))

Configuration Integration

The SMS providers are configured through the `AppConf.SmsServer` configuration objects, which contain provider-specific settings:



The routing logic in `SmsDao.sendSms()` first checks the `domain` field for VCSC and KIS providers, then falls back to checking the `type` field for FPT_SMS and ONE_SMS providers.

Sources: (<https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L42-L50>).

<src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#42-50> (<https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L42-L50>).

Error Handling and Resilience

The SMS system implements several error handling strategies:

1. **FPT SMS Token Retry:** Automatic token refresh on 401/403 errors with single retry attempt
2. **Comprehensive Logging:** All providers log request/response details for debugging
3. **Status Tracking:** FPT SMS tracks detailed status progression for audit trails
4. **Exception Propagation:** HTTP errors are logged and re-thrown for upstream handling

Sources: (<https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L183-L196>).

<src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#183-196> (<https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L183-L196>)[

<src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#70>](<https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L70-L70>)[

([%5B">https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L70-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L70-L70)

[%5B">L70">%5B](L70)).

<src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#86>](<https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L86-L86>)[

([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L86-L86)).

src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java100](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L100-L100>)
(<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L100-L100>))

FPT SMS Integration

Relevant source files

- [
src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java>
(<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java>))
- [
src/main/java/com/techx/tradex/notification/model/db/FptSmsHistory.java](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/db/FptSmsHistory.java>
(<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/db/FptSmsHistory.java>))
- [
src/main/java/com/techx/tradex/notification/model/wSDL/SendMT.java](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/wSDL/SendMT.java>
(<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/wSDL/SendMT.java>))

Purpose and Scope

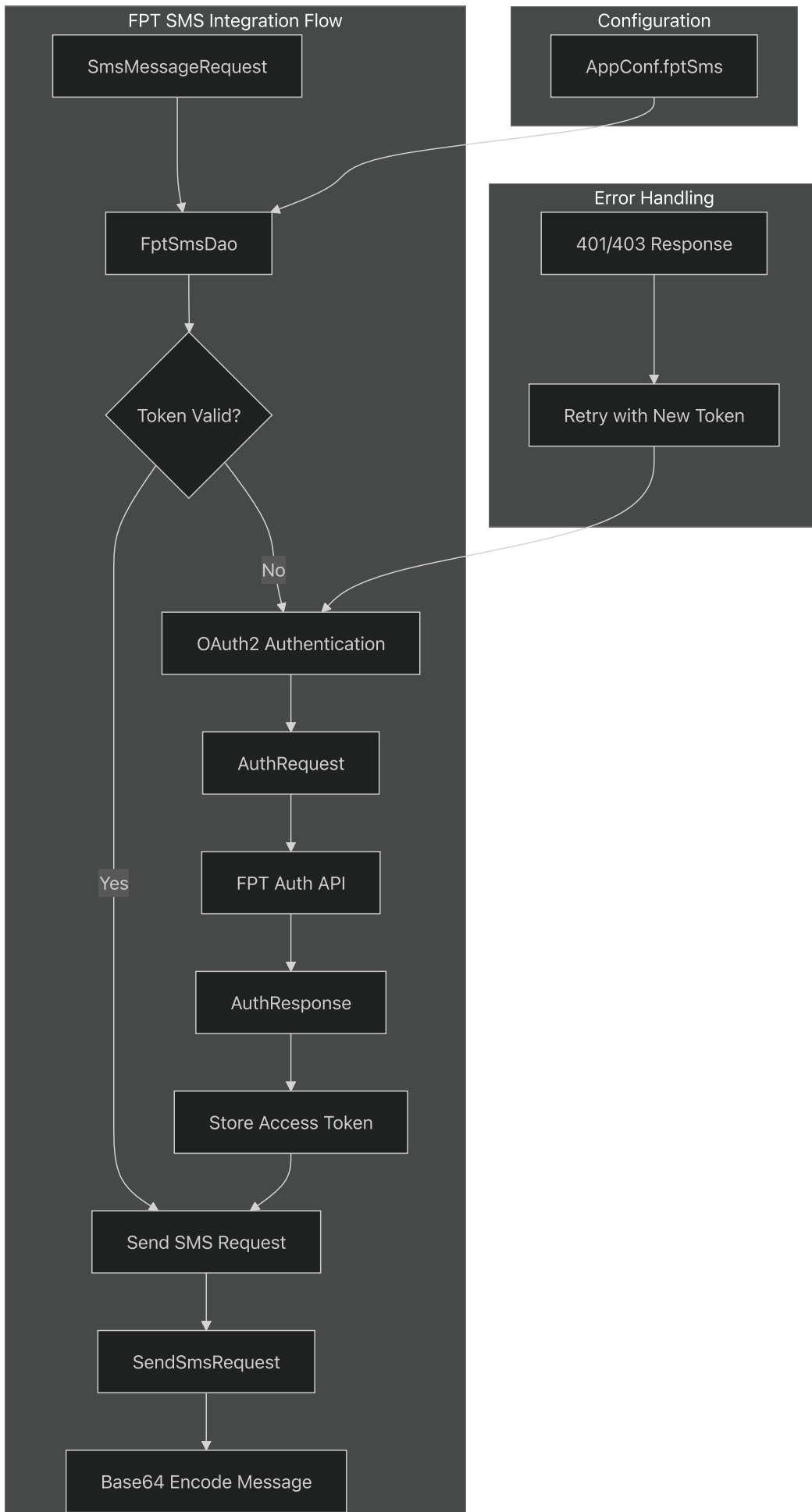
This document covers the FPT SMS gateway integration, which provides SMS delivery capabilities specifically for the Vietnamese market. The FPT SMS integration uses OAuth2 authentication and REST API communication to send SMS messages through FPT's SMS platform.

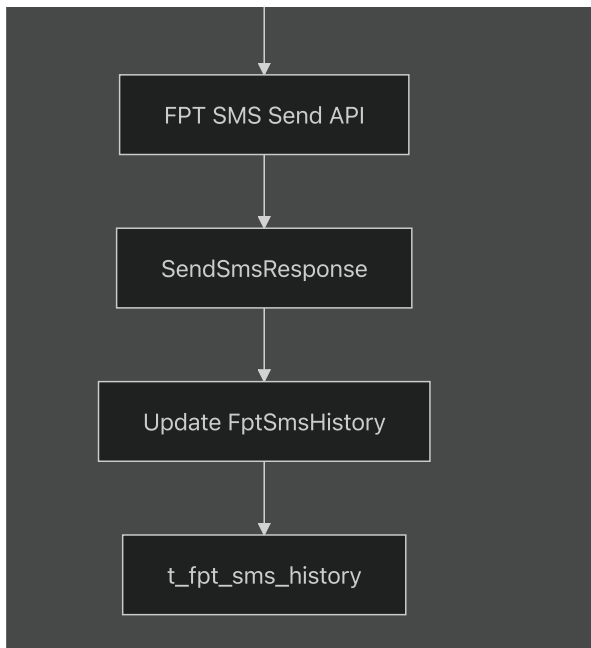
For information about other SMS providers in the multi-provider architecture, see [Multi-Provider SMS Architecture](#). For international SMS delivery options, see [OneSignal SMS Integration](#).

Architecture Overview

The FPT SMS integration implements a REST API-based communication pattern with OAuth2 client credentials authentication. Unlike other SMS providers in the system that may use SOAP or simple HTTP forms, FPT SMS requires a more sophisticated authentication flow and message encoding.

FPT SMS Integration Architecture



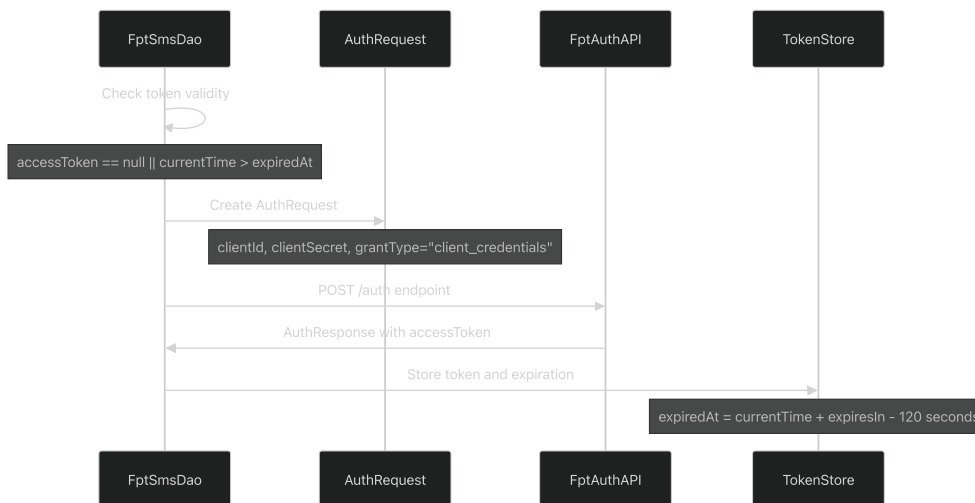


Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L77-L203> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L77-L203>)

Authentication Flow

FPT SMS uses OAuth2 client credentials grant type for authentication. The system maintains an access token with expiration tracking and automatic renewal.

OAuth2 Authentication Process



The authentication request includes the following parameters:

Parameter	Source	Description
grantType	Fixed value	"client_credentials"
clientId	appConf.getFptSms().getClientId()	FPT provided client identifier
clientSecret	appConf.getFptSms().getClientSecret()	FPT provided client secret
scope	appConf.getFptSms().getScope()	Required OAuth scope
sessionId	Generated UUID	Unique session identifier

Sources: [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L129-L146)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L129-L146](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L129-L146)

[src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java129-146](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L129-L146) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L129-L146)

[src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java34-40](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L34-L40)) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L34-L40)

SMS Sending Process

Message Preparation and Encoding

FPT SMS requires Base64 encoding of message content and supports Vietnamese language characters through Unicode handling.



The SMS quantity calculation follows these rules:

- Length ≤ 160 characters: 1 SMS
- Length ≤ 306 characters: 2 SMS
- Length ≤ 459 characters: 3 SMS
- Length > 459 characters: 4 SMS

Sources: [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L120-L120)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L120-L120](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L120-L120)

[src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java120](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L120-L120) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L120-L120)

[src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java161](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L161-L161)) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L161-L161)

Send SMS Request Structure

The `SendSmsRequest` class contains the following fields:

Field	Type	Description
accessToken	String	OAuth2 access token
sessionId	String	Unique session identifier
brandName	String	Configured brand name from AppConf
phone	String	Recipient phone number
message	String	Base64 encoded message content

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L53-L62>, [src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java53-62](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L53-L62) <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L53-L62>.

History Tracking and Status Management

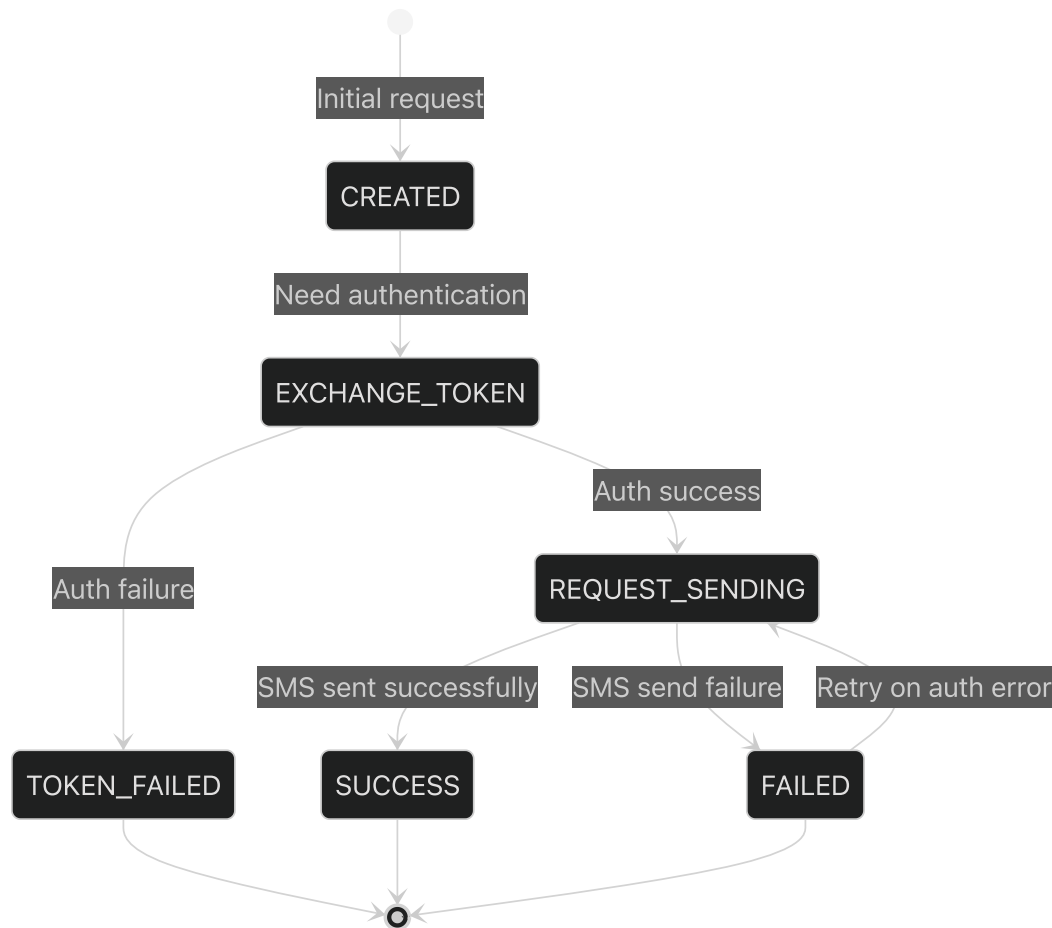
FptSmsHistory Entity

The `FptSmsHistory` entity tracks the complete lifecycle of each SMS request:

FptSmsHistory		
int	id	PK
string	brand_name	
string	telco	
string	phone_number	
string	message_type	
string	content	
string	date	
integer	qty	
string	status	
zonedDateTime	created_at	
zonedDateTime	updated_at	
string	fail_reason	

Status Lifecycle

The SMS request progresses through multiple status states:

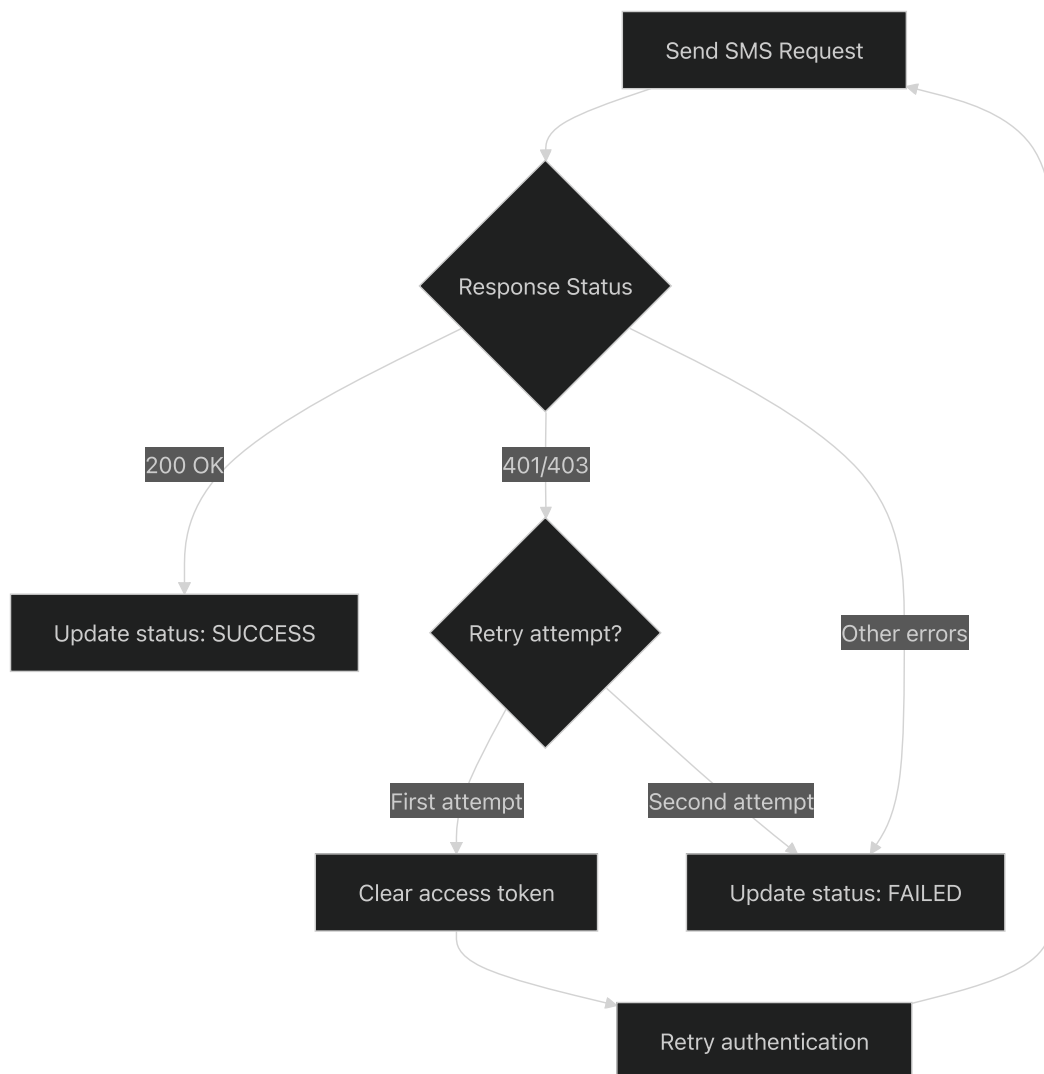


Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/db/FptSmsHistory.java#L9-L27>.
[src/main/java/com/techx/tradex/notification/model/db/FptSmsHistory.java#L9-L27](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/db/FptSmsHistory.java#L9-L27) <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/db/FptSmsHistory.java#L9-L27> [
[src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L119](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L119-L119)](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L119-L119>) [
[%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L119-L119)).
[src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L127](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L127-L127)](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L127-L127>) [
[%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L127-L127)).
[src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L155](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L155-L155)](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L155-L155>) [
[%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L155-L155)).

Error Handling and Retry Logic

Authentication Error Recovery

The system implements automatic retry logic for authentication failures:



The retry mechanism only triggers once per SMS request to prevent infinite loops. Authentication errors (`HttpStatus.UNAUTHORIZED` or `HttpStatus.FORBIDDEN`) trigger token refresh and retry, while other errors result in immediate failure.

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L183-L196>

[src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L183-L196](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L183-L196) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L183-L196>)

Configuration Requirements

The FPT SMS integration requires the following configuration parameters through `AppConf.fptSms` :

Configuration	Required	Description
<code>clientId</code>	Yes	FPT provided OAuth client ID
<code>clientSecret</code>	Yes	FPT provided OAuth client secret
<code>scope</code>	Yes	OAuth scope for SMS permissions
<code>brandName</code>	Yes	Sender brand name for SMS messages
<code>authUrl</code>	Yes	FPT OAuth authentication endpoint
<code>sendUrl</code>	Yes	FPT SMS sending endpoint

Sources: [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L130-L140)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L130-L140](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L130-L140))

[src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java130-140](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L130-L140) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L130-L140)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L130-L140](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L130-L140))

[src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java159-167](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L159-L167)]([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L159-L167)

[L167](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L159-L167) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L159-L167)

Session Management

The `FptSmsDao` maintains session state through:

- **Session ID:** UUID-based identifier with incrementing counter for request tracking
- **Access Token:** Stored with expiration timestamp (120 seconds buffer before expiry)
- **Request Counter:** Sequential numbering for debugging and tracking

The session ID format follows the pattern: `{UUID}-{incrementalNumber}`, ensuring unique identification for each request in logs and error tracking.

Sources: [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L84-L87)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L84-L87](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L84-L87))

[src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java84-87](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L84-L87) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L84-L87)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L84-L87](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L84-L87))

[src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java199-202](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L199-L202)]([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L199-L202)

[L202](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L199-L202) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/FptSmsDao.java#L199-L202)

Multi-Provider SMS Architecture

Relevant source files

- [[src/main/java/com/techx/tradex/notification/dao/SmsDao.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/SmsDao.java)]([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/SmsDao.java)
[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/SmsDao.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/SmsDao.java)
(<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/SmsDao.java>))
- [

src/main/java/com/techx/tradex/notification/dao/SmsOneSignalDao.java](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/SmsOneSignalDao.java>)

- [

src/main/java/com/techx/tradex/notification/dao/impl/KakaoDao.java](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/KakaoDao.java>)

- [

src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java>)

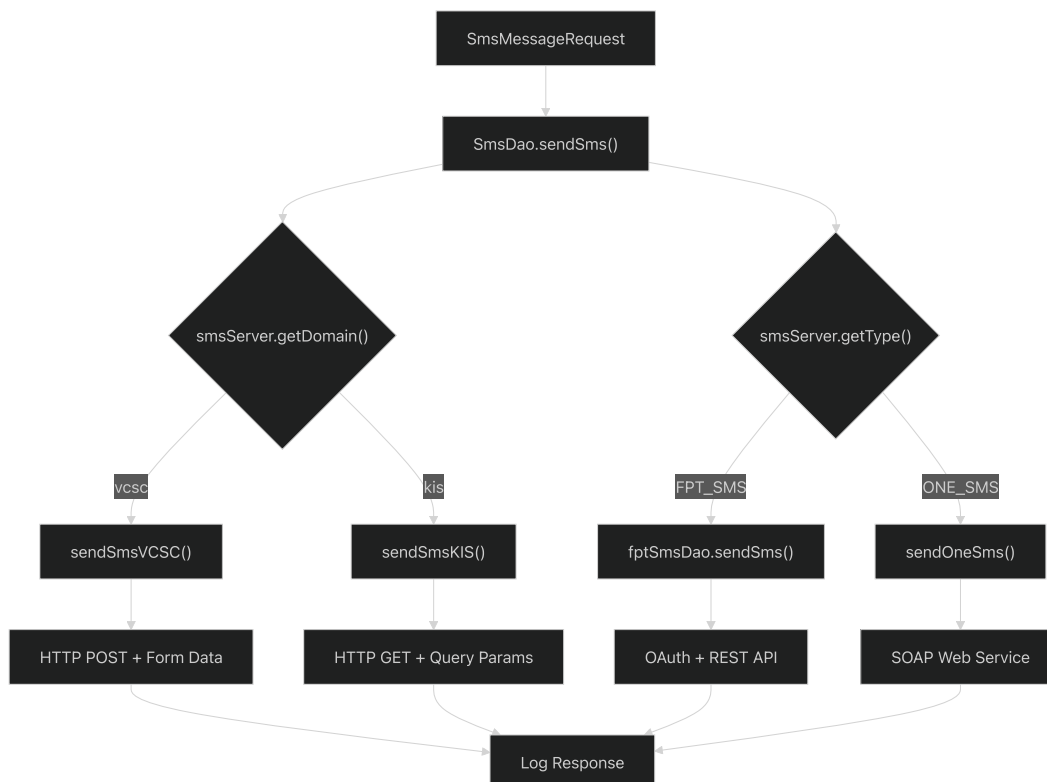
This document covers the multi-provider SMS architecture that enables the notification service to send SMS messages through multiple SMS gateway providers using different communication protocols. The system uses a unified `SmsDao` interface to abstract provider-specific implementations while supporting diverse integration patterns including HTTP REST, SOAP web services, and form-based APIs.

For OneSignal-specific SMS integration, see [OneSignal SMS Integration](#). For FPT SMS OAuth authentication details, see [FPT SMS Integration](#).

Architecture Overview

The multi-provider SMS architecture implements a strategy pattern where the `SmsDao` class acts as a central dispatcher that routes SMS requests to appropriate provider-specific implementations based on configuration. This design allows the system to support multiple SMS providers with different protocols and authentication mechanisms through a single unified interface.

Provider Selection Strategy



Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L40-L51>)

[src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java40-51](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L40-L51) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L40-L51>).

Core Components

Component	Purpose	Implementation
<code>SmsDao</code> interface	Unified SMS sending contract	[
<code>src/main/java/com/techx/tradex/notification/dao/SmsDao.java7-8</code> (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/SmsDao.java#L7-L8) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/SmsDao.java#L7-L8))		
<code>SmsDao</code> implementation	Provider routing and dispatch	[

`src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java22` (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L22-L22>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L22-L22>)) | `AppConf.SmsServer` | Provider configuration structure | Referenced in[

`src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java41` (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L41-L41>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L41-L41>)) | `SmsMessageRequest` | Standardized message format | [

`src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java4` (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L4-L4>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L4-L4>)) |

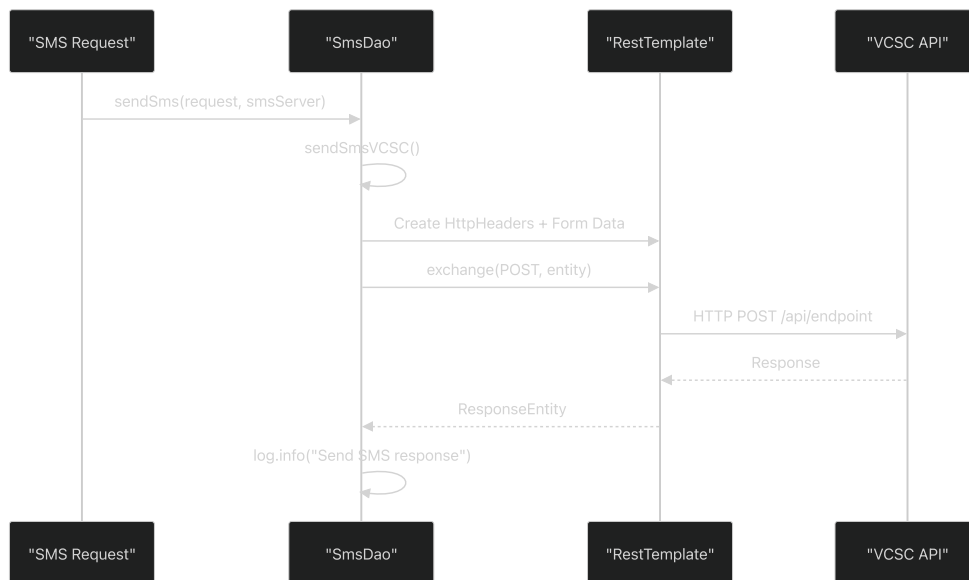
Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/SmsDao.java#L1-L9>)
[src/main/java/com/techx/tradex/notification/dao/SmsDao.java1-9](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/SmsDao.java#L1-L9) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/SmsDao.java#L1-L9>) [

`src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java1-38` (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L1-L38>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L1-L38>))

Provider Implementation Details

VCSC Provider Implementation

The VCSC provider uses HTTP POST with form-encoded data and requires an API key for authentication. The implementation creates form data with mobile number and message content.



Key Implementation Details:

- Content-Type: `application/x-www-form-urlencoded`
- API Key: `HU3uzCaqR0f0zJZ` (hardcoded)
- Form fields: `Mobile`, `Message`

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L53-L71>

[src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java53-71](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L53-L71) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L53-L71>)

KIS Provider Implementation

The KIS provider uses HTTP GET with query parameters, making it the simplest integration pattern among all providers.

Implementation characteristics:

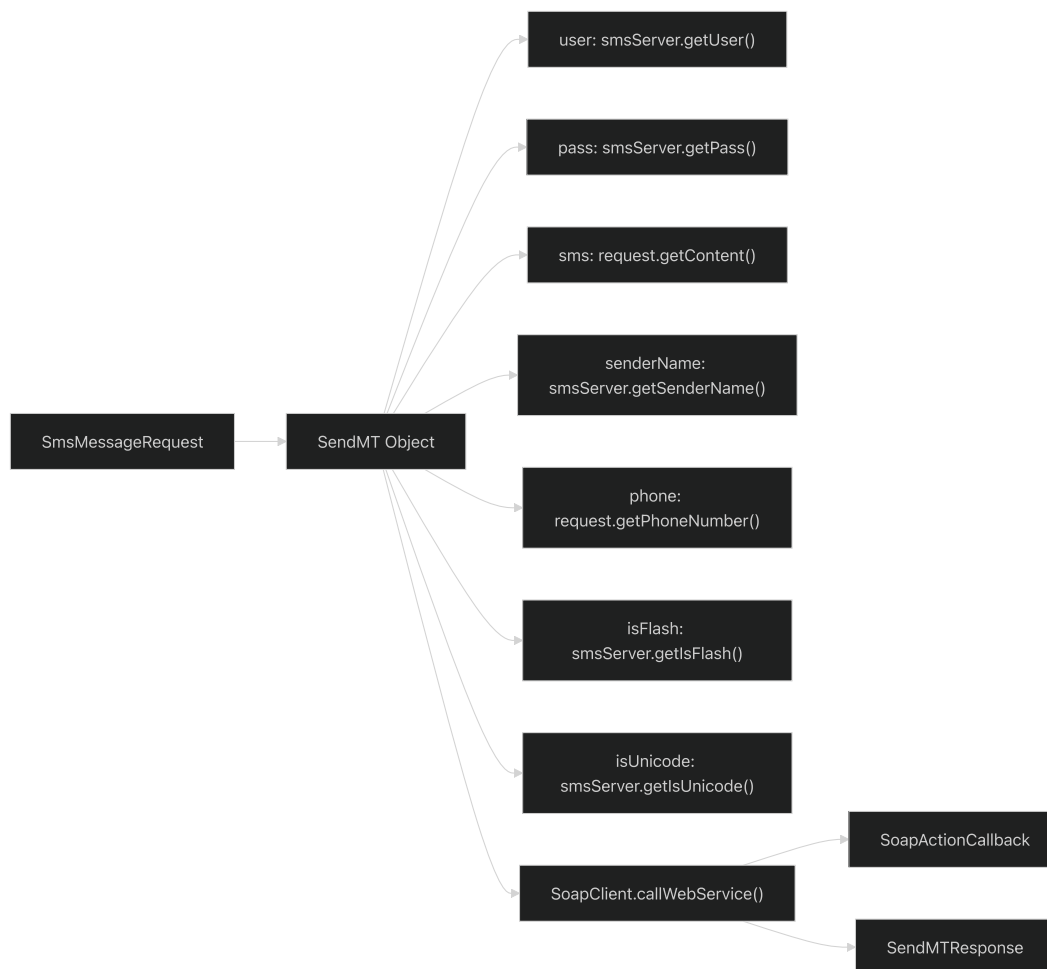
- Protocol: HTTP GET
- Parameters: `to` (phone number), `message` (content)
- Authentication: URL-based (embedded in endpoint)
- Response handling: String response logging

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L73-L87>

[src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java73-87](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L73-L87) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L73-L87>)

ONE_SMS Provider Implementation

The ONE_SMS provider uses SOAP web services with comprehensive parameter mapping including flash SMS and Unicode support.



SOAP Configuration Parameters:

- `user`, `pass` : Authentication credentials
- `senderName` : SMS sender identifier
- `isFlash` : Flash SMS flag
- `isUnicode` : Unicode message support
- `soapAction` : SOAP action header

Sources: <https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L89-L101>

<src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java89-101> <https://github.com/nhsv->

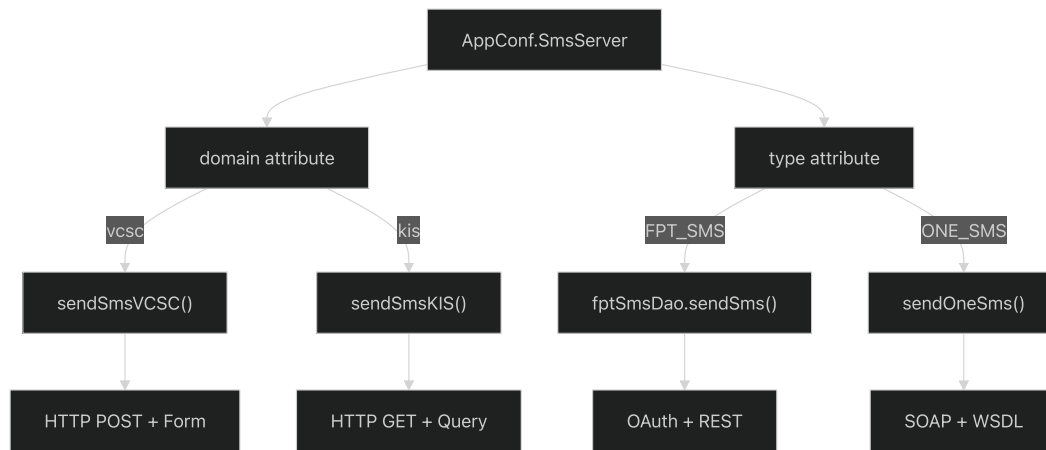
<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L89-L101>

Configuration and Routing Logic

Provider Selection Logic

The `SmsDao` implementation uses a cascading conditional structure to determine the appropriate provider based on two configuration attributes:

1. **Domain-based routing:** Uses `smsServer.getDomain()` for VCSC and KIS
2. **Type-based routing:** Uses `smsServer.getType()` for FPT_SMS and ONE_SMS



Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L42-L50> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L42-L50>)

Provider Protocol Summary

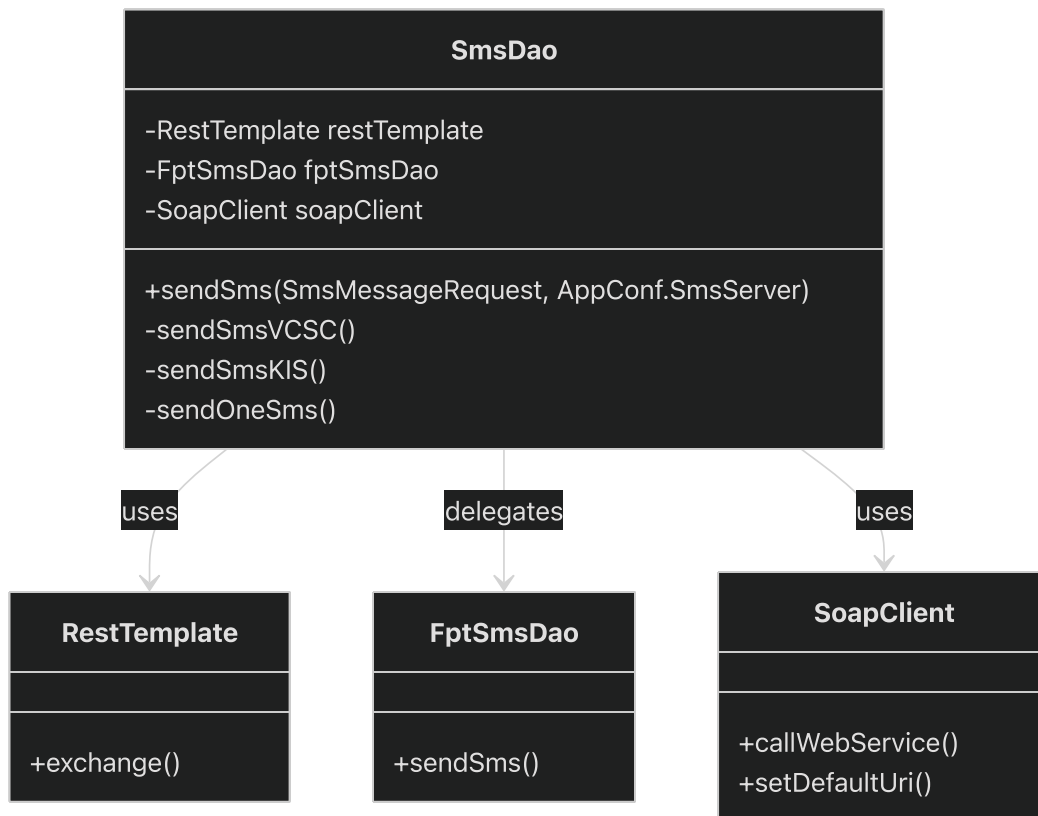
Provider	Protocol	Authentication	Content Format	Response Type
VCSC	HTTP POST	API Key Header	Form Data	Object
KIS	HTTP GET	URL-embedded	Query Parameters	String
FPT_SMS	REST API	OAuth Token	JSON	Delegated
ONE_SMS	SOAP	Username/Password	XML	SendMTResponse

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L53-L101> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L53-L101>)

Dependencies and Integration Points

Constructor Dependencies

The `SmsDao` class requires three key dependencies injected through its constructor:



Dependency Roles:

- `RestTemplate` : HTTP communication for VCSC and KIS providers
- `FptSmsDao` : Specialized OAuth-based FPT SMS handling
- `SoapClient` : SOAP web service communication for ONE_SMS

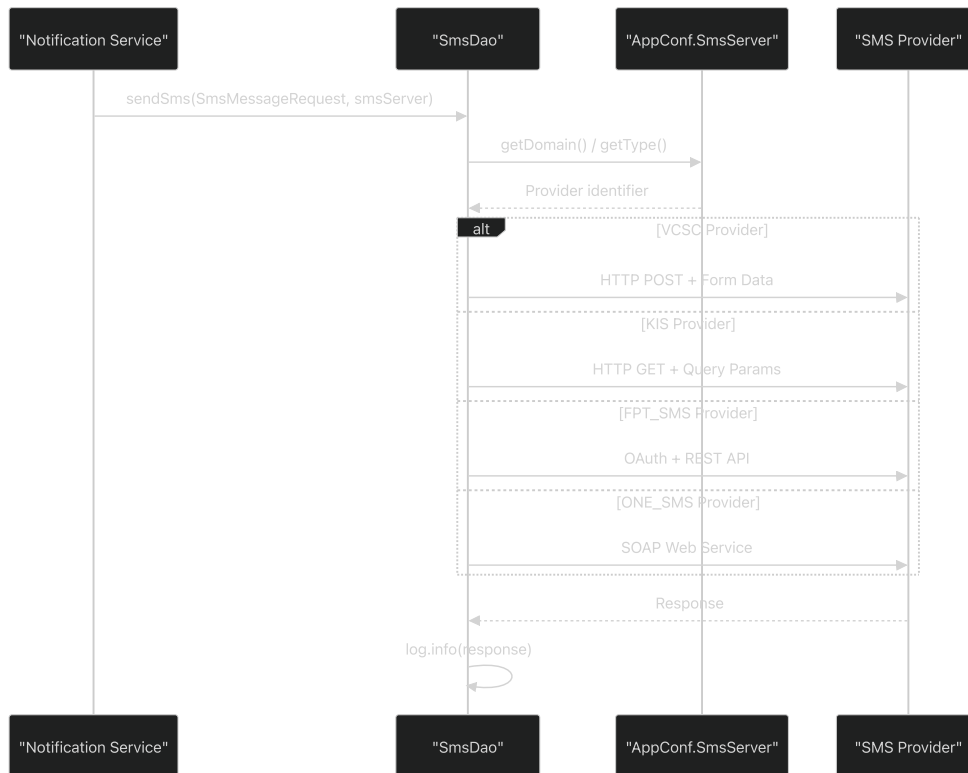
Sources: [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L25-L38)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L25-L38](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L25-L38)

[src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java25-38](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L25-L38) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L25-L38)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L25-L38](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L25-L38))

Message Flow Architecture



Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L40-L101>)

[src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java40-101](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L40-L101) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsDao.java#L40-L101>)

OneSignal SMS Integration

Relevant source files

- [
src/main/java/com/techx/tradex/notification/dao/OneSignalDao.java](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/OneSignalDao.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/OneSignalDao.java>)
- [
src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java>)
- [
src/main/java/com/techx/tradex/notification/model/SmsOneSignalRequest.java](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsOneSignalRequest.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsOneSignalRequest.java>)

Purpose and Scope

This document covers the OneSignal SMS integration within the notification service, which provides international SMS delivery capabilities through the OneSignal platform. The integration supports both phone number registration and SMS message sending operations via OneSignal's REST API.

For FPT SMS integration specific to the Vietnamese market, see [FPT SMS Integration](#). For the broader multi-provider SMS architecture, see [Multi-Provider SMS Architecture](#).

Overview

The OneSignal SMS integration provides an alternative SMS delivery channel to the notification service through OneSignal's unified messaging platform. This integration is particularly useful for international SMS delivery and leverages OneSignal's infrastructure for reliability and global reach.

The system consists of two primary operations:

- **Phone Number Registration:** Registering phone numbers as devices in OneSignal
- **SMS Sending:** Delivering SMS messages to registered phone numbers

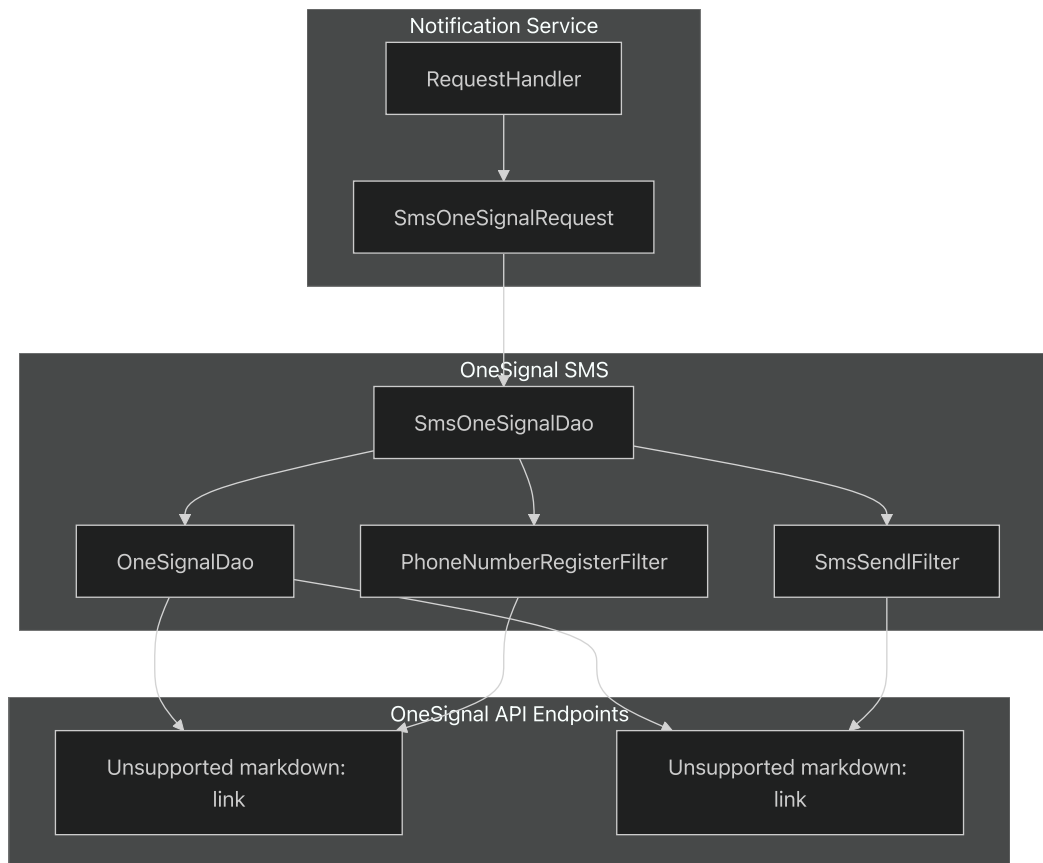
Sources: [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L1-L76)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L1-L76](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L1-L76)

[src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java1-76](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L1-L76) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L1-L76)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L1-L76](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L1-L76))

Architecture Overview



Sources: [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35)

[src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java22-35](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35))

[src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java22-35](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35))

[src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java22-35](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35))

Core Components

SmsOneSignalDao

The `SmsOneSignalDao` class serves as the primary implementation for OneSignal SMS operations. It implements the `SmsOneSignalDao` interface and provides two main functionalities:

Method	Purpose	OneSignal Endpoint
<code>sendSms(SmsOneSignalRequest)</code>	Send SMS to registered phone number	<code>/api/v1/notifications</code>
<code>registerPhoneNumber(SmsOneSignalRequest)</code>	Register phone number as OneSignal device	<code>/api/v1/players</code>

The class uses dependency injection to receive:

- `OneSignalDao` : For API communication
- `ObjectMapper` : For JSON serialization

Sources: <https://github.com/nhsv->

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35).

[src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java22-35](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L35>).

Data Models

SmsOneSignalRequest

The `SmsOneSignalRequest` class contains all necessary fields for OneSignal SMS operations:

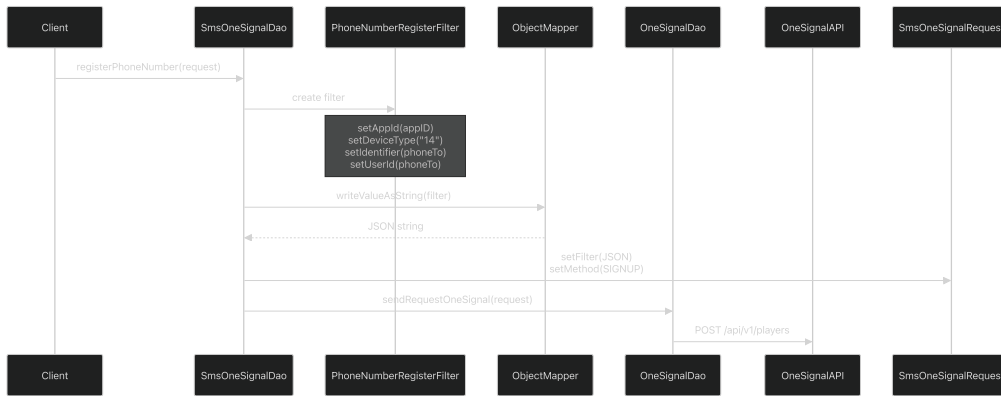
Field	Type	Purpose
<code>appID</code>	String	OneSignal application identifier
<code>appKey</code>	String	OneSignal API key for authentication
<code>method</code>	String	API endpoint URL
<code>phoneFrom</code>	String	Sender phone number
<code>phoneTo</code>	String	Recipient phone number
<code>filter</code>	String	JSON filter for API request
<code>content</code>	String	SMS message content

Sources: <https://github.com/nhsv->

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsOneSignalRequest.java#L5-L18](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsOneSignalRequest.java#L5-L18).

[src/main/java/com/techx/tradex/notification/model/SmsOneSignalRequest.java5-18](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsOneSignalRequest.java#L5-L18) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsOneSignalRequest.java#L5-L18>).

Phone Number Registration Process



The registration process creates a `PhoneNumberRegisterFilter` with the following configuration:

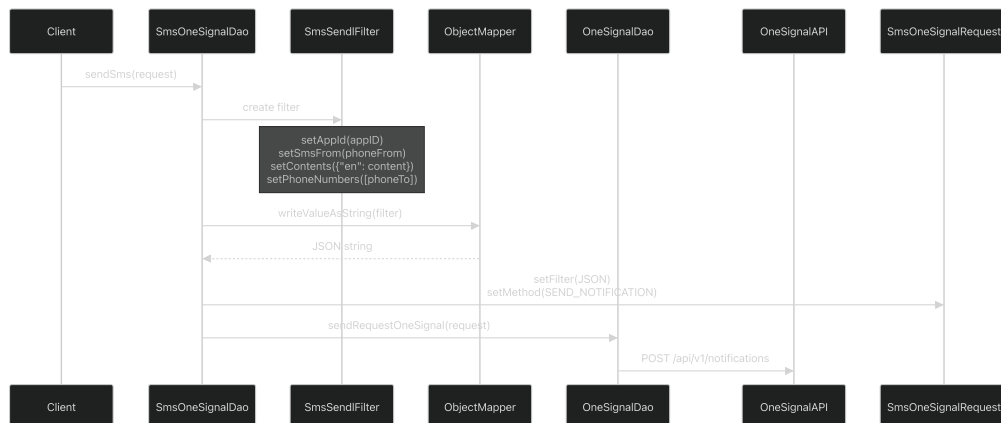
- **Device Type:** Set to "14" (SMS device type constant)
- **Identifier:** The phone number to register
- **User ID:** Same as the phone number for identification

Sources: [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L56-L72)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L56-L72](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L56-L72).

[src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java56-72](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L56-L72) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L56-L72)

SMS Sending Process



The SMS sending process constructs a `SmsSendFilter` with:

- **Content Language:** Default to English ("en")
- **Phone Numbers:** Array containing the target phone number
- **SMS From:** Sender identification
- **App ID:** OneSignal application identifier

Sources: [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L38-L53)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L38-L53](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L38-L53).

[src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java38-53](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L38-L53) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L38-L53)

API Integration Details

Constants and Configuration

The implementation defines several key constants:

Constant	Value	Purpose
SIGNUP	<code>https://onesignal.com/api/v1/players</code>	Player/device registration endpoint
SEND_NOTIFICATION	<code>https://onesignal.com/api/v1/notifications</code>	Notification sending endpoint
DEVICE_TYPE	<code>"14"</code>	SMS device type identifier

Sources: <https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L25-L27>.

<src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#25-27> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L25-L27>).

JSON Serialization

Both operations use Jackson `ObjectMapper` with pretty printing for JSON serialization:

```
ObjectWriter ow = objectMapper.writer().withDefaultPrettyPrinter();
String filterSendSMS = ow.writeValueAsString(smsSendFilter);
```

Error handling is implemented for JSON processing exceptions, particularly in the phone number registration flow where `InvalidFormatException` is thrown for serialization failures.

Sources: <https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L48-L49>.

<src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#48-49> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L48-L49>)

<src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#62-69> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L62-L69>)

Integration with Core System

The OneSignal SMS integration plugs into the notification service's multi-provider SMS architecture through the `OneSignalDao` interface. The `SmsOneSignalDao` implementation is registered as a Spring `@Repository` component, making it available for dependency injection throughout the application.

The integration follows the same pattern as other SMS providers in the system, allowing the notification service to route SMS requests to OneSignal based on configuration or routing logic defined in the broader SMS system architecture.

Sources: <https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L23>.

<src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#22-23> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/SmsOneSignalDao.java#L22-L23>)

<src/main/java/com/techx/tradex/notification/dao/OneSignalDao.java#5-7> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/OneSignalDao.java#L5-L7>)

Push Notification System

Relevant source files

- [[src/main/java/com/techx/tradex/notification/model/dto/NotificationDTO.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/dto/NotificationDTO.java)](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/dto/NotificationDTO.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/dto/NotificationDTO.java>))
- [[src/main/java/com/techx/tradex/notification/services/OneSignalService.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java)](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java>))
- [[src/main/java/com/techx/tradex/notification/test/TestService.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/test/TestService.java)](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/test/TestService.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/test/TestService.java>))

Purpose and Scope

The Push Notification System provides cross-platform mobile and web push notification capabilities through OneSignal integration. This system handles real-time notifications for user engagement, trading alerts, portfolio updates, and system notifications with multi-language support and template-based content generation.

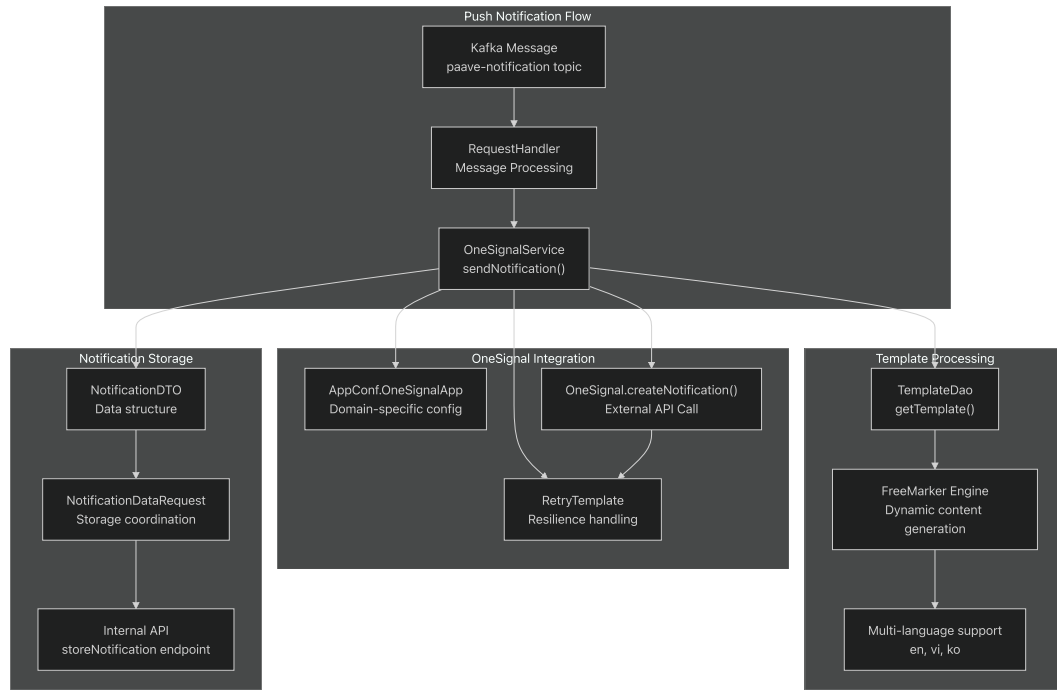
For SMS notifications, see [SMS Notification System](#). For email notifications, see [Email Notification System](#). For real-time messaging via WebSocket, see [Messaging Platform Integrations](#).

System Architecture

The push notification system is built around the `OneSignalService` class, which provides asynchronous notification delivery with template processing, multi-language support, and robust error handling.

Core Components

Component	Class/File	Purpose
Push Service	<code>OneSignalService</code>	Main service for OneSignal push notification delivery
Configuration	<code>AppConf.OneSignalApp</code>	OneSignal app configuration per domain
Templates	<code>TemplateDao</code>	Template loading and FreeMarker processing
Data Model	<code>NotificationDTO</code>	Notification data structure for storage
Request Handling	<code>RequestSender</code>	Internal message routing for notification storage



Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L40-L222>)

*[src/main/java/com/techx/tradex/notification/services/OneSignalService.java40-222](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L40-L222) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L40-L222>)

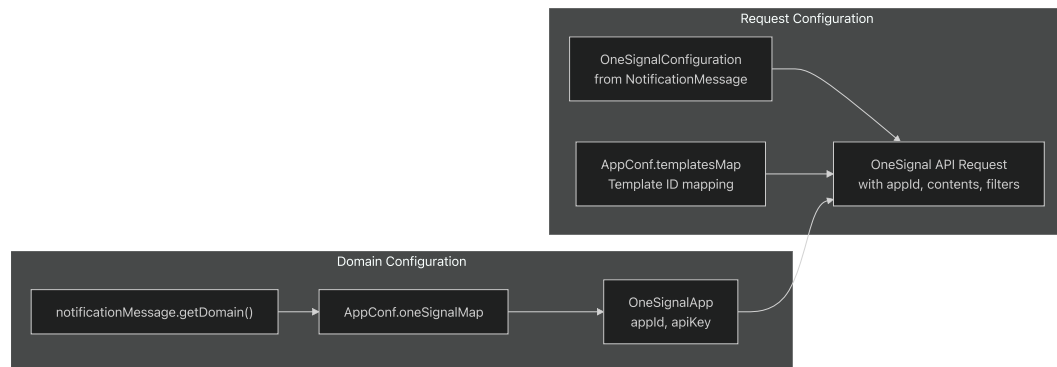
[src/main/java/com/techx/tradex/notification/model/dto/NotificationDTO.java10-30](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/dto/NotificationDTO.java#L10-L30) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/dto/NotificationDTO.java#L10-L30>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/dto/NotificationDTO.java#L10-L30>))*

OneSignal Integration

The system integrates with OneSignal platform through the `OneSignal.createNotification()` API, providing cross-platform push notification delivery to mobile and web clients.

Configuration Structure

The OneSignal configuration is managed through domain-specific settings:



Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L75-L83>)

[src/main/java/com/techx/tradex/notification/services/OneSignalService.java75-83](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L75-L83) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L75-L83>)

API Request Processing

The service builds OneSignal API requests with the following components:

Component	Description	Code Reference
App ID	Domain-specific OneSignal app identifier	<code>request.setAppId(oneSignalApp.getAppId())</code>
Contents	Multi-language notification content	<code>request.setContents(contents)</code>
Headings	Optional notification titles	<code>request.setHeadings(headings)</code>
Template ID	OneSignal template identifier	<code>request.setTemplateId(templatesMap.get(template))</code>
Data	Custom key-value payload	<code>request.setData(data)</code>
URL	Deep link or web URL	<code>request.setUrl(notificationMessage.getUrl())</code>

Sources: <https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L83-L120>

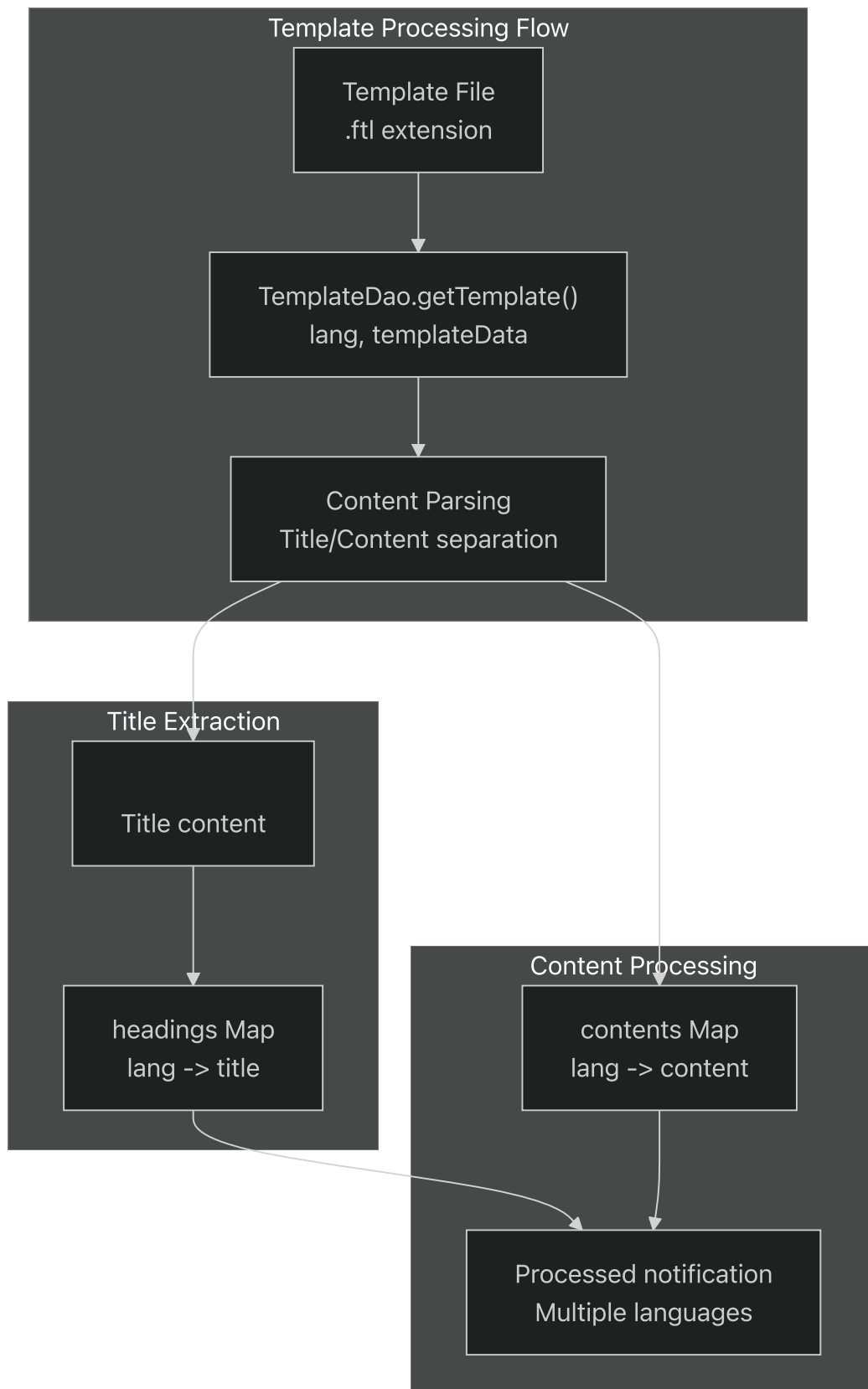
<src/main/java/com/techx/tradex/notification/services/OneSignalService.java83-120> (<https://github.com/nhsv->
<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L83-L120>)

Template Processing and Multi-Language Support

The push notification system processes templates through FreeMarker with support for English, Vietnamese, and Korean languages.

Template Content Structure

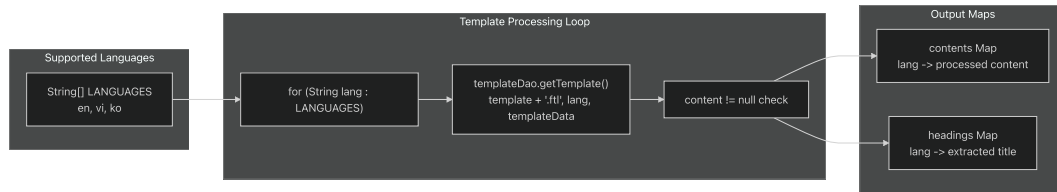
Templates can include both title and content using special tags:



Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L88-L107>)
[src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L88-L107](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L88-L107) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L88-L107>)

Language Processing

The system processes templates for each supported language:



Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L42-L101>
[src/main/java/com/techx/tradex/notification/services/OneSignalService.java42-101](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L42-L101) <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L42-L101>

Notification Targeting and Filtering

The system supports sophisticated targeting through OneSignal's filter system and special handling for different notification types.

Filter-Based Targeting

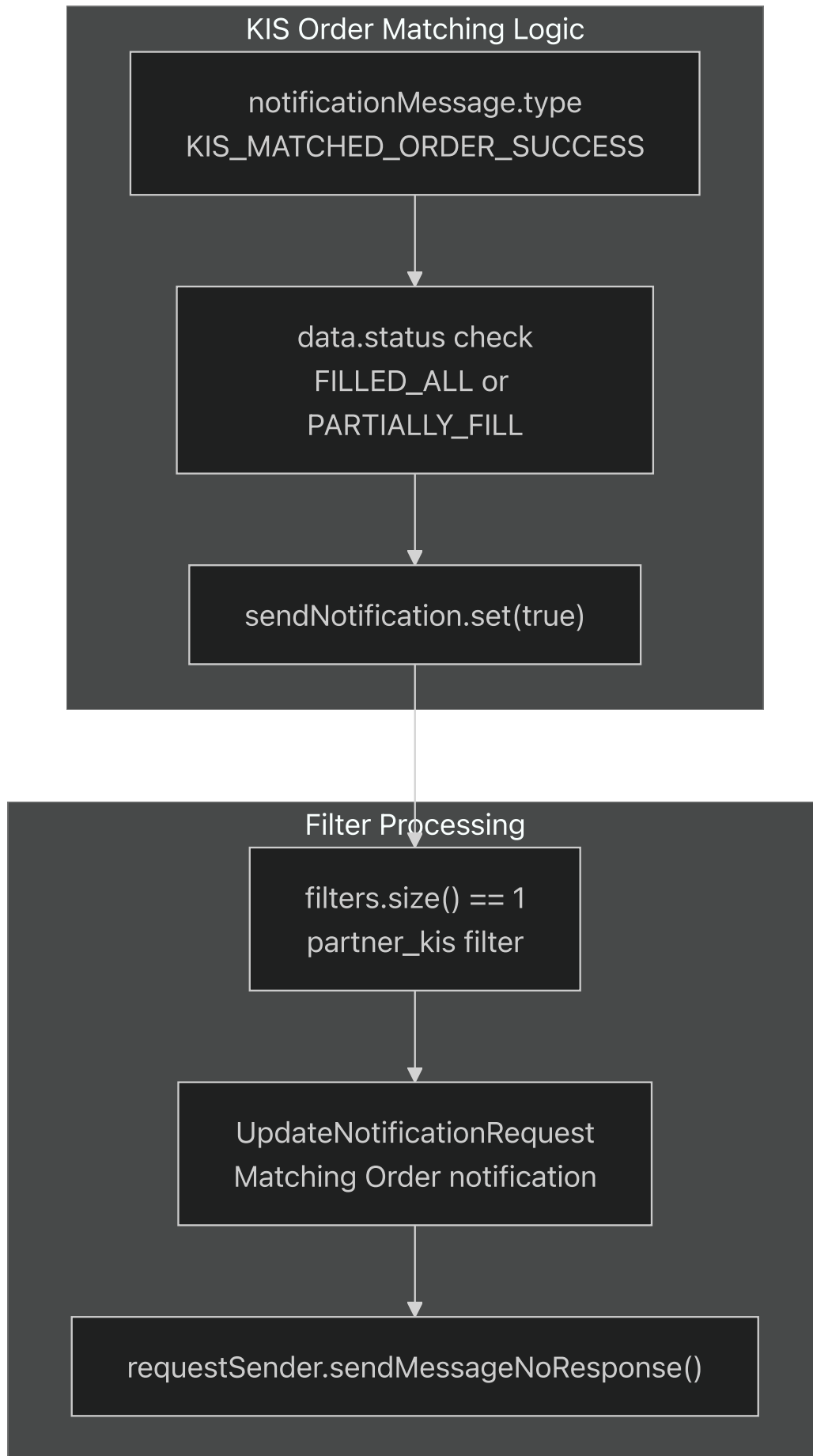
The service processes OneSignal filters for user targeting:

Filter Type	Purpose	Implementation
userid	User-specific targeting	<code>filter.getKey().equals("userid")</code>
partner_kis	KIS partner notifications	Special handling for order matching
NOT_EXISTS	Exclude certain users	Skip database storage for these notifications

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L145-L181>
[src/main/java/com/techx/tradex/notification/services/OneSignalService.java145-181](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L145-L181) <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L145-L181>

Special Notification Types

The system includes custom logic for specific notification types:

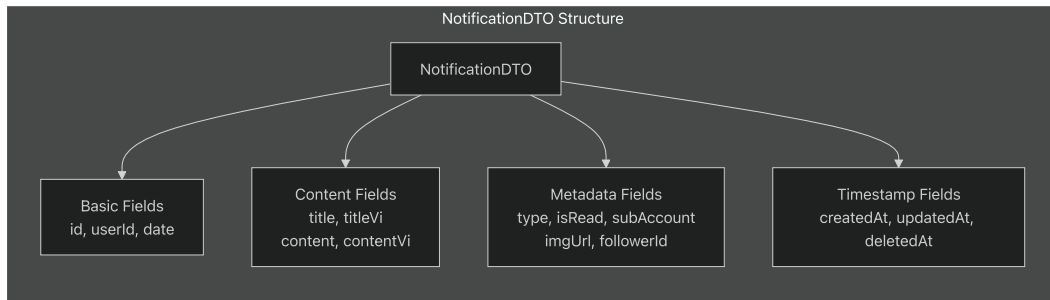


Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L113-L166>
[src/main/java/com/techx/tradex/notification/services/OneSignalService.java113-166](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L113-L166) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L113-L166>)

Notification Storage and Tracking

The system provides comprehensive notification storage for user history and tracking through the `NotificationDTO` data model.

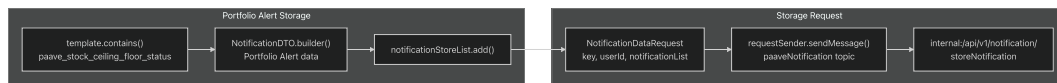
Data Model Structure



Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/dto/NotificationDTO.java#L14-L30>
[src/main/java/com/techx/tradex/notification/model/dto/NotificationDTO.java14-30](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/dto/NotificationDTO.java#L14-L30) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/dto/NotificationDTO.java#L14-L30>)

Special Notification Storage

The system includes specific storage logic for portfolio alerts:



Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L200-L221>

*[src/main/java/com/techx/tradex/notification/services/OneSignalService.java200-221](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L200-L221) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L200-L221>)

[src/main/java/com/techx/tradex/notification/services/OneSignalService.java184-191](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L184-L191) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L184-L191>)
[L191](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L184-L191) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L184-L191>))*

Configuration and Domain Management

The push notification system uses domain-based configuration to support multiple applications and environments.

Configuration Structure

Configuration Type	Purpose	Location
OneSignal Apps	<code>appConf.getOneSignalMap()</code>	Domain-specific OneSignal app credentials
Template Mapping	<code>appConf.getTemplatesMap()</code>	Maps template names to OneSignal template IDs
Notification Topics	<code>appConf.getTopics().getPaaveNotification()</code>	Kafka topics for internal messaging
Notification URI	<code>appConf.getNotificationUri()</code>	Internal API endpoints for storage

Sources: ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L75-L191)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L75-L191](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L75-L191))

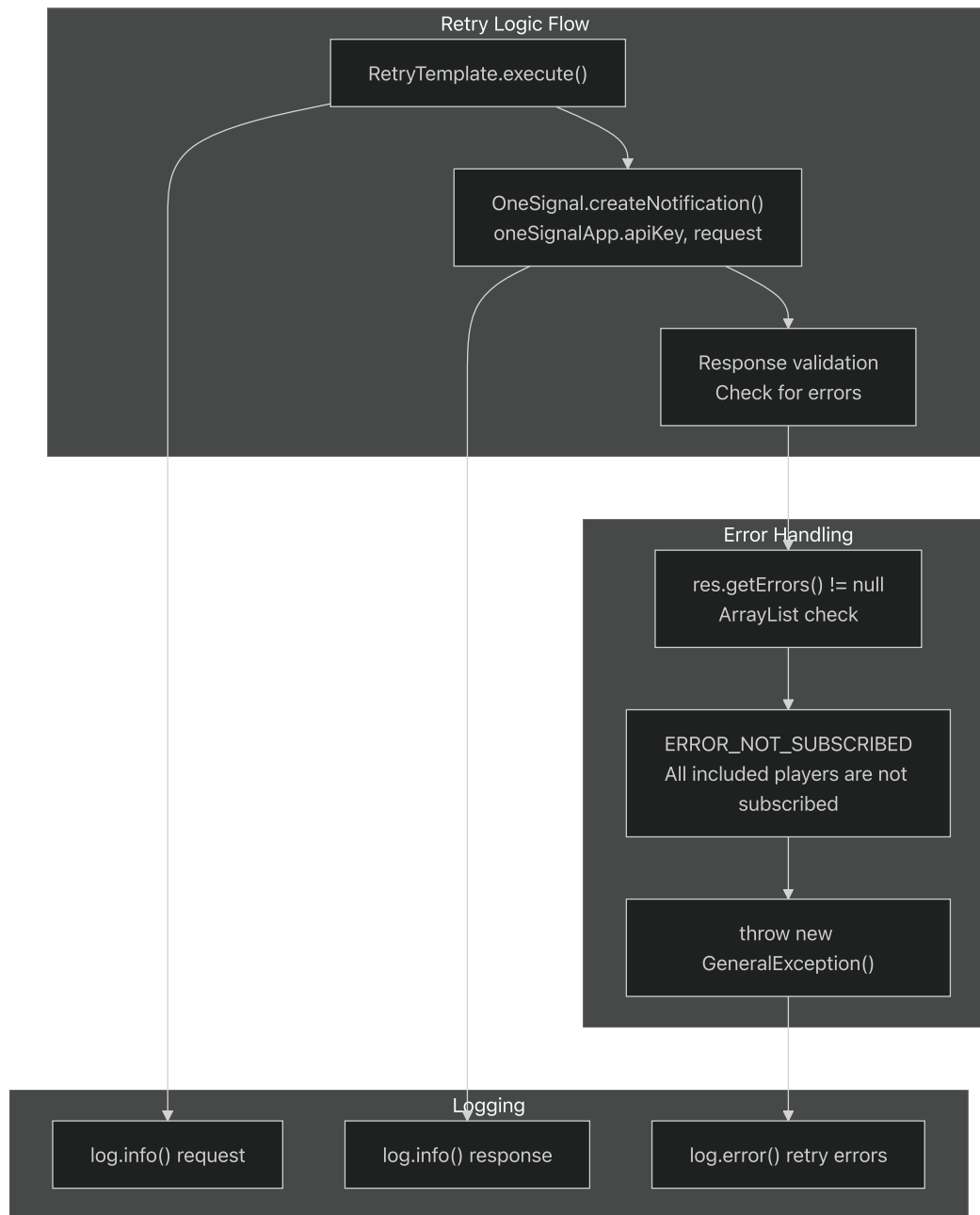
[src/main/java/com/techx/tradex/notification/services/OneSignalService.java75-191](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L75-L191) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L75-L191)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L75-L191](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L75-L191))

Error Handling and Retry Logic

The system implements robust error handling with retry capabilities for reliable notification delivery.

Retry Implementation



Sources: <https://github.com/nhsv->

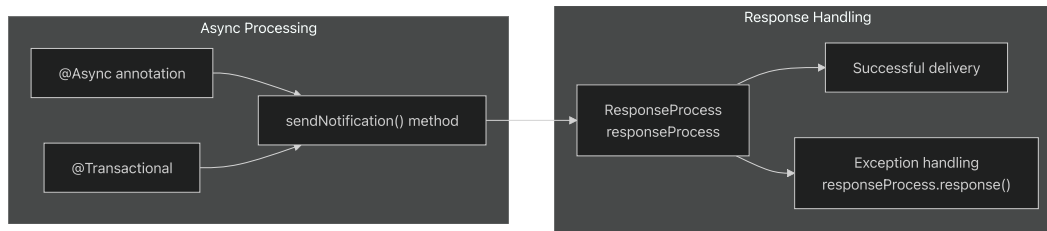
[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L122-L136](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L122-L136)

[src/main/java/com/techx/tradex/notification/services/OneSignalService.java122-136](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L122-L136) <https://github.com/nhsv->

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L122-L136](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L122-L136)

Asynchronous Processing

The service uses Spring's `@Async` annotation for non-blocking notification processing:



Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L63-L198> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/OneSignalService.java#L63-L198>)

Messaging Platform Integrations

Overview

Relevant source files

- [[src/main/java/com/techx/tradex/notification/constants/TopicMessageType.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/constants/TopicMessageType.java)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/constants/TopicMessageType.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/constants/TopicMessageType.java>)
- [[src/main/java/com/techx/tradex/notification/dao/KakaoDao.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/KakaoDao.java)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/KakaoDao.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/KakaoDao.java>)
- [[src/main/java/com/techx/tradex/notification/dao/ZaloDao.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/ZaloDao.java)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/ZaloDao.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/ZaloDao.java>)
- [[src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java>)
- [[src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java>)
- [[src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java>)
- [[src/main/java/com/techx/tradex/notification/services/KakaoService.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java>)
- [[src/main/java/com/techx/tradex/notification/services/TopicService.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java>)

(<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java>))

Purpose and Scope

This document covers the notification service's integrations with region-specific messaging platforms that provide enhanced user engagement capabilities beyond traditional SMS and email channels. The system currently supports KakaoTalk for the Korean market and Zalo Official Account for the Vietnamese market.

For email notification capabilities, see [Email Notification System](#). For SMS functionality, see [SMS Notification System](#). For push notifications, see [Push Notification System](#).

Overview

The messaging platform integrations enable the notification service to send messages through popular regional communication platforms. These integrations are implemented through dedicated DAO patterns and service layers that handle platform-specific authentication, message formatting, and delivery protocols.

Supported Platforms

Platform	Market	Integration Type	Primary Use Case
KakaoTalk	Korean	Alimtalk API	Business messaging with template support
Zalo OA	Vietnamese	Official Account API	Direct messaging to users

The messaging platforms are integrated into the bulk notification system through `TopicService` and can be used standalone through dedicated service classes.

Sources: ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L1-L162)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L1-L162](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L1-L162)).

[src/main/java/com/techx/tradex/notification/services/TopicService.java1-162](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L1-L162) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L1-L162)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L1-L162](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L1-L162))[

[src/main/java/com/techx/tradex/notification/constants/TopicMessageType.java1-6](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/constants/TopicMessageType.java#L1-L6)]([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/constants/TopicMessageType.java#L1-L6)

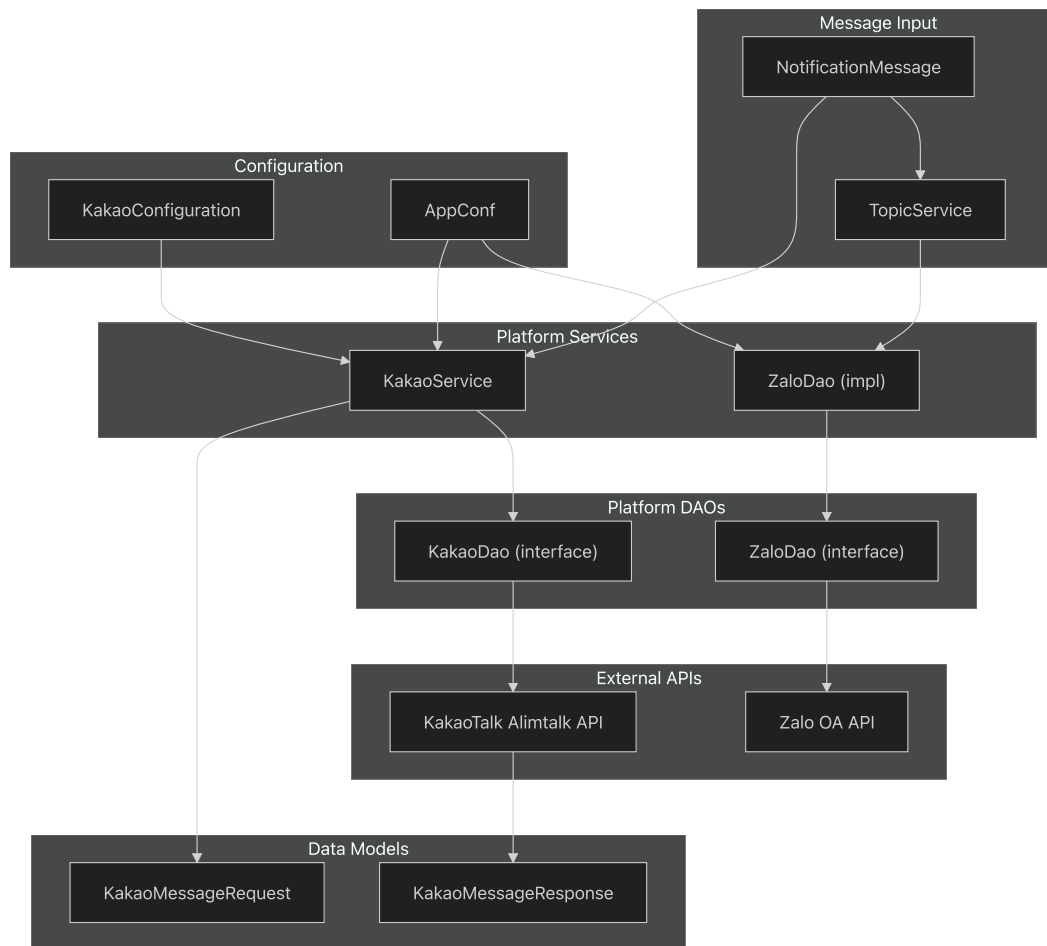
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[L6](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/constants/TopicMessageType.java#L1-L6) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/constants/TopicMessageType.java#L1-L6)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/constants/TopicMessageType.java#L1-L6](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/constants/TopicMessageType.java#L1-L6))).

System Architecture

The messaging platform integrations follow a layered architecture with dedicated DAOs, service classes, and configuration management.

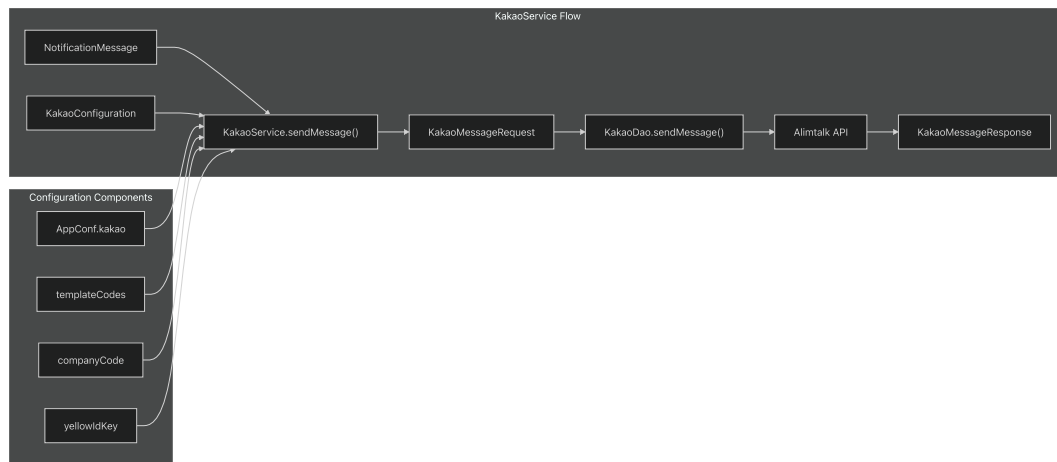


Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L1-L96>
[src/main/java/com/techx/tradex/notification/services/KakaoService.java1-96](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L1-L96) [\[https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L1-L96\]](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L1-L96)
[src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java1-63](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L1-L63) [\[https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L1-L63\]](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L1-L63)
[\[https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L1-L63\]](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L1-L63)
[src/main/java/com/techx/tradex/notification/dao/KakaoDao.java1-10](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/KakaoDao.java#L1-L10) [\[https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/KakaoDao.java#L1-L10\]](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/KakaoDao.java#L1-L10)
[\[https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/KakaoDao.java#L1-L10\]](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/KakaoDao.java#L1-L10)

KakaoTalk Integration

Service Architecture

The KakaoTalk integration is implemented through `KakaoService` and `KakaoDao` components that handle Alimtalk API communication for business messaging in the Korean market.



Message Processing Flow

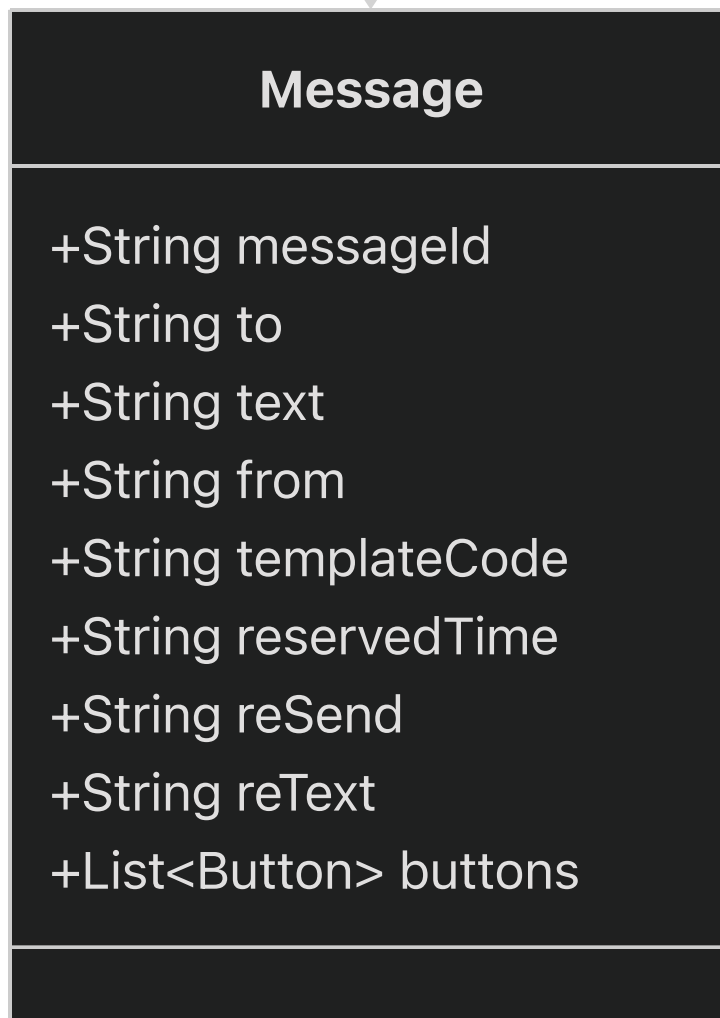
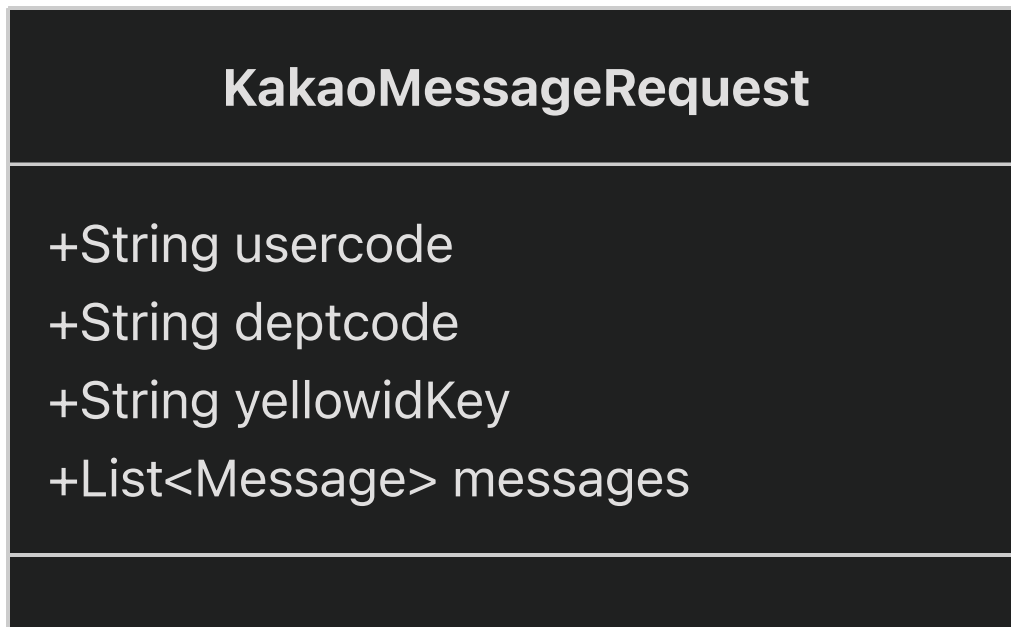
The `KakaoService.sendMessage()` method processes notification messages through the following steps:

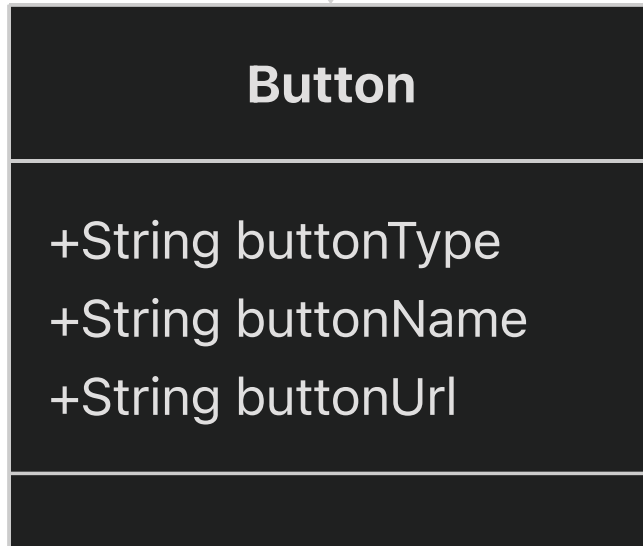
- 1. Configuration Setup:** Extracts company code, user ID, and Yellow ID key from `AppConf.kakao`
- 2. Template Processing:** Iterates through notification templates and generates `KakaoMessageRequest.Message` objects
- 3. Template Code Resolution:** Maps templates to Kakao template codes using `appConf.kakao.templateCodes`
- 4. API Communication:** Sends requests through `KakaoDao.sendMessage()` and processes responses

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L44-L94>
[src/main/java/com/techx/tradex/notification/services/KakaoService.java44-94](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L44-L94) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L44-L94>)

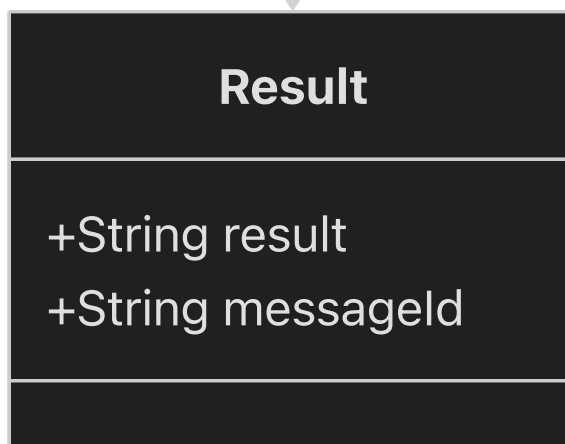
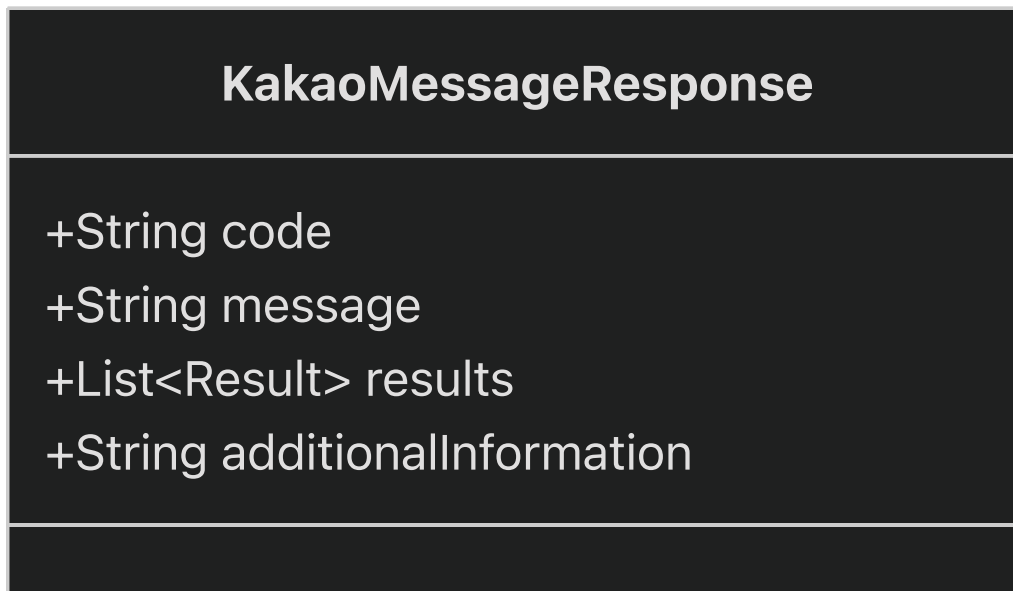
Data Models

KakaoMessageRequest Structure





KakaoMessageResponse Structure



Sources: [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java#L1-L35)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java#L1-L35](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java#L1-L35))

[src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java1-35](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java#L1-L35) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java#L1-L35)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java#L1-L35](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java#L1-L35))

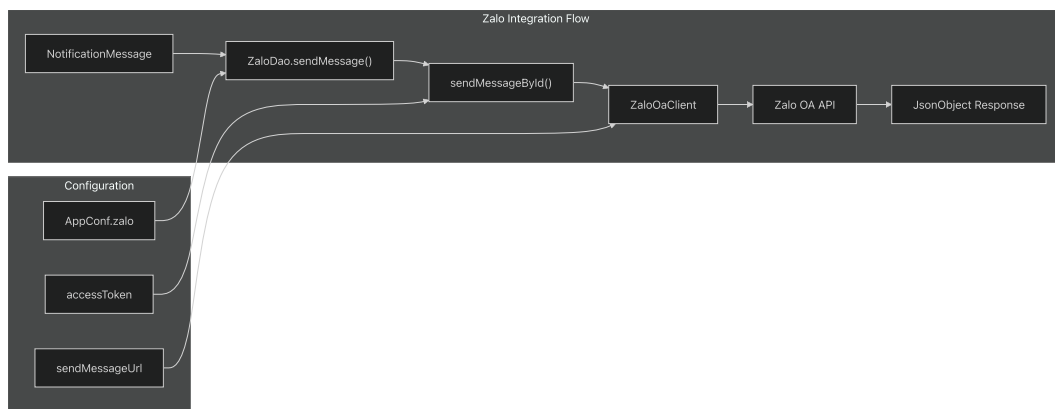
[src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java1-20](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java#L1-L20)]([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java#L1-L20)

[L20](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java#L1-L20) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java#L1-L20)

Zalo Integration

Service Architecture

The Zalo integration uses the Zalo Official Account (OA) API through the `ZaloDao` implementation for direct messaging to Vietnamese users.



Message Processing Implementation

The `ZaloDao.sendMessageById()` method handles the core message delivery:

- Parameter Setup:** Creates request parameters with access token from `appConf.zalo.accessToken`
- Message Construction:** Builds JSON payload with recipient user ID and message text
- API Request:** Uses `ZaloOaClient.executeRequest()` to send POST request to Zalo API
- Response Handling:** Parses JSON response and implements retry logic for failed messages
- Error Recovery:** Retries failed messages up to 3 times before giving up

The implementation extracts message content from the notification template:

```
notificationMessage.getTemplate().get(msgType)
```

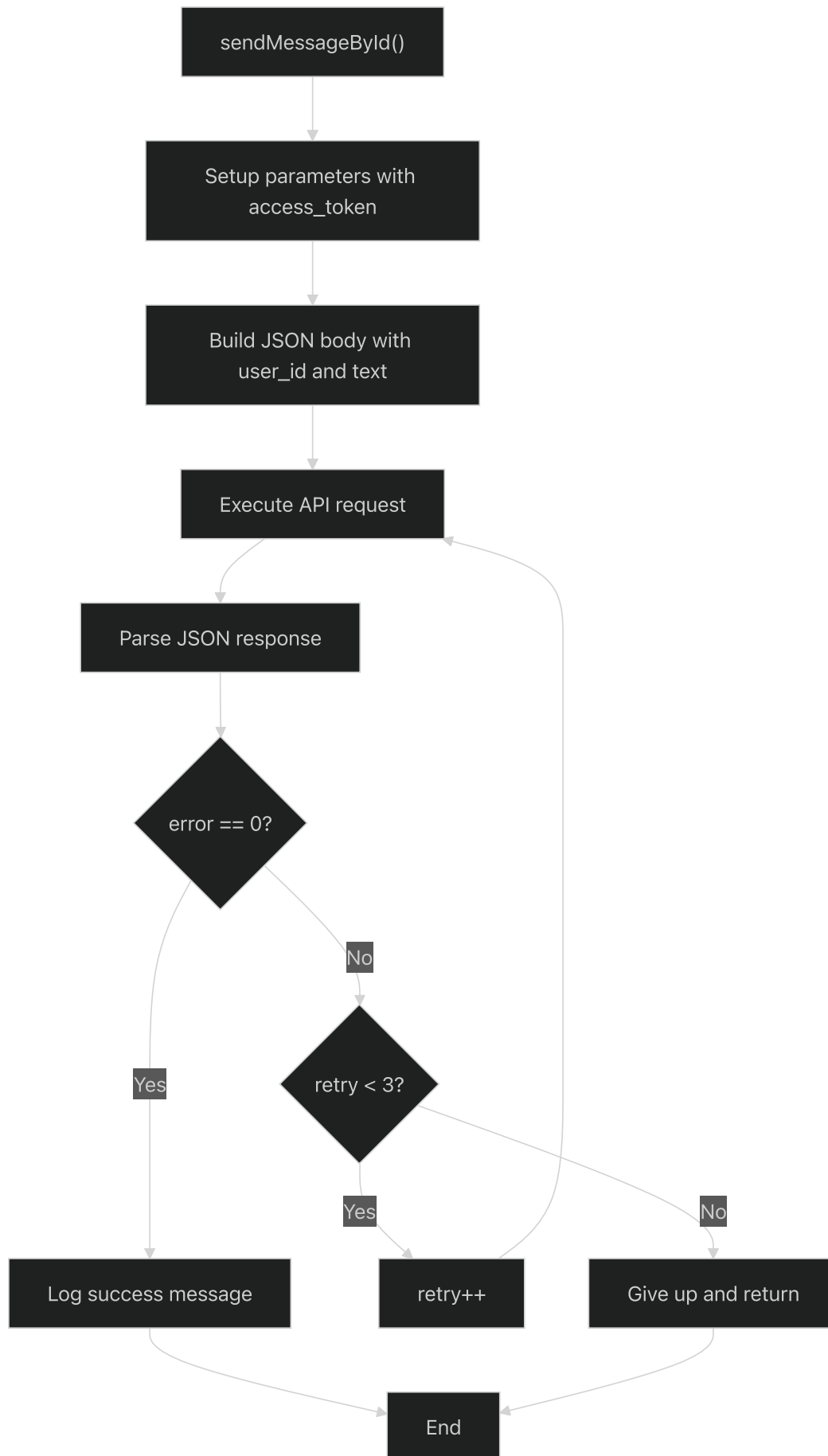
Sources: [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L36-L62)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L36-L62](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L36-L62))

[src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java36-62](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L36-L62) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L36-L62)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L36-L62](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L36-L62))

Error Handling and Retry Logic



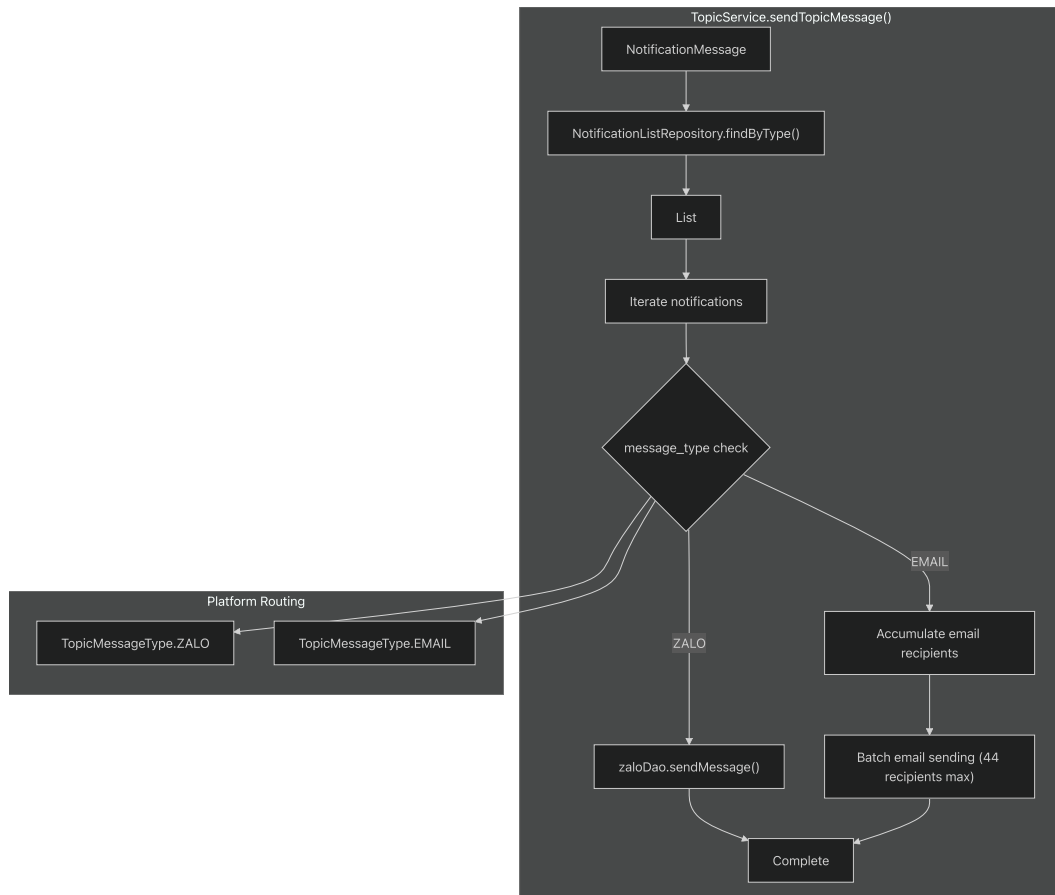
Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L51-L61>

[src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L51-61](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L51-L61) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L51-L61>).

Bulk Messaging Integration

TopicService Platform Routing

The `TopicService` integrates messaging platforms into the bulk notification system, allowing messages to be sent to multiple recipients across different platforms based on `TopicMessageType`.



Recipient Management

The `TopicService` implements recipient batching for email while sending Zalo messages individually:

- **Zalo Messages:** Sent immediately to individual recipients via `zaloDao.sendMessage(notificationMessage, item.getTo())`
- **Email Messages:** Accumulated into batches with TO/CC/BCC recipients, with a maximum of 44 recipients per batch
- **Batch Processing:** When recipient limit is reached, `doSendMessage()` is called to send the batch

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L60-L80>

[src/main/java/com/techx/tradex/notification/services/TopicService.java#L60-80](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L60-L80) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/TopicService.java#L60-L80>).

Configuration and Template Integration

Platform Configuration Structure

Both messaging platforms integrate with the central configuration system:

Platform	Configuration Class	Key Properties
KakaoTalk	KakaoConfiguration	to, from, resend, templateCode
Zalo	AppConf.zalo	accessToken, sendMessageUrl

Template Processing

Both platforms utilize the `TemplateDao` for message content generation:

- **KakaoTalk:** Uses `templateDao.getTemplate(template, notificationMessage.getLocale(), templateData)` for text content
- **Zalo:** Extracts message from notification template map structure

The template system supports localization through the `notificationMessage.getLocale()` parameter for KakaoTalk messages.

Sources: [https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L65-L65)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L65-L65](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L65-L65)

[src/main/java/com/techx/tradex/notification/services/KakaoService.java65](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L65-L65) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L65-L65)
[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L65-L65](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L65-L65))

[src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java42-44](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L42-L44) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L42-L44)
[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L42-L44](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L42-L44)
<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/impl/ZaloDao.java#L42-L44>)

KakaoTalk Integration

Relevant source files

- [[src/main/java/com/techx/tradex/notification/dao/KakaoDao.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/KakaoDao.java)] ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/KakaoDao.java)
[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/KakaoDao.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/KakaoDao.java)
<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/KakaoDao.java>)
- [[src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java)] ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java)
[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java)
<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java>)
- [[src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java)] ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java)
[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java)
<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java>)
- [[src/main/java/com/techx/tradex/notification/services/KakaoService.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java)] ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java)
[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java)
<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java>)

Purpose and Scope

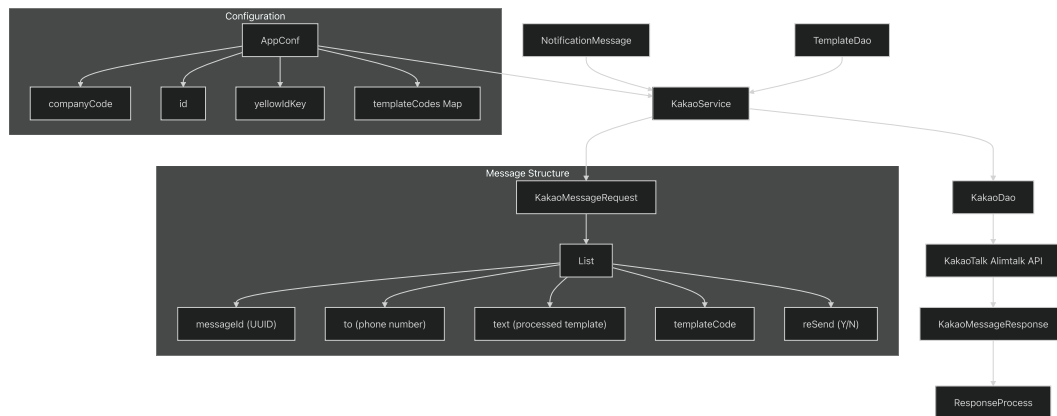
The KakaoTalk integration provides messaging capabilities for the Korean market through the KakaoTalk Alimtalk API. This system enables sending template-based messages to KakaoTalk users as part of the notification service's multi-channel messaging strategy.

For SMS messaging capabilities, see [SMS Notification System](#). For other messaging platform integrations, see [Messaging Platform Integrations](#).

System Overview

The KakaoTalk integration consists of three primary components: the `KakaoService` for business logic, the `KakaoDao` interface for API communication, and data models for request/response handling. The system integrates with the notification service's template system and configuration management to deliver localized messages to Korean users.

KakaoTalk Integration Architecture



Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L1-L95>)

[src/main/java/com/techx/tradex/notification/services/KakaoService.java1-95](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L1-L95) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L1-L95>)

[src/main/java/com/techx/tradex/notification/dao/KakaoDao.java1-10](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/KakaoDao.java#L1-L10) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/KakaoDao.java#L1-L10>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/dao/KakaoDao.java#L1-L10>)

Configuration System

The KakaoTalk integration relies on `AppConf` to provide essential configuration parameters stored in the `kakao` configuration section:

Configuration Property	Purpose	Code Reference
<code>companyCode</code>	Department code for API authentication	[
src/main/java/com/techx/tradex/notification/services/KakaoService.java47](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L47-L47)(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L47-L47)		
<code>id</code>	User code for API authentication	[

src/main/java/com/techx/tradex/notification/services/KakaoService.java48](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L48-L48>)(<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L48-L48>) | | `yellowIdKey`
 | Yellow ID key for API authentication | [

src/main/java/com/techx/tradex/notification/services/KakaoService.java49](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L49-L49>)(<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L49-L49>) | |
`templateCodes` | Map of template names to KakaoTalk template codes | [

src/main/java/com/techx/tradex/notification/services/KakaoService.java76-79](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L76-L79>)(<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L76-L79>) |

The system validates that required template codes exist in the configuration before attempting to send messages.

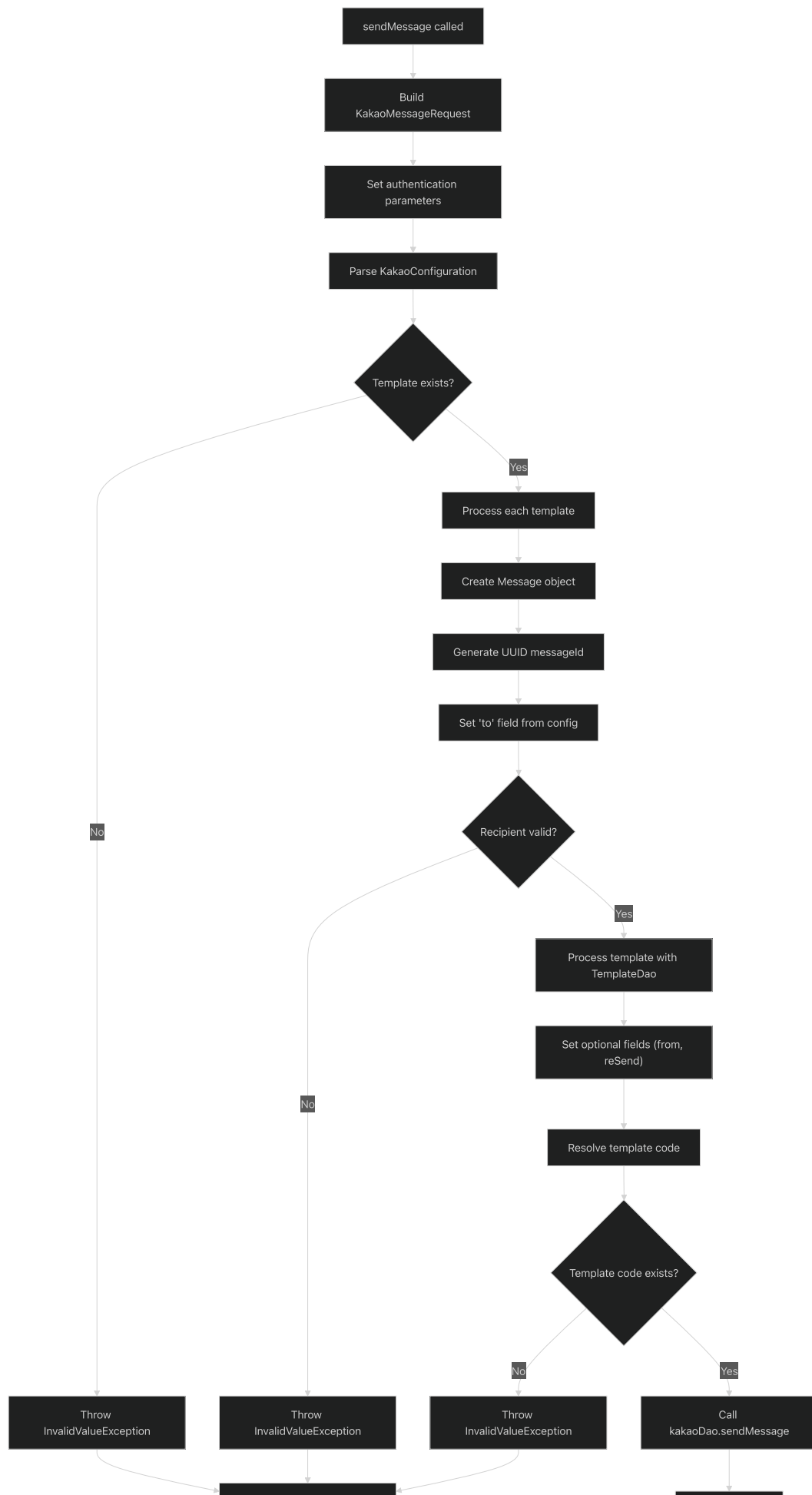
Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L47-L49>),
[src/main/java/com/techx/tradex/notification/services/KakaoService.java47-49](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L47-L49) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L47-L49>)[

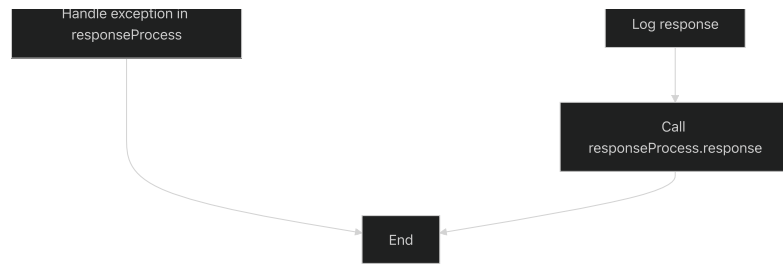
src/main/java/com/techx/tradex/notification/services/KakaoService.java76-79](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L76-L79>)(<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L76-L79>)

Message Processing Flow

The `KakaoService.sendMessage` method implements an asynchronous message processing workflow:

Message Processing Workflow



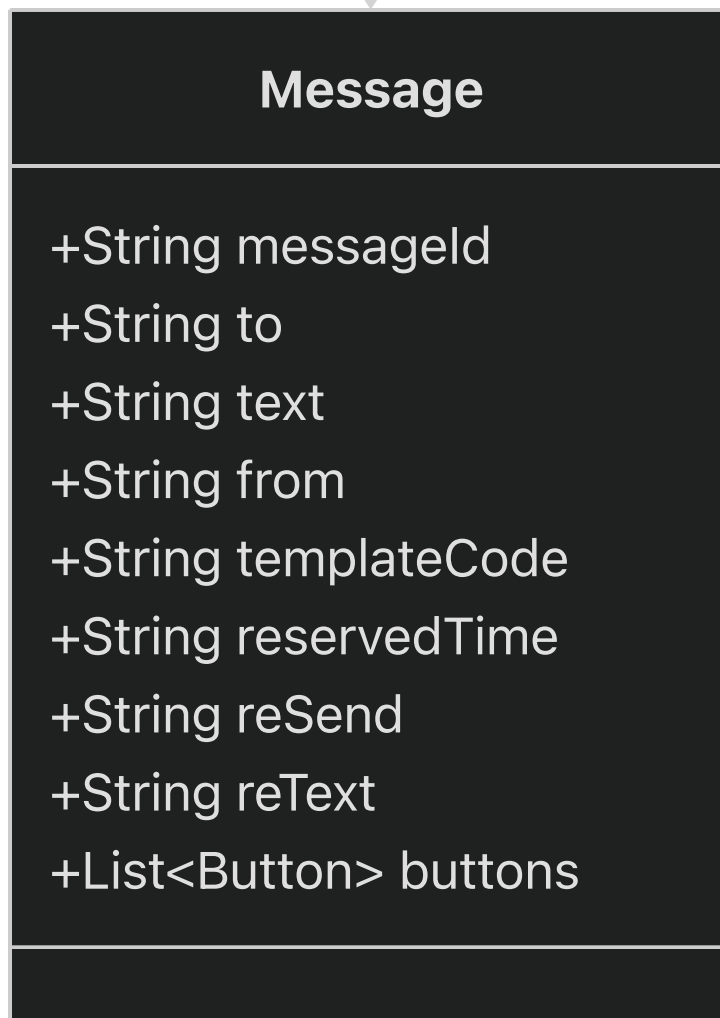
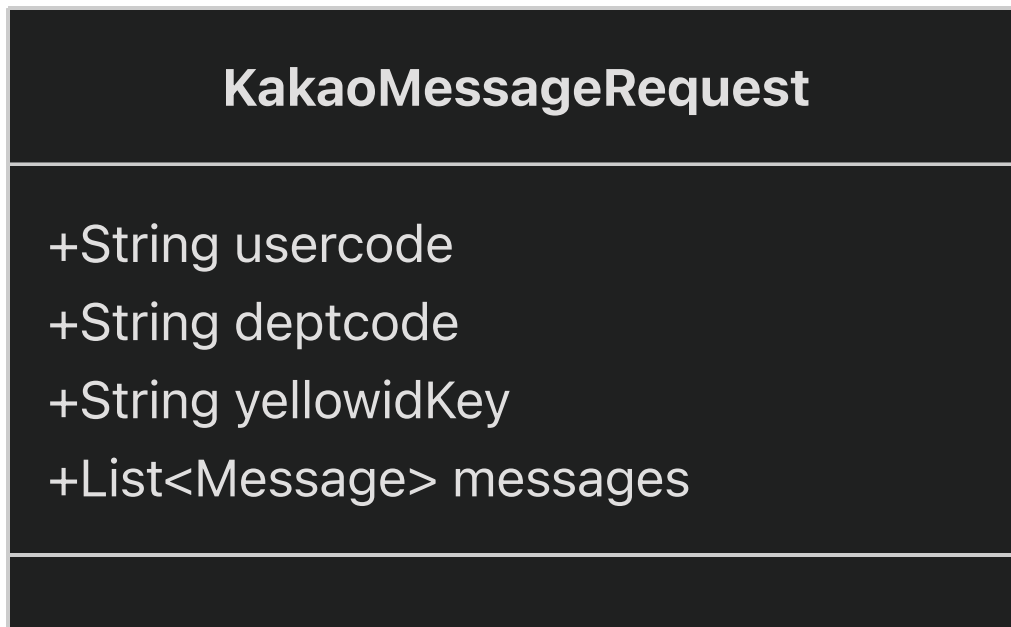


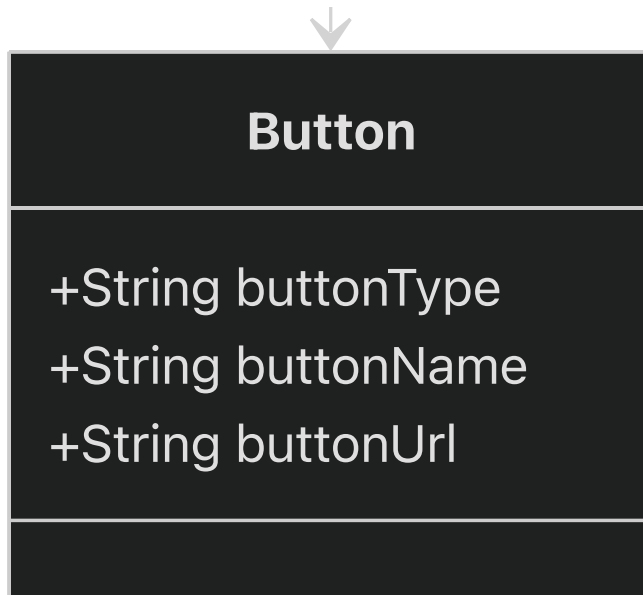
Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L44-L94> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L44-L94>)

Data Models

KakaoMessageRequest Structure

The `KakaoMessageRequest` class defines the structure for API requests:





Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java#L1-L35>)
[src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java1-35](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java#L1-L35) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageRequest.java#L1-L35>)

KakaoMessageResponse Structure

The `KakaoMessageResponse` class handles API responses:

Field	Type	Purpose
<code>code</code>	String	Response status code
<code>message</code>	String	Response message
<code>results</code>	List	Individual message results
<code>additionalInformation</code>	String	Additional response data

Each `Result` object contains:

- `result` : Status of individual message
- `messageId` : Corresponding message identifier

Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java#L1-L20>)
[src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java1-20](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java#L1-L20) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/KakaoMessageResponse.java#L1-L20>)

Template Integration

The KakaoTalk service integrates with the notification system's template engine through the `TemplateDao` :

1. **Template Processing:** The service calls `templateDao.getTemplate(template, notificationMessage.getLocale(), templateData)` to process templates with localization support (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L65-L65>).

[src/main/java/com/techx/tradex/notification/services/KakaoService.java65](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L65-L65) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L65-L65>).

2. **Template Code Resolution:** The system resolves template codes either from the `KakaoConfiguration` or from the `AppConf.kakao.templateCodes` `map`. (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L73-L80>).

[src/main/java/com/techx/tradex/notification/services/KakaoService.java73-80](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L73-L80) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L73-L80>).

3. **Multi-template Support:** The service processes multiple templates in a single notification message by iterating through the template map. (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L55-L81>).

[src/main/java/com/techx/tradex/notification/services/KakaoService.java55-81](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L55-L81) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L55-L81>).

Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L55-L81>).

[src/main/java/com/techx/tradex/notification/services/KakaoService.java55-81](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L55-L81) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L55-L81>).

Error Handling and Validation

The service implements comprehensive validation and error handling:

- **Template Validation:** Throws `InvalidValueException("template")` if templates are null or empty[
[src/main/java/com/techx/tradex/notification/services/KakaoService.java52-54](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L52-L54)](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L52-L54>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L52-L54>)).
- **Recipient Validation:** Throws `InvalidValueException("configuration.to")` if recipient is empty[
[src/main/java/com/techx/tradex/notification/services/KakaoService.java61-63](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L61-L63)](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L61-L63>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L61-L63>)).
- **Template Code Validation:** Throws `InvalidValueException("templateCode")` if template code mapping is missing[
[src/main/java/com/techx/tradex/notification/services/KakaoService.java76-78](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L76-L78)](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L76-L78>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L76-L78>)).
- **Exception Propagation:** All exceptions are caught and passed to the `ResponseProcess` callback for consistent error handling[
[src/main/java/com/techx/tradex/notification/services/KakaoService.java88-93](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L88-L93)](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L88-L93>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L88-L93>)).

The asynchronous design ensures that exceptions don't block the main notification processing thread while still providing proper error reporting through the callback mechanism.

Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L52-L94>).

[src/main/java/com/techx/tradex/notification/services/KakaoService.java#52-94](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L52-L94) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/services/KakaoService.java#L52-L94>).

Template System

Overview

Relevant source files

- [[src/main/resources/templates/email_otp_verify_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html)] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html)
- [[src/main/resources/templates/email_otp_verify_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html)] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html)
- [[src/main/resources/templates/paave_email_otp_verify_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html)] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html)
- [[src/main/resources/templates/reset_password_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html)] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html)
- [[src/main/resources/templates/reset_password_ko.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html)] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html)
- [[src/main/resources/templates/welcome_to_paave_ekyc_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html)] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html)
- [[src/main/resources/templates/welcome_to_paave_ekyc_ko.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_ko.html)] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_ko.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_ko.html)
- [[src/main/resources/templates/welcome_to_paave_ekyc_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_vi.html)] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_vi.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_vi.html)
- [[src/main/resources/templates/welcome_to_paave_kis_acc_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_en.html)] (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_en.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_en.html)

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src/main/resources/templates/welcome_to_paave_kis_acc_ko.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_ko.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_ko.html)])

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src/main/resources/templates/welcome_to_paave_kis_acc_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_vi.html)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_vi.html)])

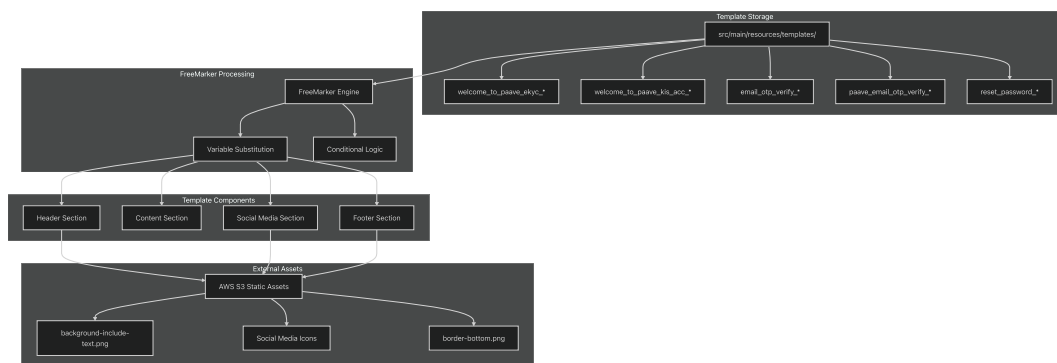
Purpose and Scope

The Template System provides a comprehensive FreeMarker-based templating framework for generating multi-language email notifications across various business contexts. This system manages HTML email templates for user onboarding, authentication, warnings, and account management scenarios, supporting English, Vietnamese, and Korean languages.

For information about email delivery mechanisms, see [Email Notification System](#). For template-specific content by category, see [Template Structure and Variables](#) and [Multi-Language Support](#).

Template Architecture

The template system is built on FreeMarker templating engine and organizes templates by functionality and language, with standardized variable substitution and asset integration.



Sources: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L1-L188).

src/main/resources/templates/welcome_to_paave_ekyc_en.html#L1-L188 (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L1-L188)

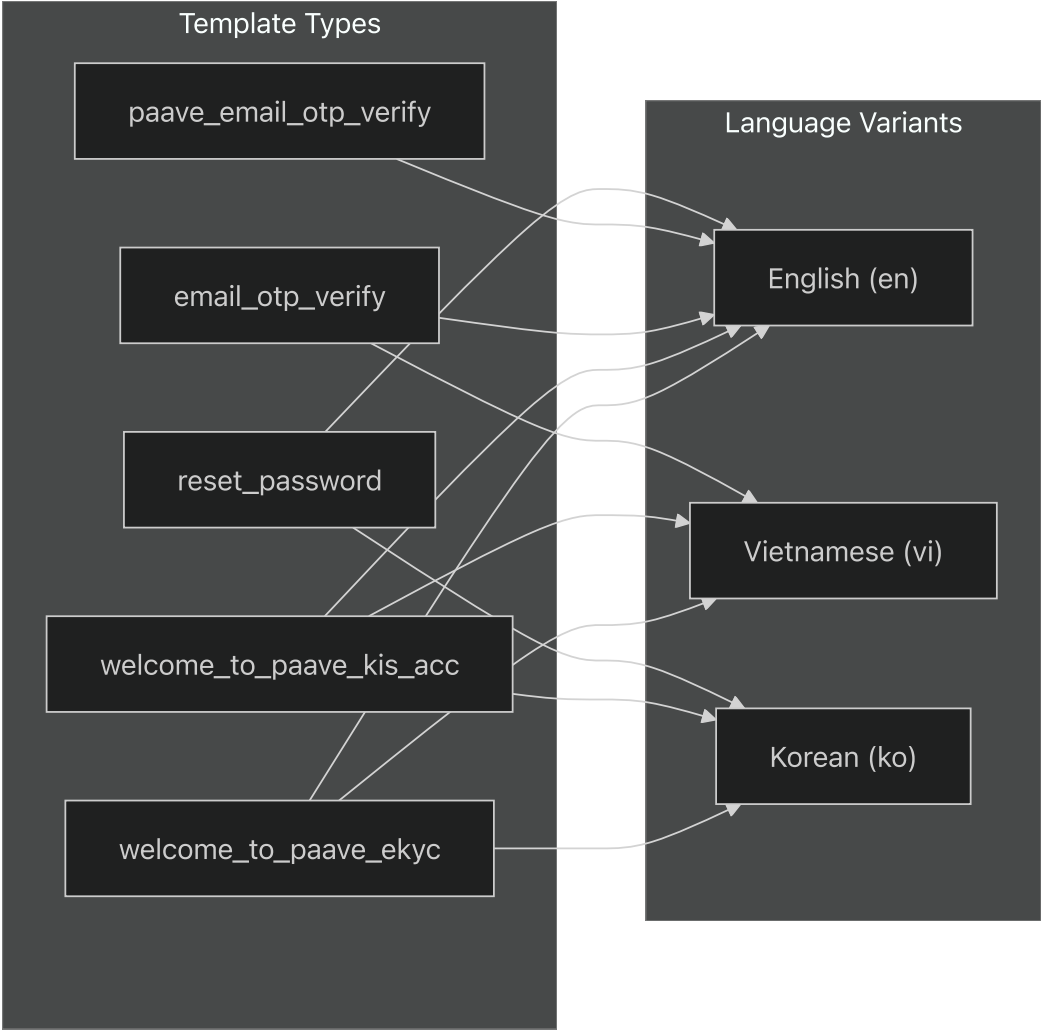
src/main/resources/templates/email_otp_verify_vi.html#L1-L193](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L1-L193)
([%5B">https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L1-L193](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L1-L193)%)5B).

src/main/resources/templates/reset_password_en.html#L1-L117](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L1-L117)
(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L1-L117)])

Template Categories and Language Support

The system organizes templates into distinct categories, each supporting multiple languages with standardized naming conventions.

Template Category	Purpose	Language Support	File Pattern
Welcome eKYC	User onboarding for Paave eKYC	en, vi, ko	welcome_to_paave_ekyc_{lang}.html
Welcome KIS Account	KIS account registration welcome	en, vi, ko	welcome_to_paave_kis_acc_{lang}.html
OTP Verification	Email OTP for authentication	en, vi	email_otp_verify_{lang}.html
Paave OTP	Paave-specific OTP verification	en	paave_email_otp_verify_{lang}.html
Password Reset	Password reset workflow	en, ko	reset_password_{lang}.html



Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L1-L188).

[src/main/resources/templates/welcome_to_paave_ekyc_en.html1-188](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L1-L188) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L1-L188)

[src/main/resources/templates/welcome_to_paave_kis_acc_ko.html1-174](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_ko.html#L1-L174) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_ko.html#L1-L174)

([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_kis_acc_ko.html#L1-L174))
src/main/resources/templates/email_otp_verify_vi.html1-193](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L1-L193)
([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L1-L193))

Variable System and FreeMarker Integration

The template system uses FreeMarker syntax for dynamic content generation, supporting both simple variable substitution and conditional logic.

Core Variables

Variable	Usage	Templates	Example
<code>\${username}</code>	User login name	Welcome eKYC	Account credentials display
<code>\${password}</code>	User password	Welcome eKYC	Account credentials display
<code>\${fullname}</code>	User full name	Welcome KIS	Personalized greeting
<code>\${otp}</code>	OTP code	OTP verification	Authentication code
<code>\${action}</code>	Action type	OTP verification	Conditional message text
<code>\${expiredInMinute}</code>	OTP expiration	OTP verification	Time-sensitive information
<code>\${resetPasswordUrl}</code>	Reset URL	Password reset	Action link

FreeMarker Conditional Logic

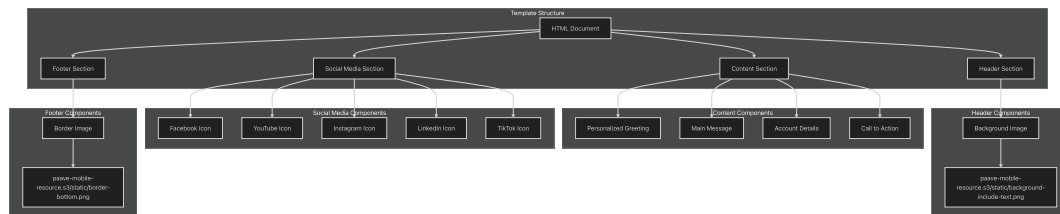
The system implements conditional rendering for action-specific messaging:

```
<#if action == "REGISTER">Dang Ky
<#elseif action == "RESET_PASSWORD">Quen Mat Khau
<#elseif action == "NEW_DEVICE">Dang Nhap
<#elseif action == "VERIFY">Xac Thuc
<#else>${action}
</#if>
```

Sources: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L96-L96)
src/main/resources/templates/email_otp_verify_vi.html96 ([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L96-L96))
src/main/resources/templates/welcome_to_paave_ekyc_en.html54-57](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L54-L57)
([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L54-L57))
src/main/resources/templates/paave_email_otp_verify_en.html44](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L44-L44)
([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L44-L44))

Template Structure and Components

All templates follow a consistent four-section structure with standardized styling and responsive design.



Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L14-L21

src/main/resources/templates/welcome_to_paave_ekyc_en.html#L14-L21 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L14-L21

src/main/resources/templates/welcome_to_paave_ekyc_en.html#L115-L172 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L115-L172 [%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L115-L172)

src/main/resources/templates/welcome_to_paave_ekyc_en.html#L177-L184 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L177-L184

Asset Management and External Resources

The template system integrates with AWS S3 for static asset delivery, ensuring consistent branding and performance across all email communications.

AWS S3 Asset Integration

Asset Type	S3 Path	Usage	Dimensions
Header Background	<code>background-include-text.png</code>	Email header branding	587px × 231px
Social Media Icons	<code>facebook-icon.png</code>	Social media links	30px × 30px
Social Media Icons	<code>youtube-icon.png</code>	Social media links	30px × 30px
Social Media Icons	<code>instagram-icon.png</code>	Social media links	30px × 30px
Social Media Icons	<code>linkedin-icon.png</code>	Social media links	30px × 30px
Social Media Icons	<code>tiktok-icon.png</code>	Social media links	30px × 30px
Footer Border	<code>border-bottom.png</code>	Email footer styling	587px width

Asset URL Pattern

All assets follow the pattern: `https://paave-mobile-resource.s3.ap-southeast-1.amazonaws.com/static/{asset-name}`

Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L17-L18

src/main/resources/templates/welcome_to_paave_ekyc_en.html#L17-L18 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L17-L18

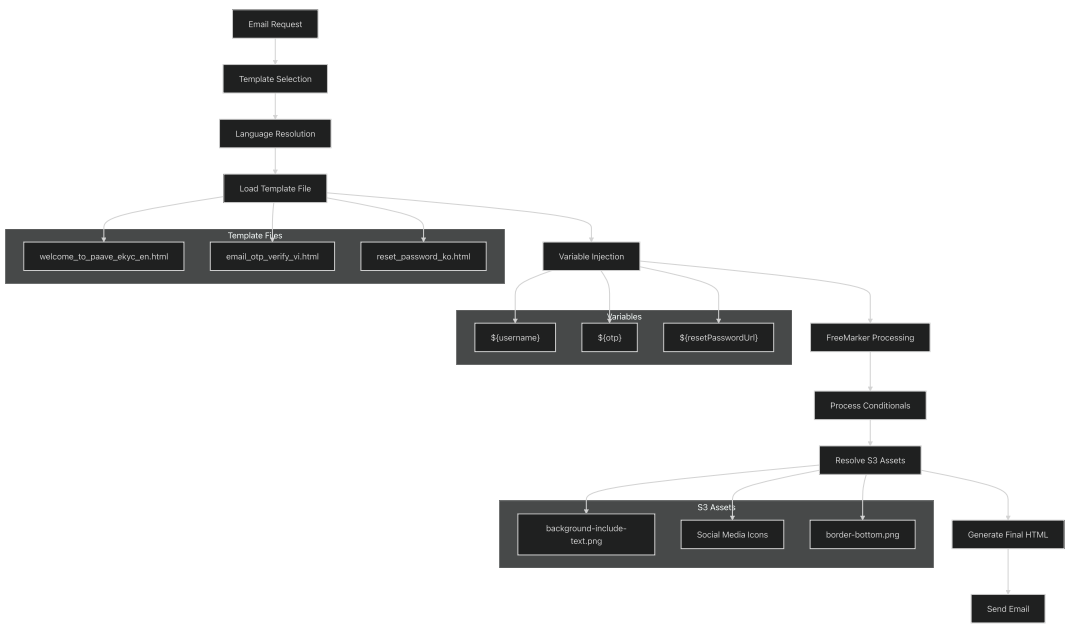
src/main/resources/templates/welcome_to_paave_ekyc_en.html#L134-L164 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L134-L164

([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L134-L164)).

src/main/resources/templates/welcome_to_paave_ekyc_en.html180-181](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L180-L181)(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/welcome_to_paave_ekyc_en.html#L180-L181))

Template Processing Flow

The template system processes email generation through a standardized workflow that handles language selection, variable injection, and asset resolution.



Sources: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L96-L96)

src/main/resources/templates/email_otp_verify_vi.html#L96(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L96-L96)[

src/main/resources/templates/paave_email_otp_verify_en.html42-67](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L42-L67)([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L42-L67)).

src/main/resources/templates/reset_password_en.html54-64](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L54-L64)(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L54-L64)

Template Structure and Variables

Relevant source files

- [
src/main/resources/templates/email_otp_verify_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html)(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html))
- [
src/main/resources/templates/email_otp_verify_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html)(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html))
- [

src/main/resources/templates/email_verify_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_en.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_en.html))

- [

src/main/resources/templates/email_verify_vi.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_vi.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_vi.html))

- [

src/main/resources/templates/paave_email_otp_verify_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html))

- [

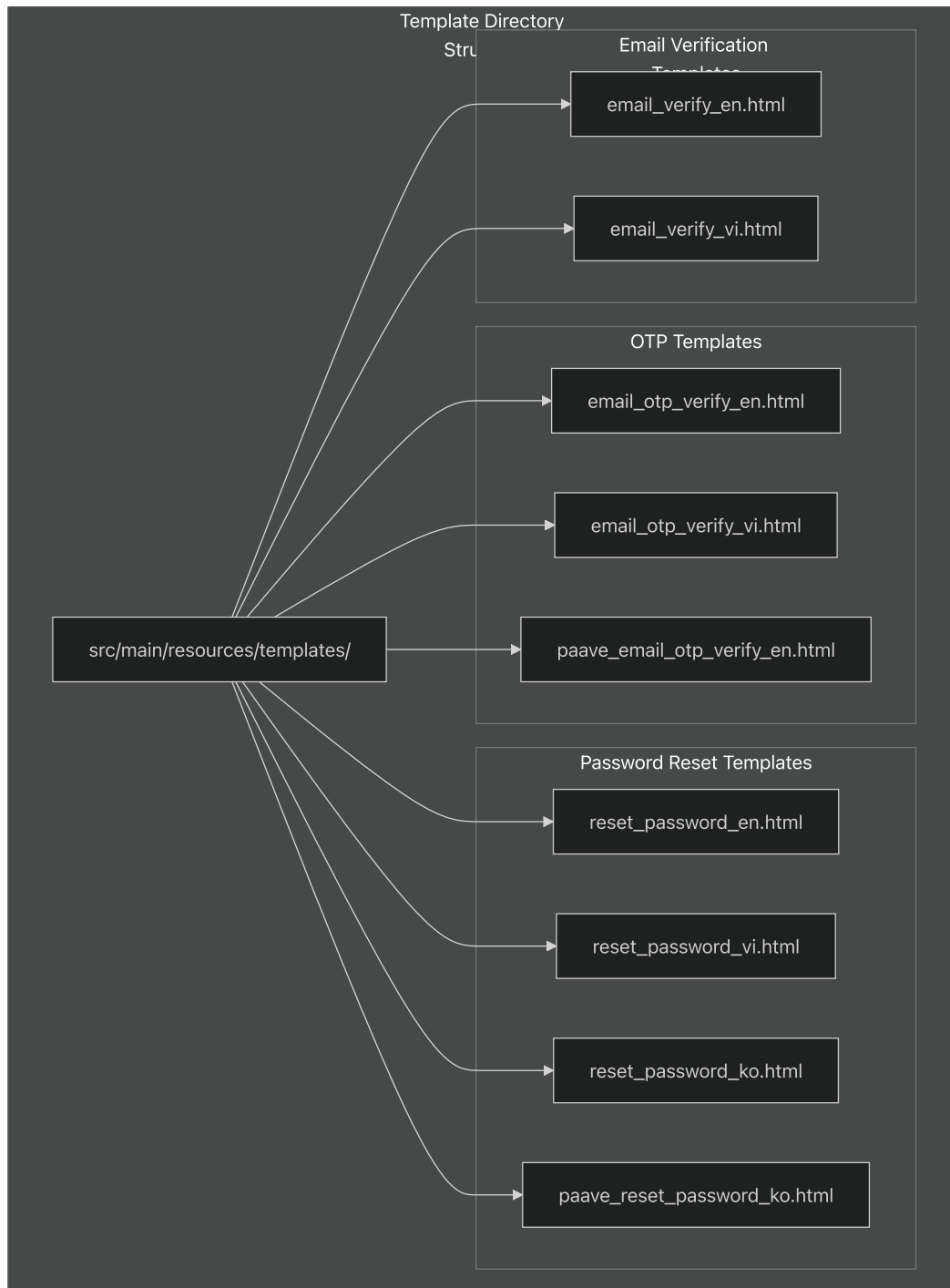
src/main/resources/templates/paave_reset_password_ko.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_ko.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_ko.html))

This document covers the structure and variable system used in email templates within the notification service. The template system uses FreeMarker as the template engine to generate dynamic HTML email content with multi-language support and brand-specific customizations.

For information about multi-language template selection, see [Multi-Language Support](#). For email delivery mechanisms, see [Email Notification System](#).

Template File Organization

The email templates are organized in the `src/main/resources/templates/` directory with a consistent naming convention that indicates the template purpose, brand, and language.



Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html#L1-L195

[src/main/resources/templates/email_otp_verify_en.html1-195](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html#L1-L195) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html#L1-L195

[src/main/resources/templates/email_otp_verify_vi.html1-193](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L1-L193) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L1-L193 [%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L1-L193)

[src/main/resources/templates/paave_email_otp_verify_en.html1-154](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L1-L154) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L1-L154 [%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L1-L154)

[src/main/resources/templates/paave_reset_password_ko.html1-115](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_ko.html#L1-L115) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_ko.html#L1-L115

([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_ko.html#L1-L115))

src/main/resources/templates/email_verify_en.html#L1-L114](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_en.html#L1-L114)([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_en.html#L1-L114))

src/main/resources/templates/email_verify_vi.html#L1-L116](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_vi.html#L1-L116)([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_vi.html#L1-L116))

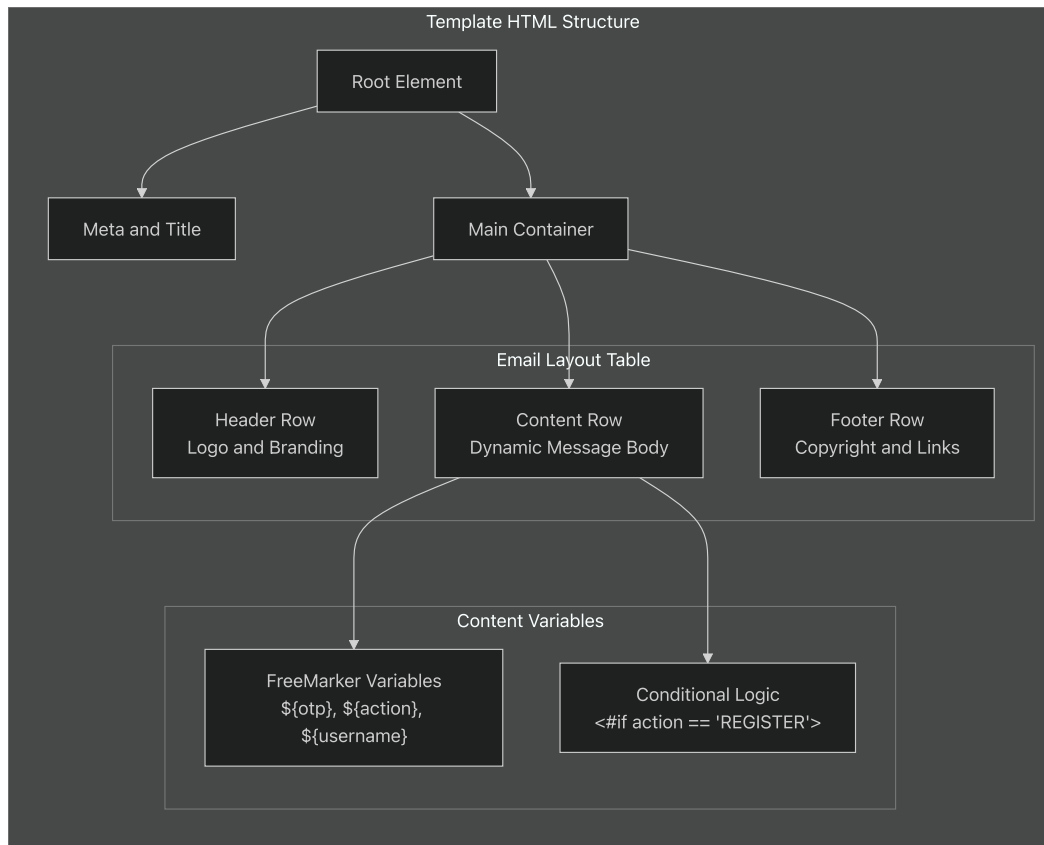
Common Template Structure

All email templates follow a consistent HTML structure with responsive design elements and brand-specific styling.

Standard Template Components

Component	Purpose	Example Location
Header Section	Logo and branding	Header <code><tr></code> with logo image
Main Content	Dynamic message content with variables	Content <code><div></code> with FreeMarker variables
Footer Section	Copyright and unsubscribe links	Footer <code><tr></code> with brand information
Social Media	Links to social platforms (Paave only)	Social media <code><div></code> with platform icons

Basic HTML Structure Pattern



Sources: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L1-L25)

[src/main/resources/templates/paave_email_otp_verify_en.html#L1-L25](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L1-L25) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L1-L25)

[src/main/resources/templates/email_verify_en.html#L10-L30](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_en.html#L10-L30) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_en.html#L10-L30)

[src/main/resources/templates/paave_reset_password_ko.html#L10-L35](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_ko.html#L10-L35) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_ko.html#L10-L35)

FreeMarker Variable System

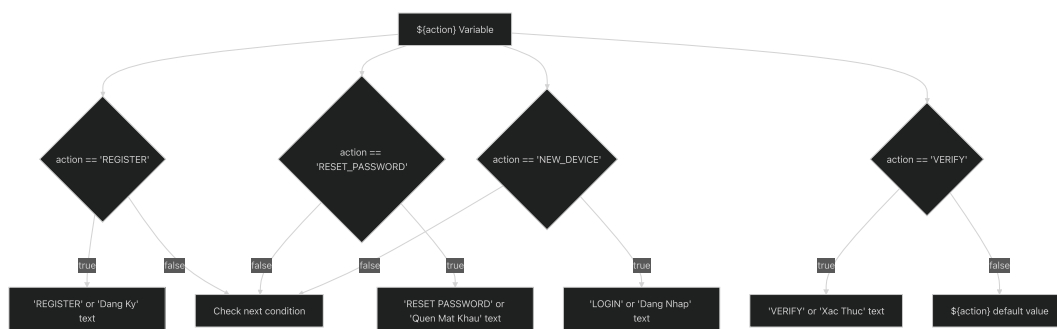
The template system uses FreeMarker template engine with a standardized set of variables that are populated by the notification service.

Core Variable Types

Variable Category	Variables	Purpose	Example Usage
Authentication	<code>\${otp}</code> , <code>\${activationCode}</code>	Security codes for verification	<code>\${otp}</code> in OTP emails
User Information	<code>\${username}</code> , <code>\${fullName}</code>	Personalization data	Greeting messages
Action Context	<code>\${action}</code>	Determines email purpose	Conditional message content
Time/Expiration	<code>\${expiredInMinute}</code> , <code>\${expirationTime}</code>	Validity periods	Code expiration warnings
URLs	<code>\${resetPasswordUrl}</code> , <code>\${baseUrl}</code>	Action links	Password reset buttons

FreeMarker Conditional Logic

The templates use FreeMarker's conditional statements to display different content based on the action type:



Sources: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L96-L97)

[src/main/resources/templates/email_otp_verify_vi.html#L96-L97](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L96-L97) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_vi.html#L96-L97)

[src/main/resources/templates/email_otp_verify_en.html#L97-L98](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html#L97-L98) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html#L97-L98)

Variable	Usage	Location Example
<code>\${otp}</code>	Display the OTP code prominently	Main content area with large font
<code>\${expiredInMinute}</code>	Show validity period	Warning text about expiration
<code>\${action}</code>	Contextualize the OTP purpose	Subject line and body text

Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html#L97-L97

src/main/resources/templates/email_otp_verify_en.html#L97 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html#L97-L97[(

https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html#L105-L105](
([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_otp_verify_en.html#L105-L105))

src/main/resources/templates/paave_email_otp_verify_en.html#L44](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L44-L44
([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_email_otp_verify_en.html#L44-L44)))

Password Reset Templates

Password reset templates use URL variables for action buttons:

Variable	Usage	Location Example
<code>\${resetPasswordUrl}</code>	Generate reset action button	Primary CTA button href
<code>\${resetPasswordUrl}</code>	Fallback text link	Sub-copy section for accessibility

Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_ko.html#L53-L53

src/main/resources/templates/paave_reset_password_ko.html#L53 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_ko.html#L53-L53[(

https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_ko.html#L63-L63](
([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_ko.html#L63-L63))

src/main/resources/templates/paave_reset_password_ko.html#L82](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_ko.html#L82-L82
([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/paave_reset_password_ko.html#L82-L82)))

Email Verification Templates

Email verification templates combine codes and URLs for account activation:

Variable	Usage	Location Example
<code>\${activationCode}</code>	Display verification code	Button text and fallback display
<code>\${username}</code>	Construct activation URL	URL path parameter
<code>\${baseUrl}</code>	Base URL for activation link	URL construction
<code>\${expirationTime}</code>	Show code validity	Footer warning text

Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_en.html#L59-L66

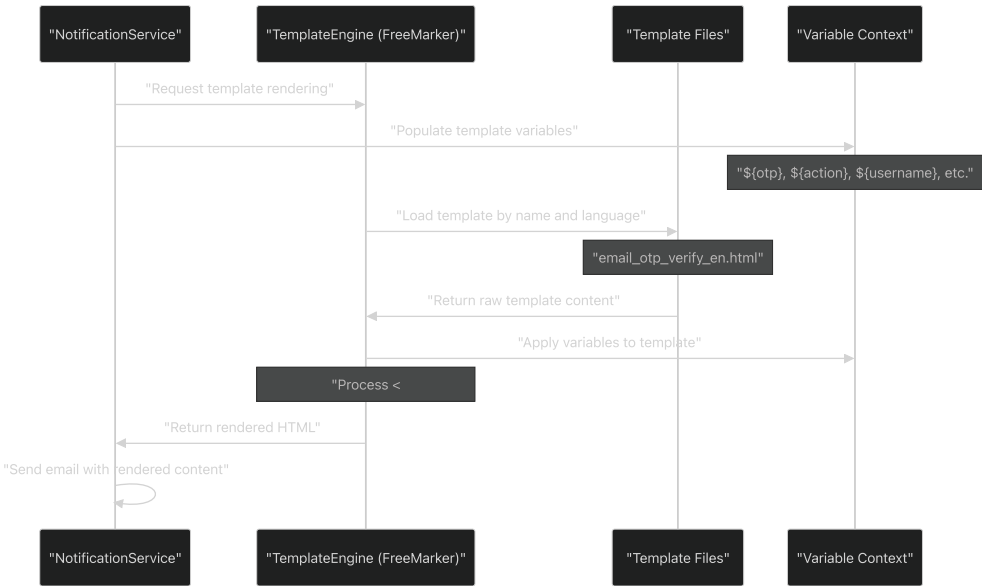
src/main/resources/templates/email_verify_en.html#L59-L66 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_en.html#L59-L66[(

[src/main/resources/templates/email_verify_en.html#L80-L81\)](src/main/resources/templates/email_verify_en.html#L80-L81) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_en.html#L80-L81[(

[src/main/resources/templates/email_verify_vi.html#L59-L66\)](src/main/resources/templates/email_verify_vi.html#L59-L66) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_vi.html#L59-L66[(

[src/main/resources/templates/email_verify_vi.html#L80-L82\)](src/main/resources/templates/email_verify_vi.html#L80-L82) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_vi.html#L80-L82[(

Template Rendering Process



Sources: Based on template variable patterns observed across all template files

Multi-Language Support

Relevant source files

- [src/main/resources/templates/email_reset_password_ko.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_ko.html)

(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_ko.html)

- [
src/main/resources/templates/email_verify_ko.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html)
- [
src/main/resources/templates/reset_password_en.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html)
- [
src/main/resources/templates/reset_password_ko.html](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html)

Purpose and Scope

This document covers the multi-language internationalization approach implemented in the notification service template system. The system supports English, Vietnamese, and Korean languages across all email notification templates, enabling localized communication for different regional markets. For information about template structure and dynamic variables, see [Template Structure and Variables](#).

Supported Languages

The notification service provides comprehensive multi-language support for three primary markets:

Language	Code	Template Suffix	Primary Market
English	en	_en.html	International
Vietnamese	vi	_vi.html	Vietnam
Korean	ko	_ko.html	South Korea

Sources: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L1-L117)

[src/main/resources/templates/reset_password_en.html#L1-L117](#) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L1-L117)

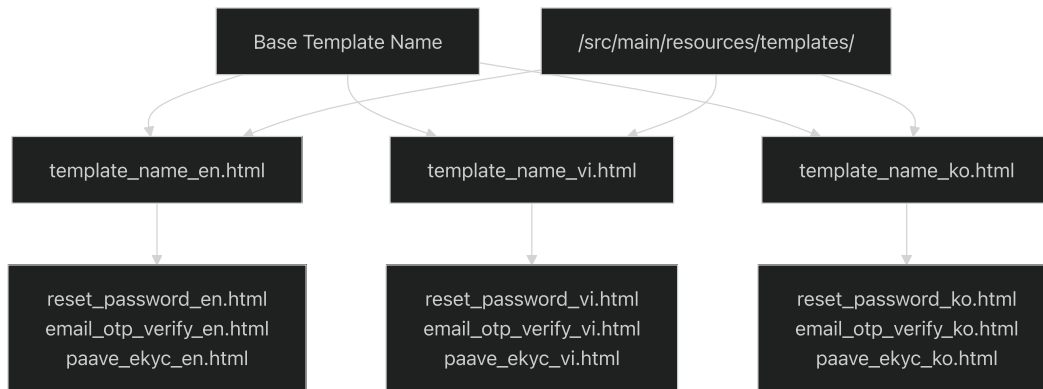
[src/main/resources/templates/reset_password_ko.html#L1-L115](#) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L1-L115) ([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L1-L115))

[src/main/resources/templates/email_reset_password_ko.html#L1-L110](#) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_ko.html#L1-L110) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_ko.html#L1-L110)

Template Naming Convention

Language-Specific Template Files

The multi-language system uses a consistent file naming pattern where each template has language-specific variants:



Template Categories with Language Support

Different template categories support varying language combinations based on business requirements:



Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L1-L7

[src/main/resources/templates/reset_password_en.html#L1-L7](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L1-L7) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L1-L7

[src/main/resources/templates/reset_password_ko.html#L1-L7](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L1-L7) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L1-L7 [%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L1-L7)

[src/main/resources/templates/email_reset_password_ko.html#L1-L7](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_ko.html#L1-L7) https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_ko.html#L1-L7 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_ko.html#L1-L7

Variable System and Localization

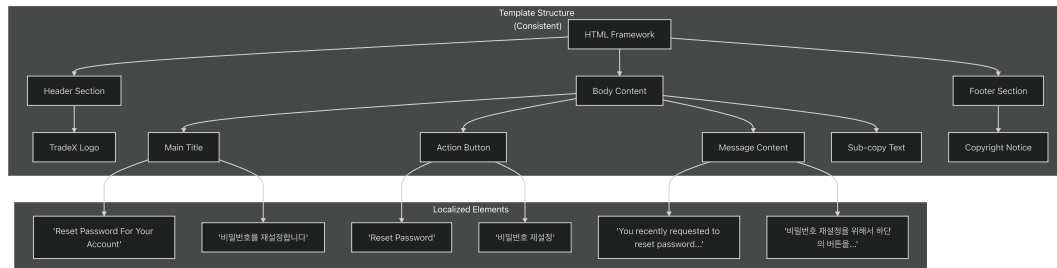
Language-Agnostic Variables

All template variables remain consistent across languages, ensuring the FreeMarker template engine can populate content regardless of the selected language:

Variable Category	Common Variables	Usage
Authentication	<code>\${resetPasswordUrl}</code> , <code>\${resetCode}</code> , <code>\${otp}</code>	Password reset and verification
User Data	<code>\${username}</code> , <code>\${fullName}</code> , <code>\${email}</code>	Personalization
System URLs	<code>\${baseUrl}</code> , <code>\${confirmLink}</code> , <code>\${activationCode}</code>	Navigation and activation
Timestamps	<code>\${expirationTime}</code> , <code>\${time}</code>	Time-sensitive operations

Localized Content Structure

Templates maintain identical HTML structure while localizing text content:



Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L35-L36

src/main/resources/templates/reset_password_en.html#L35-L36 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L35-L36

src/main/resources/templates/reset_password_ko.html#L35-L36 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L35-L36 [%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L35-L36)

src/main/resources/templates/reset_password_en.html#L63-L64 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L63-L64 [%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L63-L64)

src/main/resources/templates/reset_password_ko.html#L63-L63 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L63-L63 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L63-L63

Language-Specific Content Examples

Password Reset Templates

The password reset functionality demonstrates comprehensive localization:

English Version https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L35-L42

src/main/resources/templates/reset_password_en.html#L35-L42 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L35-L42

```
<h1>Reset Password For Your Account</h1>
<p>You recently requested to <b>reset password</b> for your <b>TradeX</b> account.
Please click this below button and follow our instructions to reset your password.</p>
```

Korean Version ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L35-L41)

[tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L35-L41](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L35-L41))

[src/main/resources/templates/reset_password_ko.html#L35-L41](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L35-L41) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L35-L41)
[tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L35-L41](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L35-L41))):

```
<h1>비밀번호를 재설정합니다.</h1>
<p>비밀번호 재설정을 <b>위해서</b> 하단의 버튼을 <b>클릭한 후</b>
설명에 따라 진행해 주실 것을 부탁드립니다.</p>
```

Email Verification Templates

Korean email verification maintains cultural appropriateness:

Korean Email Verification ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L35-L38)

[tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L35-L38](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L35-L38))

[src/main/resources/templates/email_verify_ko.html#L35-L38](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L35-L38) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L35-L38)
[tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L35-L38](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L35-L38))):

```
<h1>이메일 주소를 확인합니다.</h1>
<p>이레이즈리스크에 가입해 주셔서 감사합니다!</p>
```

Sources: (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L35-L42).

[src/main/resources/templates/reset_password_en.html#L35-L42](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L35-L42) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L35-L42)
[tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L35-L42](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L35-L42))

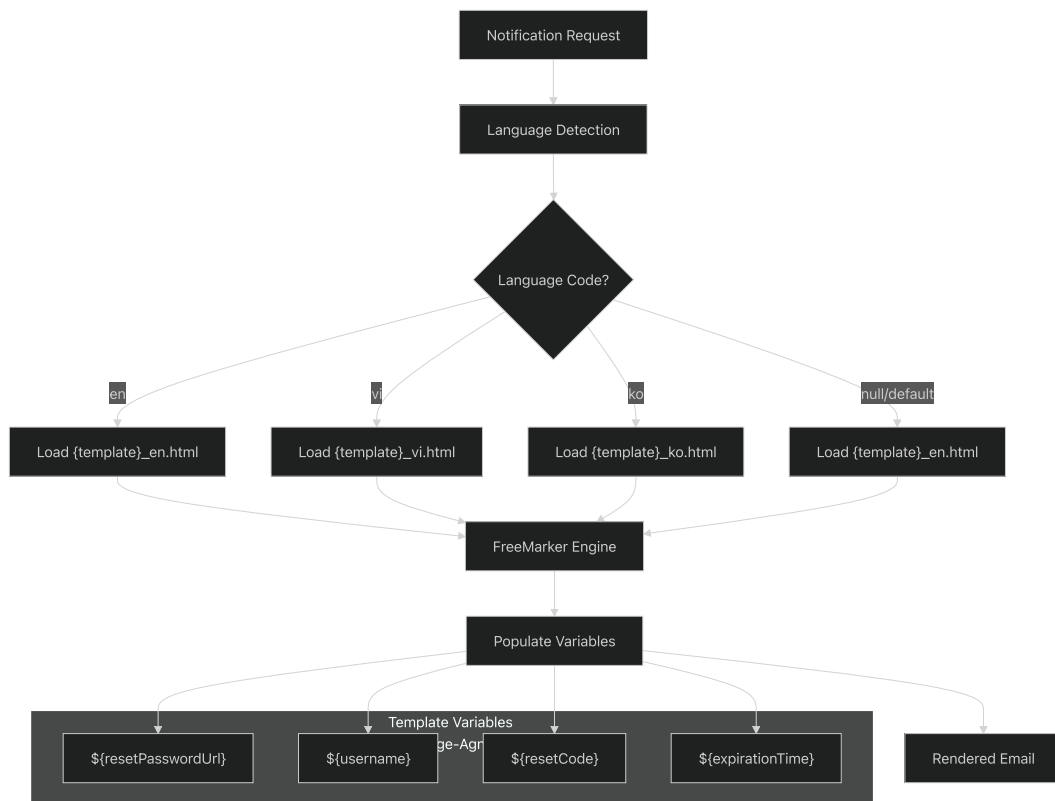
[src/main/resources/templates/reset_password_ko.html#L35-L41](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L35-L41) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L35-L41)
[tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L35-L41](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L35-L41)) [[%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L35-L41)]

[src/main/resources/templates/email_verify_ko.html#L35-L38](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L35-L38) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L35-L38)
[tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L35-L38](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L35-L38) (https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L35-L38))

Template Resolution Process

Language Selection Logic

The system resolves templates using a language-aware selection mechanism:



Template Path Resolution

Templates are resolved from the classpath using the pattern:

- Base path: `/src/main/resources/templates/`
- Template pattern: `{template_name}_{language_code}.html`
- Fallback: `{template_name}_en.html` for unsupported languages

Sources: https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L1-L117

src/main/resources/templates/reset_password_en.html#L1-L117 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L1-L117

src/main/resources/templates/email_reset_password_ko.html#L1-L110 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_ko.html#L1-L110 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_ko.html#L1-L110%5B

src/main/resources/templates/reset_password_ko.html#L1-L115 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L1-L115 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L1-L115%5B

src/main/resources/templates/email_verify_ko.html#L1-L112 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L1-L112 https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L1-L112%5B

Character Encoding and Fonts

UTF-8 Support

All templates specify UTF-8 character encoding to support international characters:

```
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8"/>
```

Korean templates properly display Hangul characters, Vietnamese templates support diacritical marks, and the system maintains font compatibility across all supported languages using the Arial font family.

Sources: [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L6-L6\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L6-L6)

src/main/resources/templates/reset_password_en.html#L6-L6 [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L6-L6\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L6-L6) [\[](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_en.html#L6-L6)

src/main/resources/templates/reset_password_ko.html#L6-L6 [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L6-L6\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L6-L6) [\[](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L6-L6) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L6-L6\)%5B\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/reset_password_ko.html#L6-L6)

src/main/resources/templates/email_reset_password_ko.html#L6-L6 [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_ko.html#L6-L6\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_ko.html#L6-L6) [\[](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_ko.html#L6-L6) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_ko.html#L6-L6\)%5B\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_reset_password_ko.html#L6-L6)

src/main/resources/templates/email_verify_ko.html#L6-L6 [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L6-L6\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L6-L6) [\[](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L6-L6) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L6-L6\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/templates/email_verify_ko.html#L6-L6)

Configuration and Deployment

Relevant source files

- [\[](#) [README.md](#) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md\)\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md)
- [\[](#) [src/main/java/com/techx/tradex/notification/configurations/AppConf.java](#) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java\)\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java)
- [\[](#) [src/main/resources/application.yaml](#) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml\)\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml)

Purpose and Scope

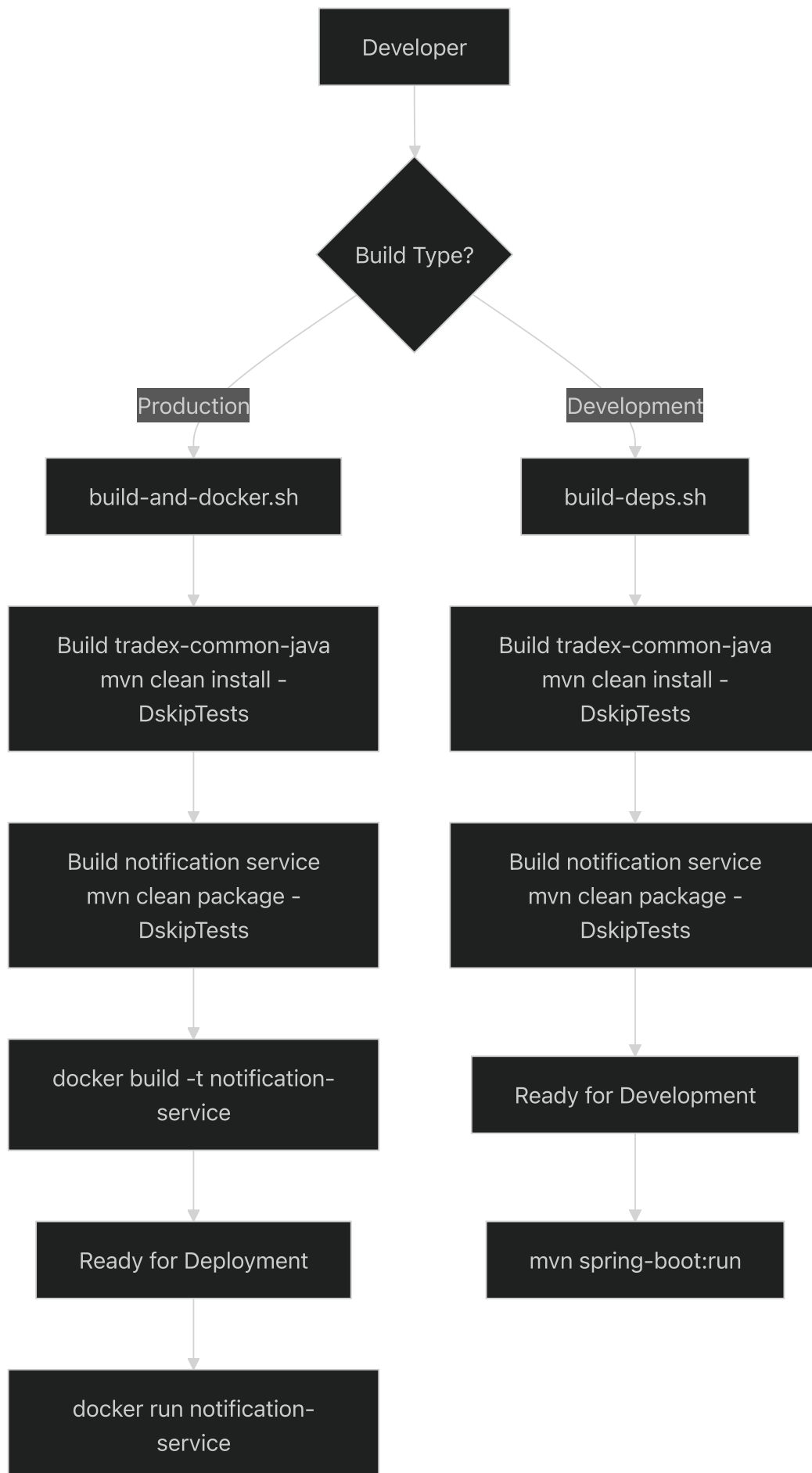
This document covers the configuration management, build processes, and deployment procedures for the notification service. It explains how to set up development and production environments, configure external service integrations, and deploy the service using various methods including Docker containers and local development setups.

For information about specific external service integrations, see [Push Notification System](#), [Messaging Platform Integrations](#), or [SMS Notification System](#).

Build Process and Dependencies

The notification service requires building the `tradex-common-java` dependency before compiling the main application. The build process is automated through shell scripts that handle dependency management and Docker image creation.

Build Scripts Architecture



Sources: [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L118-L148\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L118-L148)

[README.md118-148](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L118-L148) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L118-L148\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L118-L148)

[README.md69-76](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L69-L76) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L69-L76\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L69-L76)
[%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L69-L76)
[%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L69-L76)

[README.md94-107](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L94-L107) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L94-L107\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L94-L107)
[_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L94-L107\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L94-L107)

Manual Build Process

For manual builds without the automated scripts:

Step	Command	Purpose
1	<code>cd ../tradex-common-java && mvn clean install -DskipTests</code>	Build shared dependency library
2	<code>cd ../notification && mvn clean package -DskipTests</code>	Compile notification service
3	<code>docker build -t notification-service .</code>	Create container image (optional)

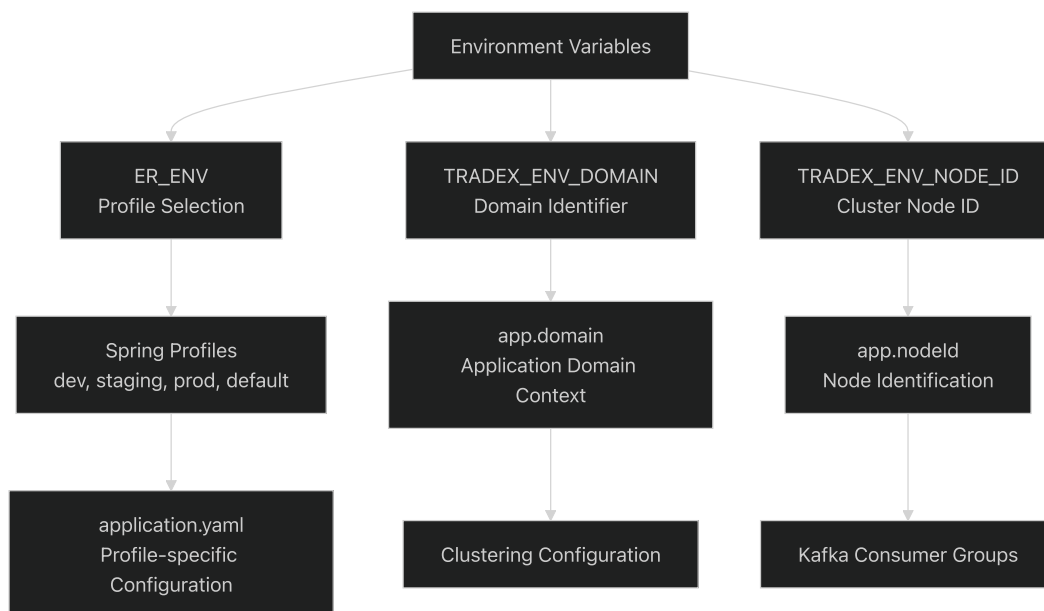
Sources: [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L134-L148\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L134-L148)

[README.md134-148](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L134-L148) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L134-L148\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L134-L148)

Environment Configuration

The application uses Spring profiles and environment variables to manage configuration across different deployment environments.

Environment Variables



Sources: [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L174-L178\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L174-L178)

[README.md174-178](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L174-L178) [_\(https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L174-L178\)](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L174-L178)

src/main/resources/application.yaml4-5](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L4-L5>)([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L4-L5)).

src/main/resources/application.yaml31-33](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L31-L33>)(<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L31-L33>))

Core Environment Setup

Variable	Default	Description
ER_ENV	default	Spring profile activation
TRADEX_ENV_DOMAIN	tradex	Domain context for multi-tenant support
TRADEX_ENV_NODE_ID	1	Node identifier for Kafka consumer groups

Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L79-L83>)

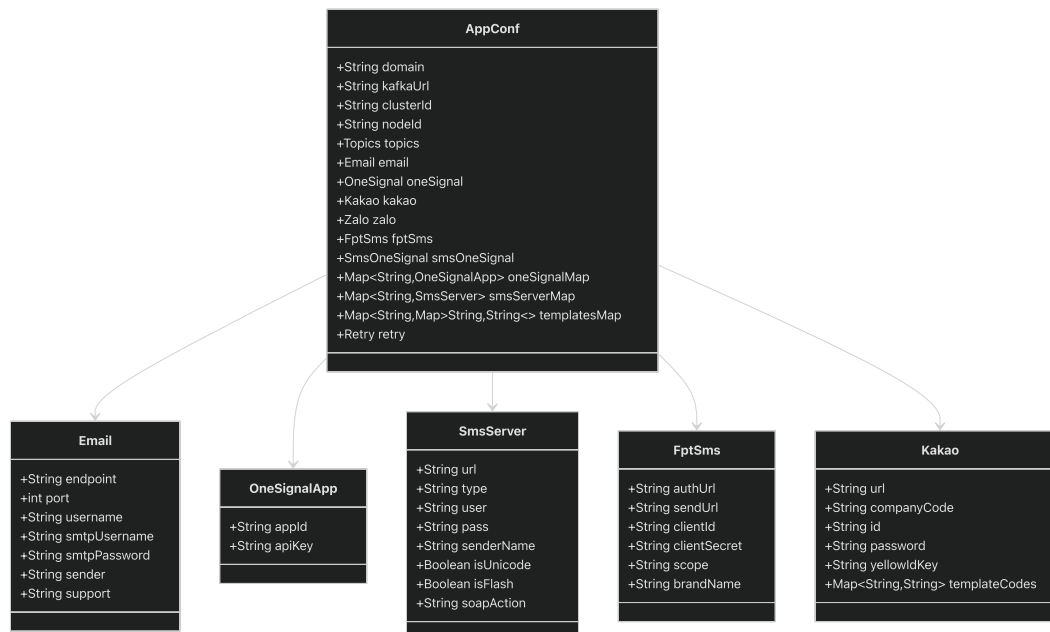
[README.md79-83](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L79-L83) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L79-L83>)

src/main/resources/application.yaml31-33](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L31-L33>)(<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L31-L33>))

Application Configuration Structure

The configuration system uses a hierarchical structure mapped through the `AppConf` class, which centralizes all external service configurations and application settings.

Configuration Class Hierarchy



Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L10-L138>)

[src/main/java/com/techx/tradex/notification/configurations/AppConf.java10-138](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L10-L138) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L10-L138>)

Database and Infrastructure Configuration

Component	Configuration Location	Key Properties
MySQL Database	<code>spring.datasource</code>	URL, username, password, driver
Kafka	<code>app.kafkaUrl</code>	Bootstrap servers, topic configuration
Logging	<code>logging</code>	File output, log levels per package
JPA/Hibernate	<code>spring.jpa</code>	DDL auto-generation settings

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L14-L18>.

[src/main/resources/application.yaml#L14-L18](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L14-L18) [\[https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L14-L18\]](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L14-L18)

[src/main/resources/application.yaml#L34](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L34-L34) [\[https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L34-L34\]](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L34-L34) [\[https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L34-L34\)%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L34-L34)

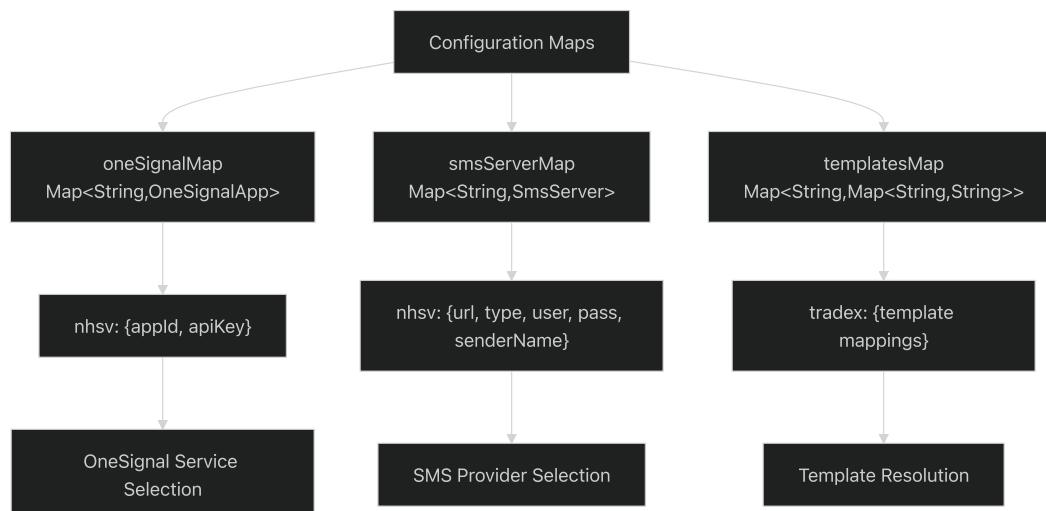
[src/main/resources/application.yaml#L20-28](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L20-L28) [\[https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L20-L28\]](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L20-L28) [\[https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L20-L28\)%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L20-L28)

[src/main/resources/application.yaml#L11-13](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L11-L13) [\https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L11-L13 [\[https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L11-L13\)\]](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L11-L13)

External Service Configuration

The system integrates with multiple external services, each requiring specific configuration parameters managed through the `AppConf` class structure.

Multi-Provider Configuration Pattern



Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L19-L21>.

[src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L19-21](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L19-L21) [\[https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L19-L21\]](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L19-L21)

src/main/resources/application.yaml60-75](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L60-L75> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L60-L75>))

Service Configuration Examples

Email Configuration (AWS SES)

```
app:
  email:
    endpoint: email-smtp.us-west-2.amazonaws.com
    port: 587
    username: ses-smtp-user.tradex_support
    sender: noreply@tradex.vn
    support: admin@tradex.vn
```

SMS Server Configuration (OneSMS)

```
app:
  smsServerMap:
    nhsv:
      url: https://gbapi.onesms.vn/wspartners/service.asmx
      type: ONE_SMS
      user: nhsv
      senderName: NHSV
      soapAction: http://lsms.vn/SendMT
```

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L41-L48>.

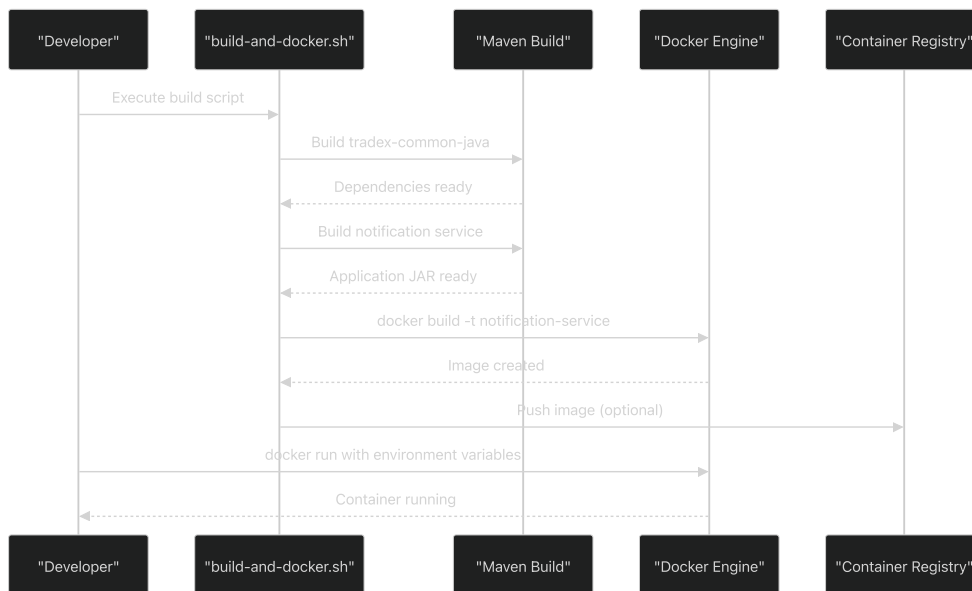
[src/main/resources/application.yaml41-48](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L41-L48) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L41-L48>)

src/main/resources/application.yaml65-75](<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L65-L75> (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L65-L75>))

Deployment Methods

The notification service supports multiple deployment approaches to accommodate different environments and operational requirements.

Docker Deployment



Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L94-L116>

[README.md94-116](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L94-L116) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L94-L116>)

[README.md127-132](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L127-L132) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L127-L132>)

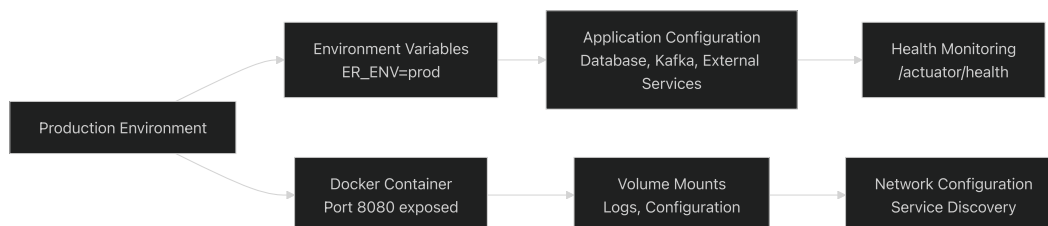
Local Development Setup

Step	Command	Environment Variables
1. Clone repository	<code>git clone <repo-url></code>	-
2. Build dependencies	<code>./build-deps.sh</code>	-
3. Set environment	<code>export ER_ENV=dev</code>	<code>ER_ENV</code> , <code>TRADEX_ENV_DOMAIN</code> , <code>TRADEX_ENV_NODE_ID</code>
4. Configure database	Edit <code>application.yaml</code>	MySQL credentials
5. Run application	<code>mvn spring-boot:run</code>	All environment variables

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L62-L91>

[README.md62-91](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L62-L91) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L62-L91>)

Production Deployment Configuration



Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L109-L116>

[README.md109-116](https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L109-L116) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L109-L116>)

README.md217-222](<https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L217-L222>
(<https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L217-L222>))

Configuration Management

Application Properties Structure

The configuration follows a hierarchical structure with clear separation of concerns for different service types and environments.

Configuration Section	Purpose	Key Classes
<code>app.email</code>	SMTP and email provider settings	<code>AppConf.Email</code>
<code>app.oneSignal</code>	Push notification configuration	<code>AppConf.OneSignal</code>
<code>app.kakao</code>	KakaoTalk messaging settings	<code>AppConf.Kakao</code>
<code>app.zalo</code>	Zalo messaging configuration	<code>AppConf.Zalo</code>
<code>app.fptSms</code>	FPT SMS gateway settings	<code>AppConf.FptSms</code>
<code>app.retry</code>	Retry mechanism configuration	<code>AppConf.Retry</code>

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L13-L32>
[src/main/java/com/techx/tradex/notification/configurations/AppConf.java13-32](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L13-L32) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L13-L32>)

Configuration Validation and Defaults

The system provides sensible defaults and validation through Spring Boot's configuration property binding:

```
app:
  retry:
    maxAttempts: 5
    maxDelay: 1000
  socketCluster:
    hostname: localhost
    port: 8000
    autoReconnection: true
    usingKafka: false
```

Sources: <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L80-L83>
[src/main/resources/application.yaml80-83](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L80-L83) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L80-L83>)
[src/main/java/com/techx/tradex/notification/configurations/AppConf.java122-133](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L122-L133) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L122-L133>)
(<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/configurations/AppConf.java#L122-L133>)

Health Monitoring and Operational Configuration

The service includes built-in health monitoring and logging configuration for operational observability:

Monitoring Aspect	Configuration	Default Value
Health endpoint	<code>/actuator/health</code>	Enabled
Log file	<code>logging.file</code>	<code>application.log</code>
Root log level	<code>logging.level.root</code>	WARN
Application log level	<code>logging.level.com.techx.tradex</code>	INFO

Sources: (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L20-L28>).

[src/main/resources/application.yaml#L20-L28](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L20-L28) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/resources/application.yaml#L20-L28>)

README.md#L219-L227 (<https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L219-L227>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/README.md#L219-L227>)

Data Models and APIs

Relevant source files

- [[src/main/java/com/techx/tradex/notification/model/EmailAddRes.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/EmailAddRes.java)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/EmailAddRes.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/EmailAddRes.java>)
- [[src/main/java/com/techx/tradex/notification/model/NotificationDataRequest.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/NotificationDataRequest.java)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/NotificationDataRequest.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/NotificationDataRequest.java>)
- [[src/main/java/com/techx/tradex/notification/model/PhoneNumberAddReq.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddReq.java)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddReq.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddReq.java>)
- [[src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java>)
- [[src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java)] (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java>) (<https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java>)

SMS Message Models

The `SmsMessageRequest` class provides the basic structure for SMS notifications:

Field	Type	Purpose
<code>phoneNumber</code>	<code>String</code>	Target phone number for SMS delivery
<code>content</code>	<code>String</code>	Message content to be sent

This model is used by SMS providers throughout the system for direct message delivery.

Registration Models

For external service registration, the system uses specialized request and response models:

- `PhoneNumberAddReq` extends `DataRequest` and contains phone number data for OneSignal SMS registration
- `PhoneNumberAddRes` provides response data with a static factory method `fromNotification(String phoneNumber)`
- `EmailAddRes` follows the same pattern for email registration responses

Sources: ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java#L6-L9)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java#L6-L9](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java#L6-L9)).

[src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java6-9](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java#L6-L9) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java#L6-L9)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java#L6-L9](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java#L6-L9)) [

[src/main/java/com/techx/tradex/notification/model/PhoneNumberAddReq.java8-10](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddReq.java#L8-L10)] ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddReq.java#L8-L10)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddReq.java#L8-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddReq.java#L8-L10)

[L10](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddReq.java#L8-L10)) [([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddReq.java#L8-L10)

[%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddReq.java#L8-L10))

[src/main/java/com/techx/tradex/notification/model/EmailAddRes.java6-14](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/EmailAddRes.java#L6-L14)] ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/EmailAddRes.java#L6-L14)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/EmailAddRes.java#L6-L14](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/EmailAddRes.java#L6-L14)) [

([%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/EmailAddRes.java#L6-L14)

[%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/EmailAddRes.java#L6-L14)).

[src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java6-14](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java#L6-L14)] ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java#L6-L14)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java#L6-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java#L6-L14)

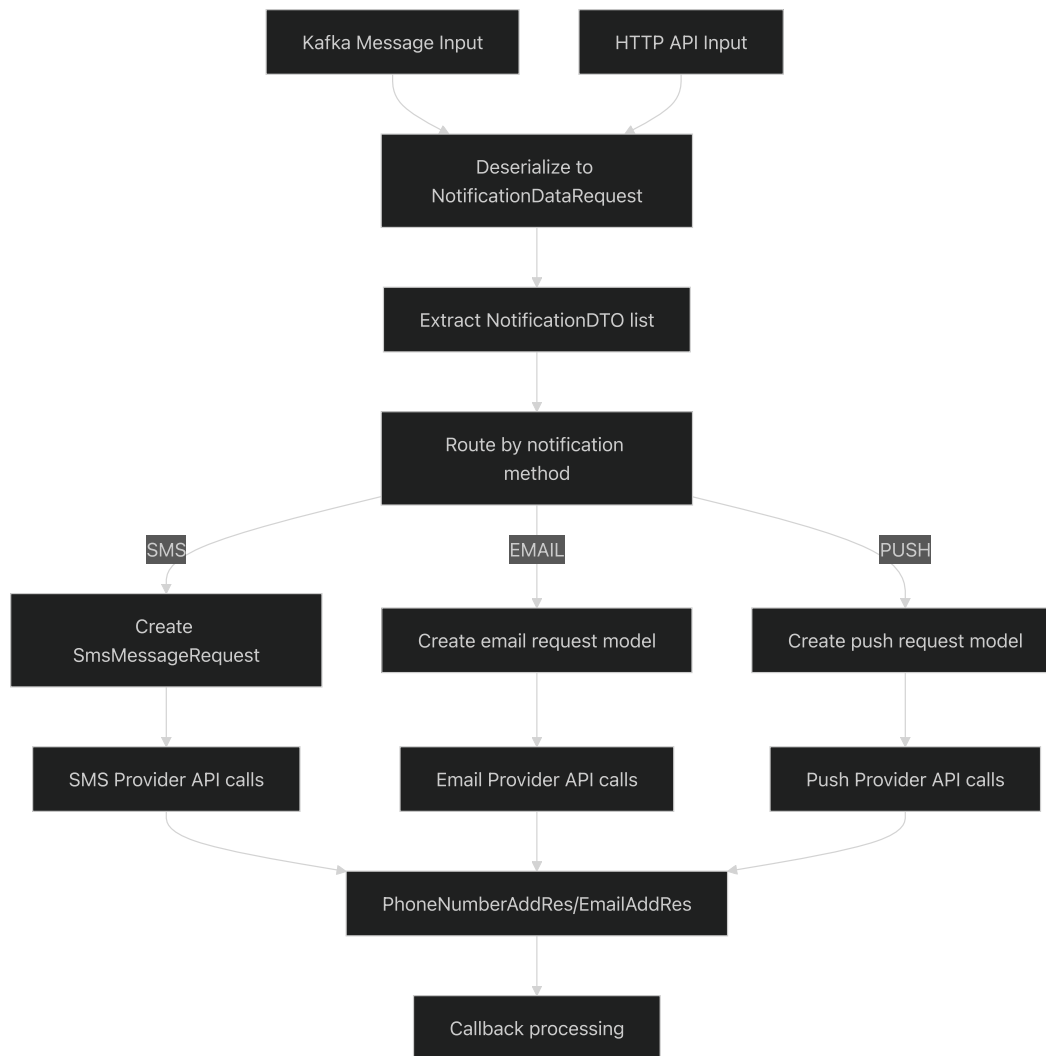
[L14](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java#L6-L14) ([https://github.com/nhsv-](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java#L6-L14)

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java#L6-L14](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java#L6-L14)))

API Data Flow Architecture

The notification service processes data through several stages, with different models used at each stage of the pipeline.

Notification Processing Pipeline



Sources: <https://github.com/nhsv->

[tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/NotificationDataRequest.java#L13-L16](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/NotificationDataRequest.java#L13-L16)

[src/main/java/com/techx/tradex/notification/model/NotificationDataRequest.java#L13-L16](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/NotificationDataRequest.java#L13-L16) <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/NotificationDataRequest.java#L13-L16>

[src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java#L6-L9](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java#L6-L9) <https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java#L6-L9>

Request and Response Model Patterns

Static Factory Pattern

Both `EmailAddRes` and `PhoneNumberAddRes` implement a static factory pattern for object creation:

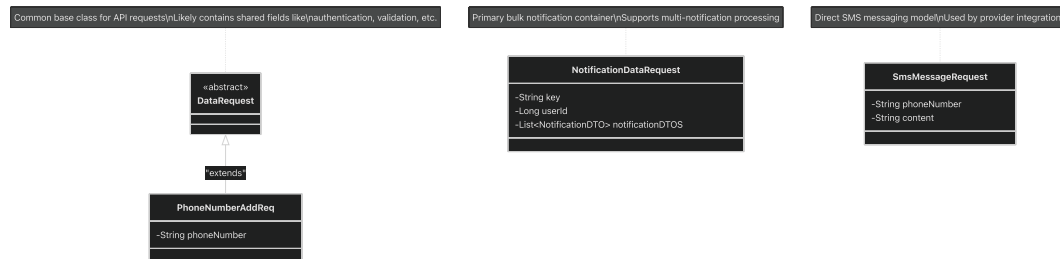
```
public static EmailAddRes fromNotification(String email)
public static PhoneNumberAddRes fromNotification(String phoneNumber)
```

This pattern provides a clean API for creating response objects from notification data, ensuring consistent object construction across the system.

Inheritance Hierarchy

The `PhoneNumberAddReq` class extends `DataRequest`, indicating a common base class pattern for request objects. This suggests a standardized request structure that may include common fields like authentication tokens, request IDs, or validation metadata.

Model Inheritance Structure



Sources: <https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddReq.java#L8-L8>).

<src/main/java/com/techx/tradex/notification/model/PhoneNumberAddReq.java8> (<https://github.com/nhsv->
<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddReq.java#L8-L8>)

<src/main/java/com/techx/tradex/notification/model/EmailAddRes.java9-13>](<https://github.com/nhsv->
<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/EmailAddRes.java#L9-L13>)[
[%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/EmailAddRes.java#L9-L13)).

<src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java9-13>](<https://github.com/nhsv->
<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java#L9-L13> (<https://github.com/nhsv->
<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java#L9-L13>))

Integration with External Services

These data models serve as the bridge between the notification service's internal processing and external service APIs. The models are designed to support the multi-provider architecture by providing standardized data structures that can be adapted to different provider requirements.

Provider-Agnostic Design

The `SmsMessageRequest` model provides a provider-neutral interface that can be used across different SMS providers (FPT SMS, OneSignal SMS, VCSC SMS, etc.) without requiring provider-specific modifications to the core notification processing logic.

Response Model Consistency

The static factory methods in response models (`fromNotification`) ensure consistent response formatting regardless of which external service provider is used, supporting the system's multi-provider fallback capabilities.

Sources: <https://github.com/nhsv->

<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java#L1-L11>)

<src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java1-11> (<https://github.com/nhsv->
<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/SmsMessageRequest.java#L1-L11>)

<src/main/java/com/techx/tradex/notification/model/EmailAddRes.java9-14>](<https://github.com/nhsv->
<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/EmailAddRes.java#L9-L14>)[
[%5B](https://github.com/nhsv-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/EmailAddRes.java#L9-L14)).

<src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java9-14>](<https://github.com/nhsv->
<tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java#L9-L14>)

[L14](https://github.com/nhsy-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java#L9-L14) (<https://github.com/nhsy-tradex/notification/blob/c316bffb/src/main/java/com/techx/tradex/notification/model/PhoneNumberAddRes.java#L9-L14>)
